ALL-IN-ONE

Hybrid Inverter

KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS

KR-L8K-SPUS	8 kW
KR-L10K-SPUS	10 kW
KR-L12K-SPUS	12 kW
KR-L15K-SPUS	15 kW

- 120/240 V split phase, 60 Hz
- 3 MPPT solar inputs
- 51.2 V lithium or lead-acid battery
- Built-in PV, battery and load breakers



User Manual

Distributed and supported by Fantastech Energy Solution LLC · Goshen, Indiana

Read Before Operating

KOOR All-in-One Hybrid Inverter · KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS

READ THE INSTRUCTIONS COMPLETELY BEFORE OPERATING THE EQUIPMENT.

1. Check the utility voltage before you turn the unit ON.
2. Verify the inverter's programmed grid type before you connect it to the utility.
3. By default, the unit is programmed for 120/240 V split phase at 60 Hz.
4. Disregarding these instructions could result in permanent damage to the unit.

⚠ DISCLAIMER

UNLESS SPECIFICALLY AGREED TO IN WRITING:

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KOOR

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All-in-One Hybrid Inverter — Quick Start

KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS | 120/240 V split phase, 60 Hz

Keep this sheet with the inverter. It covers the steps you use on the job. Full detail, settings and fault codes are in the **User Manual** in this same book. **Qualified electrical personnel only.**

1 WHAT IS ON THE UNIT

Front panel — indicator LEDs	
1	**Green** — ON: the inverter is running. Flashing: standby.
2	**Blue** — ON: communication with the battery BMS is normal.
3	**Yellow** — ON: the inverter is in EPS (backup) mode.
4	**Red** — the inverter is in fault mode.
Buttons	
5	**UP** — go to the previous selection
6	**DOWN** — go to the next selection
7	**ENTER** — confirm the selection
8	**ESC** — exit setting mode

Outside of the unit	
9	PV DC disconnect
10	LCD screen
11	BAT breaker
12	Parallel RJ45 ports
13	Input pinouts for sensors and accessories
14	**BMS RJ45 ports** (RS485 / CAN)
15	Load breaker
16	Wi-Fi port (for the Wi-Fi logger stick)
17	ON/OFF button

Inside the wiring compartment	
18	Battery terminals
19	I-LOAD terminals
20	3 PV inputs (MPPT trackers), 2 strings each
21	GEN (generator) terminals
22	LOAD terminals
23	Ground busbar
24	GRID terminals

Parallel systems

Up to **6 units**, **90 kW**. Same model and **same firmware** on all (ARM, INV, DCDC on Device Info). **One shared battery bank.** All inputs and outputs shared, except PV.

2 WHERE IT CAN HANG

Site limits	
Max. temp	+60 °C (derates above +45 °C)
Min. temp	-25 °C
Humidity	100%, no condensation
Altitude	4000 m (derates above 2000 m)
Enclosure	NEMA 4X, sealed
Clearance	500 mm free on every side

Yes

- Solid vertical wall, unit straight up and down
- Out of direct sunlight, rain and snow
- Out of reach of children
- Dry, ventilated, 500 mm clear all around

No

- No tilting forward or backward, and never lay it down
- No water pipes or cables buried in the wall behind it
- No flammable material, no corrosive or explosive area
- No direct sun, rain or snow on the unit

3 MOUNTING — 5 STEPS

1. Mark the bracket holes on the wall.
2. Drill **10 mm** holes.
3. Fasten the bracket to the wall with the M8 anti-collision bolts.
4. Lift the inverter onto the bracket.
5. Lock it down with the mounting screws, two at each side.

4 GENERATOR — FANTASTECH RULES · WIRE COLORS

Generator	Land it on
120/240 V split phase, **under 15 kW**	**GEN port** (63 A)
120/240 V, **15 kW or bigger**	**GRID port**
120/208 V (two legs of a three-phase generator)	**GRID port — never the GEN port**
Three-phase power	**NEVER to the GEN port**

Generator charge time: Start 1 minute after End (End **00:00**, Start **00:01**), so the battery **charges whenever the generator runs**. More: User Manual 6.8 and 9.9.

Wire colors — every job, every drawing	
L1	Black
L2	Red
L3	Blue (120/208 V only)
Neutral	White (drawn grey in the manual)
Ground	Green or bare copper
DC +	Red — battery and PV positive
DC –	Black — battery and PV negative

Check the box and the unit for damage before you sign for it. Box contents list: User Manual 4.2.

Off-grid jobs — Fantastech setup

- **ACID Fallback = Yes** (9.7): keeps power on 30 min if the battery stops talking.
- **GRID Warn = No** (9.13): no false grid alarms.
- Gen on the GRID port: **Gen Connect To Grid = Yes** (9.9).
- OFF from the app: set **Auto Power On = No** first (9.6), or it restarts in 10 s.

Quick Start — Wiring, Start-Up and Checks

KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS | 120/240 V split phase, 60 Hz

5 CABLES

#	Cable	Type	Size
1	**Grounding**	Single-core green (or green/yellow) copper, outdoor rated	6 mm² (10 AWG)
2	**AC output**	Two or three different-color multi-core copper	2/0–4/0 AWG
3	**PV input**	PV cable (such as PV1-F)	12–10 AWG
4	**Battery**	Red and black multi-core copper	2/0–4/0 AWG
5	**Communication**	CAT5e	—

Conductor	Strip length	Finish the end with
PV (12–10 AWG)	16–18 mm (5/8–11/16 in)	Wire ferrules if stranded
Battery (2/0–4/0 AWG)	6–8 mm (1/4–5/16 in)	Crimped OT ring lugs
GRID and LOAD	8–10 mm (5/16–3/8 in)	Crimped ring lugs (2/0–4/0 AWG)

Terminal sizes (User Manual 6.2.2): ring lug for up to 4/0 AWG — 1 in [25.4 mm] wide, 3/8 in [10 mm] hole; ferrule for up to 4 AWG (GEN and I-LOAD wire, 6–4 AWG) — 5/8 in [16 mm]; ferrule for up to 10 AWG — 5/8 in [16 mm]; ferrule for up to 20 AWG — 1/4 in [6.35 mm]. Strip lengths are from User Manual 6.5, 6.6.1 and 6.7.3.

Before you touch a terminal

- **Every switch OFF** — PV, battery, grid, load, generator. Never work on it with power on.
- The lithium battery has its **own** power line and comm line — use the ones that came with it.
- The inverter ships with the basic comm line — use it directly.
- **Do not reverse battery polarity.** It damages the system and voids the warranty.
- Never land the grid on the **LOAD** output terminals.

PV check before you land it

Meter every string open-circuit before you connect it. **550 Vdc** is the absolute maximum; size every string so its cold-corrected Voc is **at or below 525 Vdc**.

PV+ = RED · PV− = BLACK. Crimp the pin, push the cable in firmly and press the catch until it clicks.

6 COMMUNICATION PORTS

	Port	What it is for
A	NTC	Temperature sensor for a lead-acid battery
B	CT interface	Current sensor clamps
C	GEN (NO)	Generator auto-start dry contact. When the GEN signal is active, the contact closes — **no voltage output**.
D	±12 V RSD	Power for the RSD PLC transmitter (500 mA max.)
E	EPO	Reserved for an external emergency stop
F	Meter RS485	Energy meter communication
G	ATS	240 Vac output port, live when the inverter is on
H	**Battery comm**	CAN and RS485 to the battery BMS
I	Parallel comm	CAN and RS485 between inverters
J	DIP switches (BMS CAN / parallel CAN resistors)	Switch 1 always ON; switches 2–4 ON on the first and last inverter, OFF in between and on a single inverter (6.9.1)

BMS RJ45 pinout (battery port)

Pin	Signal	Pin	Signal
1	BAT RS485 A	5	**BAT CAN L**
2	BAT RS485 B	6	NC
3	NC	7	BAT RS485 B
4	**BAT CAN H**	8	BAT RS485 A

Only the **master** battery gets a comm cable to the inverter. Set Bat Type, Comm Type and Bat Protocol on the screen (User Manual 9.7). Comms come up in 1–2 minutes. Damage from the wrong protocol is not covered by the warranty.

7 TURN IT ON — IN THIS ORDER

1. Turn on the **battery** breaker(s). Wake the battery up if it has a power button.
2. Press the **power button** in to ON. Wait for the LED (a few minutes).
3. Turn on the **PV** switch, then the **GRID** disconnect.
4. Turn on the **LOAD** breaker and the **GEN** breaker. Set **Power On/Off** to ON on the screen (Work Mode 2/4, section 9.6), then set up the rest (chapter 9).

8 SHUT IT DOWN

1. All AC breakers off: **GRID**, **GEN**, **LOAD**.
2. **PV** switch off, then the **power button** off (wait for "OFF").
3. **Battery** breaker(s) off last. Wait at least 5 minutes before you touch anything.

Shutdown is start-up backwards. Before you power on, make sure every voltage and current is inside the inverter spec — outside it, the inverter gets damaged. Questions before you energize it? Call Fantastech at 574-218-1973.

9 POST-INSTALL CHECK

☐ Inverter mounted correctly and firmly	☐ All switches off
☐ Wi-Fi logger stick installed correctly and firmly	☐ Ground wire connection confirmed and reliable
☐ Cable runs reasonable, no broken insulation	☐ All wires correct and securely connected
☐ Cable ties trimmed, no sharp corners	☐ Exposed terminals protected, no open ports
☐ All packing and scrap material cleaned up and hauled out	

10 APP AND MONITORING

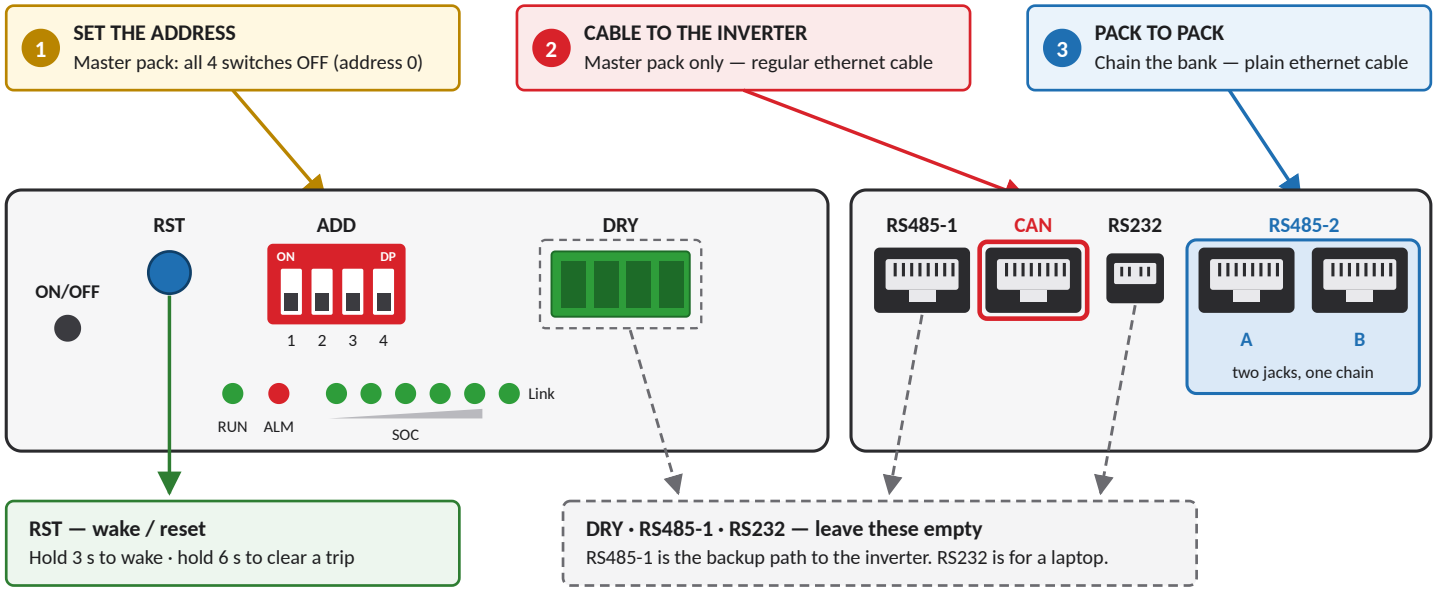
App	**SOLARMAN Smart** — App Store (iPhone) or Google Play (Android)
Add the logger	Scan the barcode on the stick label, or type its serial number (SN).
PWD	Type the stick's PWD in **lowercase** (label FEE32734 → fee32734).
Site Wi-Fi	Join the stick's own Wi-Fi, open **10.10.100.254** in the phone's browser, pick the site Wi-Fi and save.
After that	The stick uploads every 5 minutes. A new logger shows **Offline** until its first upload — wait at least 10 minutes. Full steps: User Manual chapter 8.

Appendix — KOOR batteries on a KOOR All-in-One

Where every comm cable plugs in | KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS

Four of these jacks look the same, and three of them say RS485. Only two jacks get a cable on a normal job. Go by the words stamped on the panel. This page shows the KOOR battery comm panel on the KOOR 16K Pro tower and the KOOR Active Stack S314.

THE COMM PANEL — AS YOU FACE THE BATTERY



Drawn from a photo of a real KOOR 16K Pro panel and the KOOR Active Stack S314 User Manual, section 5.2

WHAT GOES IN EACH ONE

	Stamped on the panel	What plugs in	Which packs	Cable
1	**ADD** (red DIP switch)	Nothing — it is a switch	Every pack, a different number each	**Master = address 0:** all 4 switches OFF. Next packs 1, 2, 3… no repeats. Set them before power-on.
2	**CAN** — the right-hand jack of the RS485-1/CAN pair	Goes to the inverter BMS RJ45 port	**Master pack only**	**A regular straight ethernet cable.** Both ends wired the same. Nothing crossed, nothing special built — this is how Fantastech has done it many times.
3	**RS485-2** — the pair on the far right	Pack to pack, down the line	Every pack in the bank	Plain straight ethernet, all 8 pins fine. Out of **B** on one pack, into **A** on the next. Keep it the same all the way down.
–	**DRY**, **RS485-1**, **RS232**	Nothing on a normal job	—	RS485-1 is the backup path to the inverter if CAN will not come up. RS232 is the laptop / BMS-software port. DRY is a dry contact.

Two jacks stay empty on purpose. On the master pack, the **A** jack of RS485-2 has nothing coming into it. On the last pack in the line, the **B** jack has nothing leaving it. That is how you know the chain is a line and not a ring.

16K Pro and Active Stack wire the same. Regular ethernet on both, no crossing, no custom cable. This panel is drawn from a **16K Pro**. The Active Stack has the same layout, but its manual shows the last pair stamped **RS485A / RS485B** instead of RS485-2 — go by what is on the panel in front of you.

Appendix — KOOR batteries on a KOOR All-in-One

Which jack, how the packs chain, DIP addresses | KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS

Read this first. Batteries covered: KOOR Active Stack S314 (51.2 V, 314 Ah, 16 kWh) and the KOOR 16K Pro tower. Two cables, a DIP switch and Pylon on the battery are the whole job. One **regular ethernet cable** runs from the master battery jack marked **CAN** to the inverter, and one runs from each pack to the next on **RS485-2**. Set every address before power-up, then set every pack to **Pylon** (power-up step 5).

1 WHICH JACK GOES WHERE

On the inverter	
Port	**BMS RJ45 port** — port **H** on the communication board (User Manual 6.3), item **14** in the Quick Start
Not this one	The **Parallel** RJ45 ports (Quick Start item 12, port I) are inverter-to-inverter only
Signal	Pin 4 = CAN-H · pin 5 = CAN-L · pins 1, 2, 7, 8 = RS485

On the battery	
Active Stack S314	The jack stamped **CAN** — the right one of the RS485-1/CAN pair. Pin 4 = CAN-H, pin 5 = CAN-L. **RS485-2** is the pack-to-pack pair. **RS485-1** is the backup path to the inverter.
16K Pro tower	Same — the jack stamped **CAN**. It wires the same way as the Active Stack.

The cable — a regular ethernet cable	
Type	**Straight CAT5e or CAT6, both ends wired the same**
Battery end	The jack stamped **CAN** — the right one of the RS485-1/CAN pair
Inverter end	The **BMS RJ45** port
Signals	CAN-H on pin 4, CAN-L on pin 5, straight through

Both KOOR batteries wire the same. Active Stack or 16K Pro, it is a regular ethernet cable off the CAN jack. No crossed cable, no cable to build. Fantastech has put this combination in many times.

2 HOW THE PACKS CHAIN — ONE PACK TALKS, THE REST REPORT THROUGH IT

PACK 1
MASTER
address 0 · all 4 OFF
CAN jack → inverter

→

PACK 2
address 1
RS485-2 A in from pack 1
RS485-2 B out to pack 3

→

PACK 3
address 2
RS485-2 A in from pack 2
RS485-2 B out to pack 4

→

… up to 16 packs
addresses 0–15
pack 16 = all 4 ON

- **Only the master** gets a cable to the inverter. Nothing else does.
- The rest chain **in a line** on the **RS485-2** pair: out of **B** on one pack, into **A** on the next, all the way down.
- **Never** run a cable from the last pack back to the master. It is a line, not a ring.
- The master RS485-2 **A** jack stays empty. The last pack RS485-2 **B** jack stays empty.
- **No two packs share an address.**
- Adding a pack later? Keep it **within 2 V** of the others, or it trips charge overcurrent and will not take a charge.

Set every DIP switch before anything is powered up. The BMS reads its address at power-on. Changed one after? Hold **RST 3 s** (sleep), then **3 s again** (wake) on that pack.

Master address — ADDRESS 0: all 4 switches OFF. It is the pack cabled to the inverter. The rest get 1, 2, 3 … up to 15 (16 packs in all), no repeats.

3 DIP SWITCHES — SET EACH PACK LIKE ITS PICTURE · BIGGER PICTURES ON THE NEXT PAGE

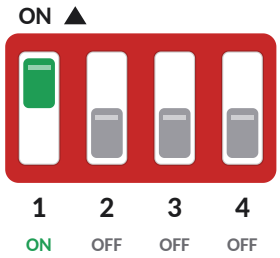


Appendix — Setting the battery address switches

One small red switch on every battery pack, marked ADD | KOOR Active Stack S314 and KOOR 16K Pro

Every battery pack gets its own number. Pack 1 is the master — the one with the cable to the inverter — and it gets **all 4 switches OFF** (address 0). Set the switches with the batteries **turned off**. Find your pack number below and make the switches look exactly like the picture.

1 HOW TO READ THE SWITCH



Slider pushed UP, toward ON
= that switch is ON

Slider pushed DOWN
= that switch is OFF

Switches are numbered 1, 2, 3, 4 from the left.
Use a small screwdriver or a pen tip.
This one reads: 1 ON, 2 OFF, 3 OFF, 4 OFF.

2 SET EACH PACK LIKE ITS PICTURE — GREEN SLIDER = PUSHED UP TO ON



Only 1 battery? It is the master — all 4 switches OFF. **No two packs may have the same picture.** If you change a switch after the battery is on, hold **RST 3 s** (sleep), then **3 s again** (wake) on that pack so it reads the new number.

Appendix — KOOR batteries on a KOOR All-in-One

Settings, power-up and troubleshooting | KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS

4 INVERTER SETTINGS — BATTERY MENU

Setting	Set it to	Why
Bat Type	**Lithium**	Other choices are Lead-acid, DC Source and No Battery.
Comm Type	**CAN**	Both KOOR batteries talk on CAN. RS485 is the fallback, not the first pick.
Bat Protocol	**Pylon**	Pick the **Pylon** entry on the list. Current screens read **PYLON LV**; older firmware may show **Pylon LV2** — both are the Pylon protocol. The inverter then matches the protocol on its own.
Activate Bat	Use it once	Wakes a sleeping pack if the inverter sees nothing.
Grid Charge	Per job	On for grid or generator charging, off for PV only.
Shared Bat	No	Single inverter: **No**. Yes only when inverters in a parallel group share one bank; then the comm cable goes to the **master** inverter (User Manual 9.7).
Max A Charge / Max A Disch	Lower of the two	Whichever is smaller, the bank limit or the inverter limit. With CAN comms up, the BMS reports its own limits.

Why Pylon. KOOR batteries speak the Pylon low-voltage protocol over CAN, and **both ends must be on it** — some packs come set to a different protocol, so set Pylon on every pack too. A KOOR All-in-One in the field came up right on the Pylon setting, and KOOR batteries on KOOR off-grid inverters use the same Pylon-over-CAN setting.

5 POWER-UP ORDER

1. **Everything off.** Set every DIP address before a single pack is powered.
2. **Cable it.** Chain the packs on RS485-2, then run a regular ethernet cable from the master CAN jack to the inverter BMS port.
3. **Recheck the DIPs.** Master on address 0 (all 4 OFF), no two packs alike.
4. **Batteries on.** DC breaker on every pack, then each power switch. Wake the master if it sits dark.
5. **Pylon on the battery.** Every pack: CAN protocol = Pylon (battery app, password 1234), then **RST 3 s**, 3 s.
6. **Inverter on.** Turn on the inverter BAT breaker, then press the power button in to ON (User Manual 7.1.4).
7. **Set the inverter menu.** Lithium → CAN → Pylon. Then finish the start-up (User Manual 7.1.4): PV, GRID, LOAD/GEN breakers on, and set **Power On/Off** to ON on the screen (Work Mode 2/4, section 9.6).

How you know it worked — three things, all of them:

- The **blue LED** on the inverter is **ON** (BMS communication normal)
- A **live SOC %** shows on the screen
- **Charge and discharge limits** show, and they change

Give it **1 to 2 minutes** after you pick the brand.

6 IT WILL NOT TALK — WORK DOWN THIS LIST

What you see	What to check
Blue LED off, no SOC	Cable in the wrong jack, or a bad cable. It goes in the jack stamped **CAN**, not RS485-1. Swap in a known-good ethernet cable.
Blue LED off, cable is right	**Is the battery on Pylon?** Set CAN protocol = Pylon on every pack (battery app), restart each pack. Then Comm Type CAN and Bat Protocol Pylon on the inverter. Wait 2 minutes.
Only some packs show	Two packs share an address, or the chain is broken. Count them on the PACKS tab of the battery screen.
Master dark	Pack asleep. Wake it at the battery, then use **Activate Bat** on the inverter.
Nothing at all	Check the master DC breaker and power switch before anything else.

Do not leave it on Lead-acid and walk away. The manual lets you pick Lead-acid when lithium will not communicate. That runs the bank **open loop** — no SOC, no BMS limits, no protection handoff. Use it to finish a day, never to finish a job. Stuck on comms? Call Fantastech at 574-218-1973 and we will get the comms right.

Damage from the wrong communication protocol is not covered by the warranty. Neither is reversed battery polarity. Meter before you land anything.

Where this came from. The KOOR All-in-One User Manual (section 6.6 and the Battery Settings menu, section 9.7); the KOOR Active Stack S314 User Manual, section 4.3; the KOOR off-grid 12K manuals, Appendix A; and a KOOR All-in-One brought up in the field on the Pylon setting. **Fantastech has installed this combination many times** — if it will not talk, the cause is almost always the wrong jack, a bad cable, a pack not set to Pylon, or a duplicate address.

Limited Warranty — Terms

KOOR All-in-One Hybrid Inverter · KR-L8K · KR-L10K · KR-L12K · KR-L15K-SPUS | sold and warranted by Fantastech Energy Solution LLC

WHAT IS COVERED AND FOR HOW LONG

KOOR All-in-One Hybrid Inverter — 10 years

KOOR batteries — 10 years

The term runs from the date of installation, but it ends no later than six months after the same term would end if counted from the date the unit left the factory.

(KOOR off-grid inverters, sold separately, carry a 5-year warranty.)

The warranty covers **defects of design, components and manufacturing**. It applies to devices bought from Fantastech and installed at the address on the Warranty Registration Card. It is non-transferable unless Fantastech agrees in writing.

WHAT IS NOT COVERED

- Incorrect installation or commissioning.
- Failure to follow the User Manual, the installation guide or the maintenance instructions.
- Unauthorized modifications, changes or attempted repairs.
- Incorrect use or inappropriate operation, including reversed battery polarity and use of the wrong battery communication protocol.
- Insufficient ventilation of the device.
- Failure to observe applicable safety regulations and electrical code.
- Breaking the product seal — opening the casing — without written approval.
- Force majeure: lightning, over-voltage, storm, flood, fire.
- Transport damage, and superficial or cosmetic defects — dents, marks, scratches — that do not affect how the inverter works.

HOW A CLAIM WORKS

- **Report it within 14 days** of the first sign of a problem. Call 574-218-1973 or e-mail derek@fantastech.org.
- Send the completed Warranty Registration Card (at the back of this manual), the purchase invoice, the installation or commissioning report, and any extended warranty certificate if you have one.
- Send pictures and the fault code or fault screen. It is the fastest way to get a unit moving.
- Fantastech decides whether the unit is repaired on site, repaired at the shop, or exchanged for a replacement device of equivalent value for model and age.
- On an exchange, the remainder of the original warranty carries over to the replacement device. A replacement does not start a new warranty term and does not come with a new certificate.
- Because of ongoing product improvement, a replacement part or unit may not match your system monitoring or other components, and the cost of changing those out is not part of this warranty.

LIMITS

- This warranty covers the equipment. It does not cover labor, travel, lift or crane rental, lost production, lost profits, or other direct or indirect costs, unless Fantastech agrees to them in writing beforehand.
- Claims beyond the rights stated here — including costs of removal and reinstallation — are not covered.
- Your rights under statutory liability and product liability law are unaffected by this warranty.

Keep a copy of your Warranty Registration Card and your invoice together. Both are needed to support a claim. If the card was never sent in, send it with the claim.

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1. About This Manual

This manual covers the product information, installation, electrical connection, commissioning, troubleshooting and maintenance of the inverter. Read the whole manual before you install or operate the product. All installers and users must know the product's features, functions and safety precautions. This manual may be updated without notice. For more product details and the latest documents, contact Fantastech (574-218-1973, derek@fantastech.org).

1.1 Applicable Models

Model	Nominal output power	Nominal output voltage
KR-L8K-SPUS	8 kW	120 Vac / 240 Vac (split phase); 208 Vac (two legs of a three-phase wye) 2L/N/PE (L1, L2, neutral and ground)
KR-L10K-SPUS	10 kW	
KR-L12K-SPUS	12 kW	
KR-L15K-SPUS	15 kW	

1.2 Target Group

This manual is for qualified electrical technicians who install and commission hybrid inverters in energy storage systems and electrical systems.

1.3 Symbol Definitions

This manual uses the following types of safety instructions and general information:

DANGER "Danger" marks a hazard with a high level of risk. If it is not avoided, it **will** cause death or serious injury.

WARNING "Warning" marks a hazard with a medium level of risk. If it is not avoided, it **could** cause death or serious injury.

CAUTION "Caution" marks a hazard with a low level of risk. If it is not avoided, it could cause minor or moderate injury.

NOTICE "Notice" gives tips and methods to solve product problems and save time.

1.4 Updates

The latest issue of this document contains all the updates made in earlier issues.

Version	Date	Changes
V00 / V01 / V02	2026-03-20 / 2026-04-20 / 2026-10-14	First / second / third issue
KOOR 1 / KOOR 2	2026-09 / 2026-11	KOOR plain-English edition / updated to issue V02

2. Safety Precautions

Follow the safety instructions in this manual at all times when you work on or operate the inverter.

2.1 General Safety

NOTICE

- The information in this manual may change because of product updates or other reasons. Unless stated otherwise, this manual does not replace the labels on the product or its safety precautions. All descriptions here are for guidance only.
- Before you install, read the Quick Start at the front of this manual. For more information, read the rest of this manual.
- All installation work must be done by trained, knowledgeable technicians who know the local codes and safety rules.
- Use insulated tools and wear personal protective equipment (PPE) when you work on the equipment. Wear anti-static gloves, clothes and wrist straps when you touch electronic parts. This protects the inverter from damage.
- Follow the installation, operation and setup instructions in this manual exactly. KOOR is not liable for equipment damage or personal injury if you do not follow the instructions. For warranty details, see the Limited Warranty — Terms page at the front of this manual.

2.2 PV String Safety

DANGER Connect the PV cables only to the inverter's PV terminals, as shown in section 6.5. KOOR is not liable for equipment damage if other connectors or terminals are used.

WARNING

- Make sure the module frames and the racking system are securely grounded.
- Make sure the DC cables are connected tightly, securely and correctly.
- Check the DC cables with a multimeter before you connect them, so the polarity is not reversed. Also check that the voltage is within the allowed range (see the Vmax PV row in section 3.3.4).
- Do not connect one PV string to more than one inverter at the same time. This can damage the inverter.
- The PV modules used with the inverter must have an IEC 61730 Class A rating.
- When light falls on the PV array, the array sends DC voltage to the inverter.

2.3 Inverter Safety

WARNING

- The voltage and frequency at the connection point must meet the grid-connection requirements.
- Extra protective devices, such as circuit breakers or fuses, are recommended on the AC side. Rate the protective device at no less than 1.25 times the AC rated output current.
- Make sure all ground connections are tight. When there are several inverters, make sure the grounding points on all the enclosures are equipotentially bonded.
- Off-grid use is not recommended if the PV system has no batteries. If you use it that way, problems with the system's power supply are not covered by the warranty.
- PV power is not steady. If no battery is connected, there is no EPS (backup) function in off-grid mode: with PV alone and no battery, there is no EPS output. When the grid is present, the EPS function is available.
- Do **NOT** connect the grid to the "LOAD" output terminals.

DANGER

- All labels and warning marks must stay visible after installation. Do not cover, write on or damage any label on the equipment.
- The warning labels on the inverter are shown below.

Label	Meaning	Label	Meaning
	DANGER: high voltage. Disconnect all incoming power and turn off the product before you work on it.		Delayed discharge. After power-off, wait 5 minutes until the components are fully discharged.
	Read the user manual before you work on this device.		Hazards exist. Wear proper PPE before any work.
	Hot surface. Do not touch the product while it is running, or you may be burned.		Grounding point.
	Do not throw the inverter away with household waste. Dispose of it as local laws and regulations require, or send it back to KOOR through Fantastech.		

2.4 Battery Safety

WARNING

- The battery used with the inverter must be approved by KOOR. For the approved battery list, contact Fantastech (574-218-1973, derek@fantastech.org).
- Before you install, read the battery's own user manual to learn about the product and its precautions. Follow its requirements exactly.
- If the battery is completely discharged, charge it exactly as that battery model's user manual says.
- Factors such as temperature, humidity and weather can limit the battery current and affect the load it can carry.
- If the battery will not start, contact Fantastech right away. Otherwise the battery may be damaged for good.
- Check the DC cables with a multimeter before you connect them, so the polarity is not reversed. Also check that the voltage is within the allowed range.
- In a single-inverter system, do not connect one battery bank to more than one inverter at the same time. This can damage the inverter. (A parallel system is different: there the master and slave inverters must share one battery bank. See section 6.9.)
- Lead-acid batteries, including but not limited to flooded lead-acid (FLA), gel and absorbent glass mat (AGM) batteries, often need special charging methods and may work best only under certain conditions. The inverter may not fully meet these needs by default. You may need to change its charge and discharge settings.
- If you plan to use one of these battery types, ask the battery maker for advice on best practices, compatibility and overall system sizing for your application.

2.5 Personnel Requirements

NOTICE

- People who install or maintain the equipment must be fully trained. They must know the safety precautions and the correct way to operate the equipment.
- Only qualified professionals or trained personnel may install, operate, maintain or replace the equipment or its parts.

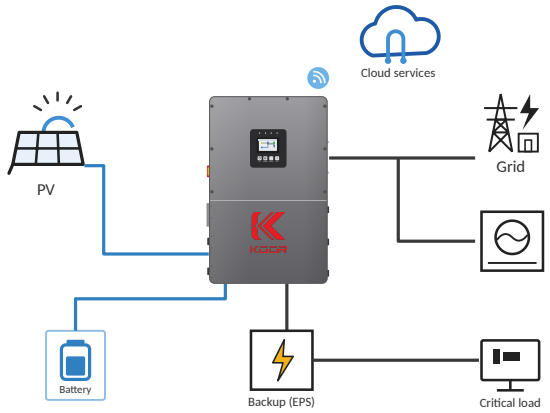
3. Product Introduction

3.1 Product Features

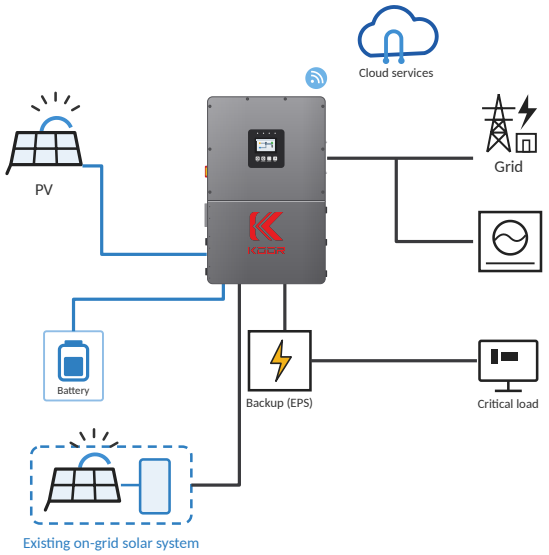
Intended use

This is a multifunction inverter. It combines an inverter, a solar charger and a battery charger in one wall-mounted unit, to give uninterruptible power. The inverter works with a phone app (SOLARMAN Smart, see chapter 8). In the app you can watch the inverter's operating status in real time, including battery information, PV generation, alarm information and more. The simple operating interface makes it easy to set the right work mode for each use, to meet many different needs.

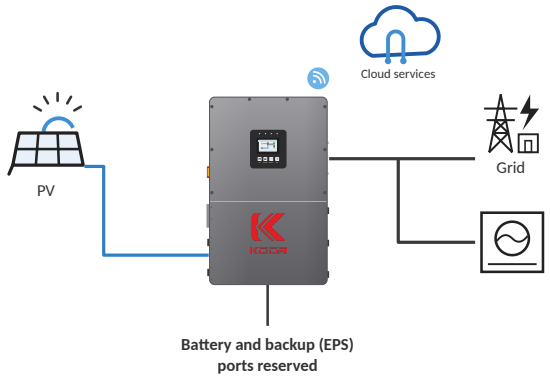
This unit and its system are suitable for the following uses:



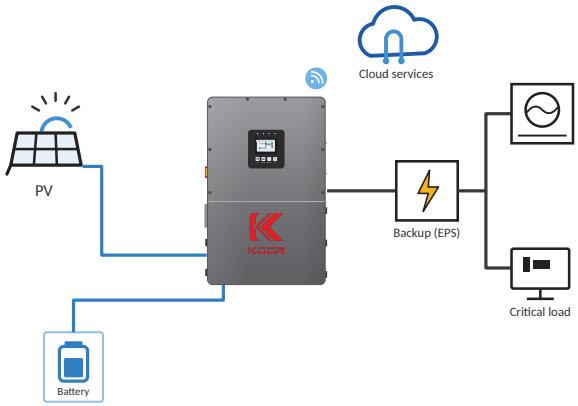
A: Solar and battery storage system



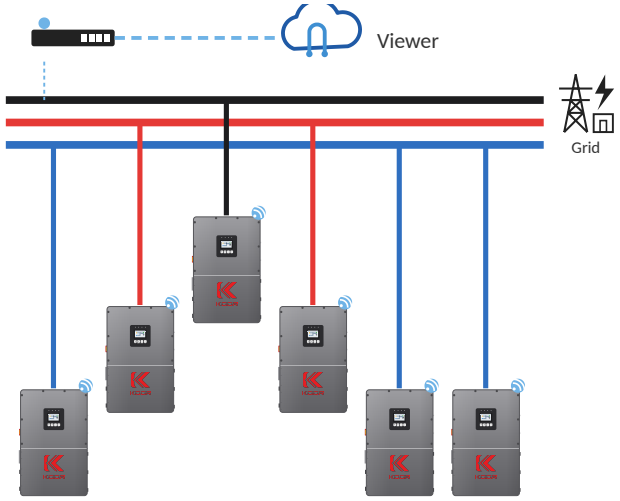
B: AC coupling with an existing on-grid solar system



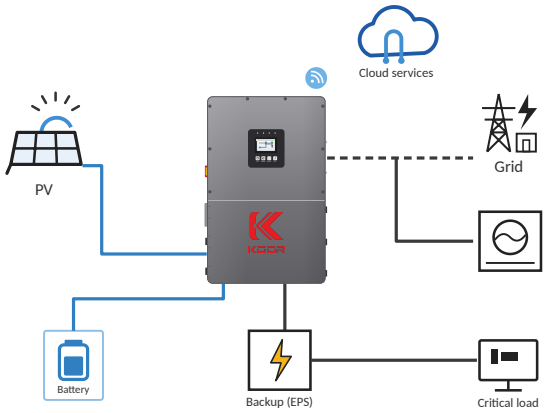
C: On-grid solar system without a battery; battery and backup (EPS) ports reserved. EPS works only while the grid is present; with PV alone and no battery there is no EPS output



D: Off-grid and backup applications



E: Single-phase and three-phase parallel system



F: Energy storage system with peak shaving

Models and serial number

This manual applies to the KR-L8K-SPUS, KR-L10K-SPUS, KR-L12K-SPUS and KR-L15K-SPUS inverters.

The serial number (SN) is built from the codes below, in this order: S + XXXXXXXX + 24 + 01 + 11 + 00001. The example values show an inverter made on 11 January 2024.

No.	Part of the SN	Code (example)	Meaning
1	Brand code	S	Brand
2	Product category	XXXXXXXX	Inverter model ID
3	Production date	24	Year of production
4	Production date	01	Month of production
5	Production date	11	Day of production
6	Production serial number	00001	Unit number

3.2 Working Modes

The KR-L8K–L15K-SPUS series hybrid inverter has the work modes below. Pick the mode that fits your system and site.

Work mode	Description
Self Use (with PV power)	Priority: load > battery > grid PV energy is used first to cover your own use. Extra energy charges the batteries, and what is left is exported to the grid.
Self Use (without PV power)	Priority: load > battery When there is no PV power, the battery discharges first to power the local loads. The grid supplies power when the battery capacity is not enough.
TOU ECO[1]	In this mode you give each time window that you set its own working strategy, to get the most savings. **Discharge:** export first. After the local loads are covered, the energy that is left is fed into the grid first. **Self Use:** the system runs on self-consumption. PV energy and stored battery energy are used on site first, to cut the power bought from the grid. **Charge:** standby storage. The system actively charges the battery from extra PV power or from grid power, to keep the battery at a high state of charge, ready for backup.

Work mode	Description
Backup Power	Priority: load > battery In this mode the system powers the home loads first. Extra PV power or grid power charges the battery until it is full. When the grid is off, the system supplies emergency power from PV or the battery to the home loads.
Grid Priority	Priority: load > grid In this mode, power from PV and from battery discharge goes to the home loads first. Extra power is fed into the grid. You can use this mode to answer grid requests at peak times.

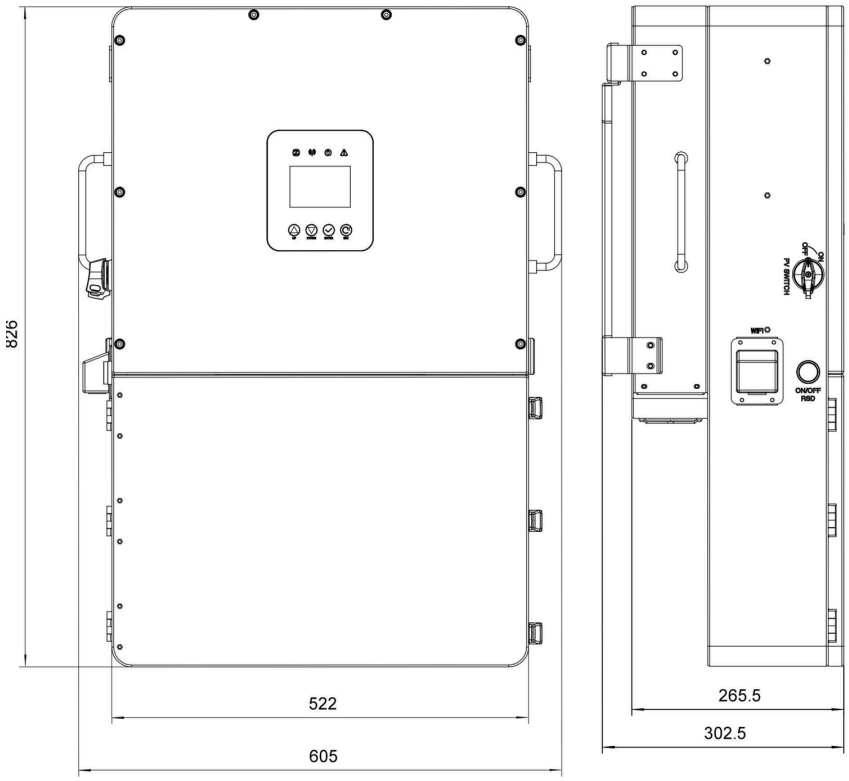
[1] TOU means time of use.

DANGER Make sure the total load power is within the EPS output rating. If it is not, the inverter shuts down with an "over load" warning. When "over load" appears, reduce the load power until it is within the EPS output range, then turn the inverter on. For nonlinear loads, check that the inrush (starting) power is also within the EPS output range.

3.3 Appearance

3.3.1 Dimensions

Keep the size of the inverter in mind when you choose a location for the system. Leave enough space above, below and on the left and right sides of the inverter so heat can escape (see section 5.1).

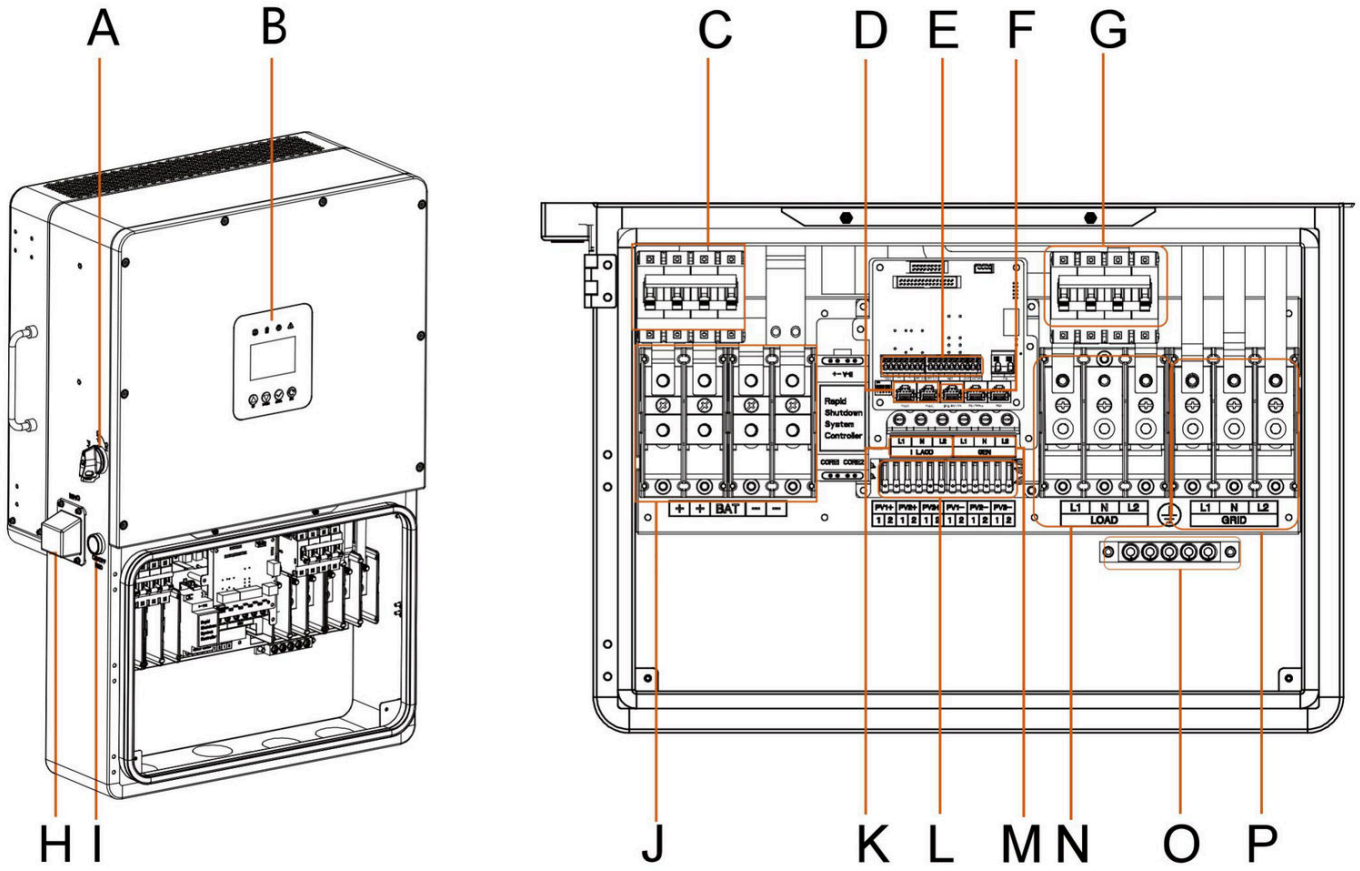


Front and side views, in mm: 522 wide (605 across the handles), 826 high, 265.5 deep (302.5 with the wall bracket)

Protect the LCD screen from direct exposure to UV light.

The inverter has a NEMA 4X enclosure. It is rated for outdoor installation, but it can also be installed indoors.

3.3.2 General Description



Parts of the inverter (left) and the wiring compartment (right); the letters match the table below

Item	Name	Item	Name
A	PV DC disconnect	I	ON/OFF button
B	LCD screen	J	BAT terminals (battery)
C	BAT breaker	K	I-LOAD terminals (intelligent load)
D	Parallel RJ45 ports	L	3 PV inputs (MPPT trackers), 2 strings each
E	Input pinouts for sensors and accessories	M	GEN terminals (generator and Smart Load)
F	BMS RJ45 ports (RS485/CAN)	N	LOAD terminals
G	LOAD breaker	O	Ground busbar
H	Wi-Fi	P	GRID terminals

The terminal block for item K is marked "I LAOD" in the drawing; the correct name is I-LOAD.

NOTICE The RSD (Rapid Shutdown System Controller) shown in the wiring compartment is a customer-procured component. It is not supplied with the inverter.

3.3.3 Nameplate

The nameplate is on the side of the inverter. It lists the model number, serial number, electrical ratings and certification marks. Check the model number on the nameplate before you install or order parts.

3.3.4 Specification

Model	KR-L8K-SPUS	KR-L10K-SPUS	KR-L12K-SPUS	KR-L15K-SPUS
Product type	Hybrid inverter			
Battery				
Battery type	Li-ion / lead-acid			
Battery voltage range	40–60 Vdc			
Rated battery voltage	51.2 Vdc			
Max. charge/discharge power	8 kW	10 kW	12 kW	15 kW
Rated charge/discharge current	167 Adc	210 Adc	250 Adc	275 Adc
Max. charge/discharge current	167 Adc	210 Adc	250 Adc	275 Adc
Reverse connection protection	Yes			
Communication interface	CAN / RS485			
Input (PV)				
Recommended max. PV array power (total per inverter)	12 kW	15 kW	18 kW	22.5 kW
Max. operating (rated) PV input current (PV1 / PV2 / PV3)	26 Adc + 26 Adc + 26 Adc (26 Adc per tracker)			
Max. PV short-circuit current, Isc (PV1 / PV2 / PV3)	39 Adc + 39 Adc + 39 Adc (39 Adc per tracker)			
Vmax PV (max. PV input voltage)	550 Vdc (design cold Voc at or below 525 Vdc)			
MPPT voltage range	108–520 Vdc			
Full-power MPPT voltage range	120–480 Vdc	135–480 Vdc	162–480 Vdc	202–480 Vdc
Start-up voltage	120 Vdc			
Number of MPP trackers	3			
Strings per MPP tracker	2			
Number of PV inputs	6			
Grid AC input and AC output				
Grid rated voltage	120 Vac / 240 Vac (split phase); 208 Vac (two legs of a three-phase wye)			
Grid rated frequency	60 Hz			
Grid rated input active power	8 kW	10 kW	12 kW	15 kW
Grid rated input apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid max. input active power	48 kW	48 kW	48 kW	48 kW
Grid max. input apparent power	48 kVA	48 kVA	48 kVA	48 kVA
Grid rated output active power	8 kW	10 kW	12 kW	15 kW
Grid rated output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid max. output active power	8 kW	10 kW	12 kW	15 kW

Model	KR-L8K-SPUS	KR-L10K-SPUS	KR-L12K-SPUS	KR-L15K-SPUS
Grid max. output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid rated input current	33.4 Aac	41.7 Aac	50 Aac	62.5 Aac
Grid max. input current	200 Aac	200 Aac	200 Aac	200 Aac
Grid rated output current	33.4 Aac	41.7 Aac	50 Aac	62.5 Aac
Grid max. output current	33.4 Aac	41.7 Aac	50 Aac	62.5 Aac
Grid power factor	0.8 leading to 0.8 lagging			
Grid input and output inrush current	500 Aac @ 3 µs			
Max. grid output fault current	67 Aac @ 0.2 µs	84 Aac @ 0.2 µs	100 Aac @ 0.2 µs	125 Aac @ 0.2 µs
Max. grid output overcurrent protection (inverter output rating)	240 Vac / 200 Aac (for the external breaker rating, see section 5.1)			
Grid input Icc (rated conditional short-circuit current)	500 Aac			
Grid input Icw (rated short-time withstand current)	500 Aac			
Total harmonic distortion (THDi, rated power)	<3%			
EPS output				
EPS rated output voltage	120 Vac / 240 Vac (split phase); 208 Vac (two legs of a three-phase wye)			
EPS rated output frequency	60 Hz			
EPS rated output active power	8 kW	10 kW	12 kW	15 kW
EPS rated output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
EPS max. output active power	8 kW	10 kW	12 kW	15 kW
EPS max. output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
EPS rated output current	33.4 Aac	41.7 Aac	50 Aac	62.5 Aac
EPS output power factor	0.8 leading to 0.8 lagging			
EPS output peak power	200% overload (<10 s)			
EPS output inrush current	200 Aac @ 10 ms			
EPS max. output fault current	200 Aac @ 10 ms			
Switch time	<10 ms			
Total harmonic distortion (THDv, linear load)	<2%			
Compatible with generator	Yes — generator input with auto-start dry contact			
Efficiency				
MPPT efficiency	99.50%			
California efficiency	96.50%			
Max. efficiency	97.50%			

Model	KR-L8K-SPUS	KR-L10K-SPUS	KR-L12K-SPUS	KR-L15K-SPUS
Standards				
Safety	UL 1741, UL 1741 SA/SB, UL 1699B, CSA C22.2, IEEE 1547, CA Rule 21, HECO Rule 14H, PREPA — see product label			
EMC	FCC Part 15 Class B, SunSpec			
General				
Degree of ingress protection	NEMA 4X			
Protection class	I			
Environment category	Indoor and outdoor			
Wet location classification	Yes			
Pollution degree	PD3			
Operating altitude	4000 m (derate above 2000 m)			
Operating ambient temperature	-25 to +60 °C (linear derating to 60% from +45 °C to +60 °C)			
Operating relative humidity	0–100% (non-condensing)			
Storage temperature	-25 to +60 °C			
Storage relative humidity	0–100% (non-condensing)			
Noise emission (typical)	<45 dB			
Overvoltage category	AC: III; PV: II			
Electrical supply system	TN, TT			
Dimensions (W × H × D)	522 × 826 × 265.5 mm			
Net weight	55 kg			
Cooling	Intelligent air cooling			
Topology	Non-isolated			
Active anti-islanding method	Active frequency drift			
Communication	RS485: yes · Wi-Fi: yes · Bluetooth: optional · LAN: optional CAN: yes · DRM: yes · Meter: optional			
Display and indicators	LCD + LED lights + app			
Warranty	10 years (see Limited Warranty — Terms at the front of this manual)			

KR-L15K battery power: 275 Adc × 51.2 Vdc (rated battery voltage) = 14.1 kW. This is the nominal battery power.

4. Check and Storage

4.1 Check Before Receiving

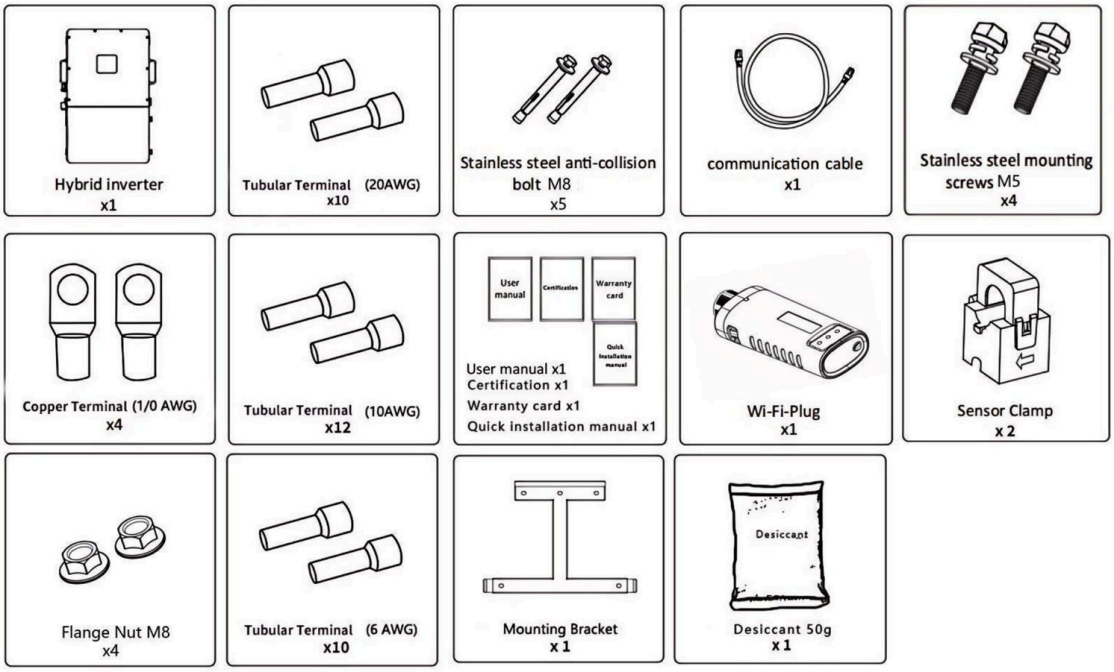
Check the following items before you accept the product.

1. Check the outer packing box for damage, such as holes, cracks, dents or other signs of damage. If you find any damage, do not unpack the box. Contact Fantastech (574-218-1973, derek@fantastech.org) as soon as possible.
2. Check the inverter model. If it is not the model you ordered, do not unpack the product. Contact Fantastech.
3. Check the delivery for the correct model, complete contents and undamaged appearance. If you find any damage, contact Fantastech as soon as possible.

4.2 What's in the Box

No.	Item	Quantity
1	Hybrid inverter (KOOR All-in-One)	1
2	Tubular terminal (wire ferrule), 20 AWG	10
3	Stainless steel anti-collision bolt, M8	5
4	Communication cable	1
5	Stainless steel mounting screw, M5	4
6	Copper terminal (ring lug), 1/0 AWG, for the battery connection	4
7	Tubular terminal (wire ferrule), 10 AWG	12
8	User manual	1
9	Certification	1
10	Warranty card (use the KOOR Warranty Registration Card at the back of this manual)	1
11	Quick installation manual (the KOOR Quick Start is at the front of this manual)	1
12	Wi-Fi plug (the Wi-Fi logger stick, see chapter 8)	1
13	Sensor clamp (CT)	2
14	Flange nut, M8	4
15	Tubular terminal (wire ferrule), 6 AWG	10
16	Mounting bracket	1
17	Desiccant, 50 g	1

The 1/0 AWG ring lugs in the box fit 1/0 AWG cable. These lugs do not fit the 2/0–4/0 AWG battery cable (section 6.2.1). Use ring lugs that match your cable size.

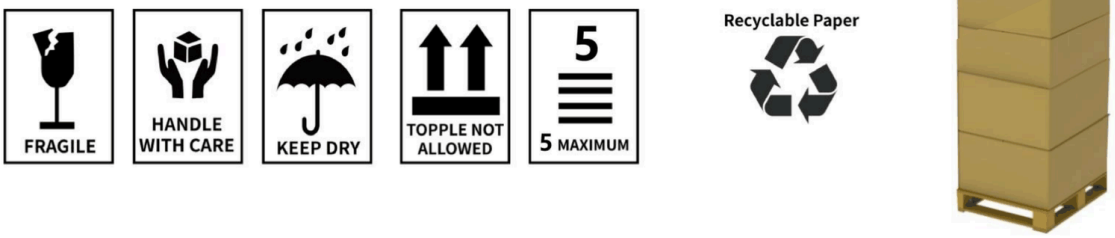


Box contents (the pictures show what each item looks like; the quantity table in this section gives the count)

If an item is missing or damaged, contact Fantastech.

4.3 Storage

If you will not install or use the equipment right away, make sure the storage place meets these requirements:



Packing box marks: fragile, handle with care, keep dry, do not tip over, stack 5 high maximum, recyclable paper; boxes stacked on a pallet

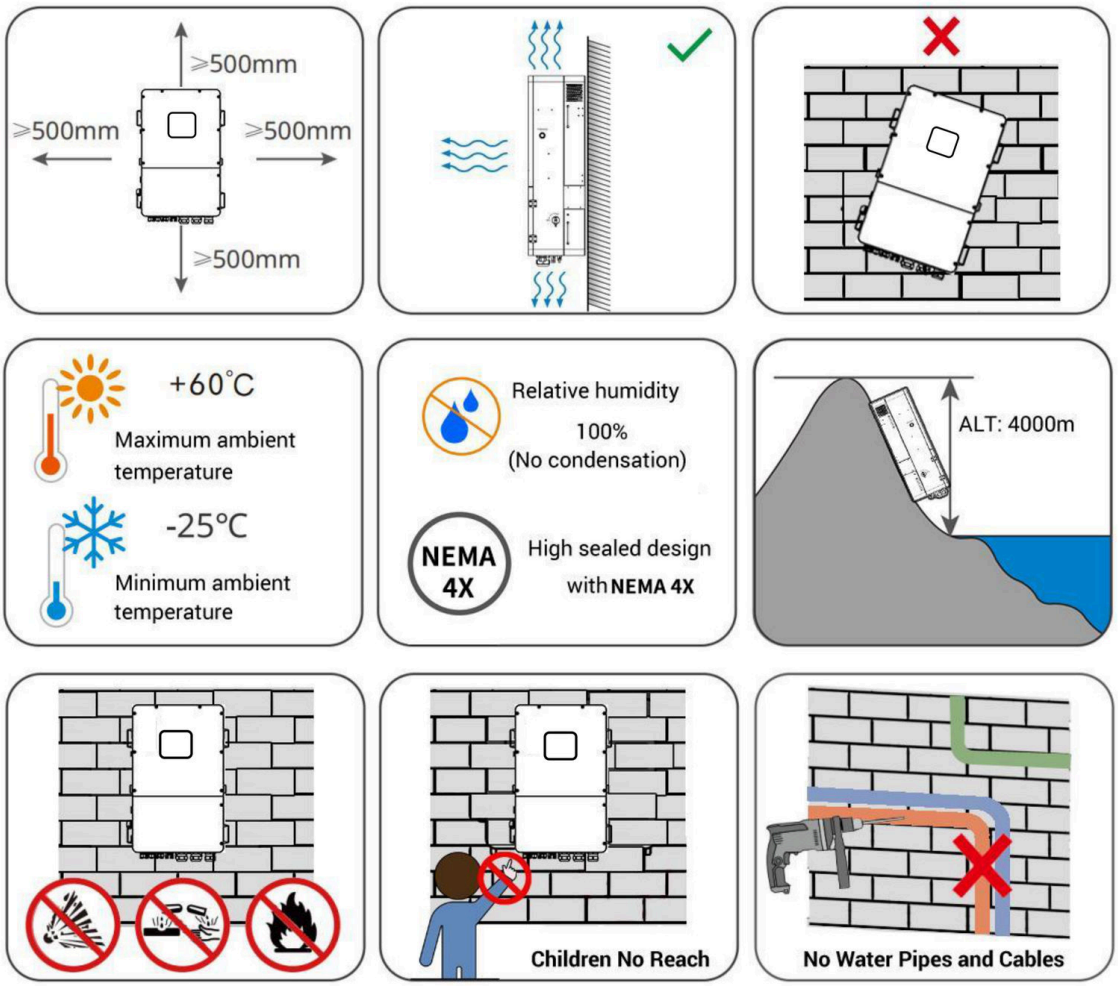
1. Do not unpack the outer package. Do not throw the desiccant away.
2. Store the equipment in a clean place. Make sure the temperature and humidity are suitable and there is no condensation.
3. Stack the inverters to the height and in the direction shown on the packing box.
4. Stack the inverters with care so they cannot fall.
5. If the inverter has been stored for a long time, have it checked by a qualified professional before you put it into use.
6. Storage temperature range: -25 to +60 °C. Storage humidity: 0–100%.
7. The box should be able to carry loads of more than 75 kg.

5. Installation

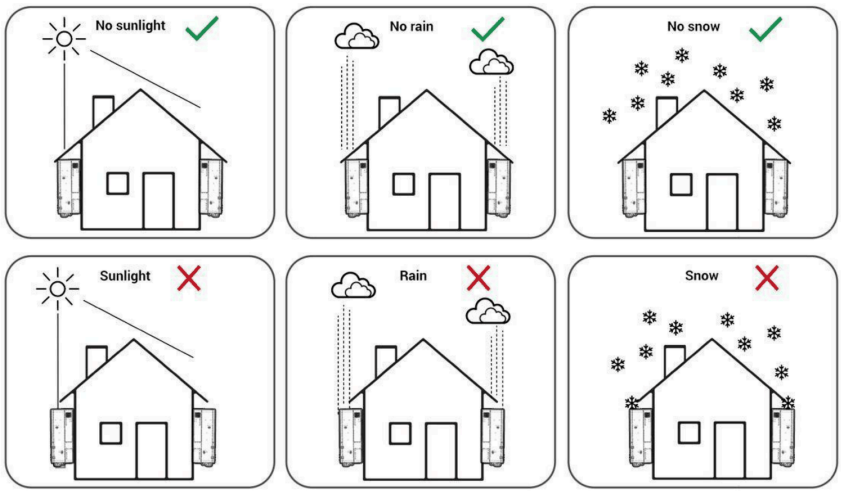
5.1 Installation Requirements

Installation Environment Requirements

1. Do not install the equipment near flammable, explosive or corrosive materials.
2. Do not install the equipment where people can easily touch it, especially within the reach of children. The equipment gets hot when it is working. Do not touch its surface, or you may be burned.
3. When you drill holes, stay clear of water pipes and cables buried in the wall.
4. Install the equipment in a sheltered place, out of direct sunlight, rain and snow. Build a sunshade if one is needed.
5. The installation place must be well ventilated so heat can escape, and large enough to work in.
6. The equipment has a high ingress protection rating (NEMA 4X). It can be installed indoors or outdoors. The temperature and humidity at the site must stay within the allowed range.
7. Install the equipment at a height that makes it easy to operate and maintain, make the electrical connections, and read the indicators and labels.
8. Install the inverter below its maximum working altitude of 4000 m.
9. The PV modules used with the inverter must have an IEC 61730 Class A rating.
10. Provide overcurrent protection before the GRID and LOAD terminals, such as an external circuit breaker rated 400 Vac / 200 Aac. (400 Vac / 200 Aac is the rating of that external breaker. The 240 Vac / 200 Aac in section 3.3.4 is the inverter's own output rating.) Make sure the installation position does not block access to the disconnecting means.
11. Make sure the inverter has enough space around it for ventilation after it is installed. Leave at least 500 mm on every side, as shown in the installation drawing below.
12. This inverter has no internal isolation transformer between the PV input and the battery / AC output circuits. Basic insulation is provided between the PV input / battery / AC output circuits and the metal enclosure / earth. Reinforced (double) insulation is provided between the PV input / battery / AC output circuits and the communication circuits (DRM / meter / Wi-Fi / RS485).
13. Install the equipment away from electromagnetic interference. If radio stations or wireless communication equipment working below 30 MHz are near the installation site, install the equipment as follows:
    - Add a multi-turn ferrite core on the DC input line or the AC output line of the inverter, or add a low-pass EMI filter.
    - Keep the inverter more than 30 m away from the wireless equipment that causes the interference.



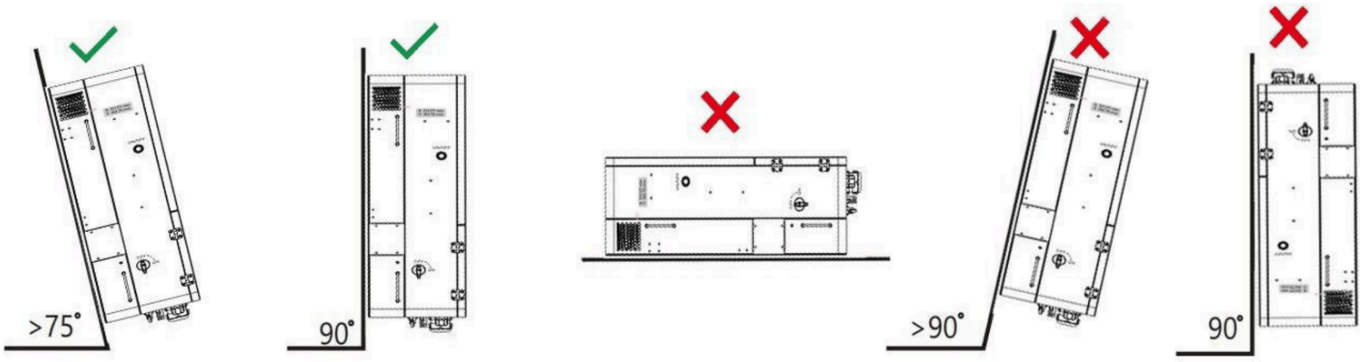
Top row: leave at least 500 mm on every side; mount upright with air flow around it; do not mount tilted sideways. Middle row: ambient temperature -25 to +60 °C; relative humidity up to 100% (no condensation), NEMA 4X sealed design; altitude up to 4000 m. Bottom row: keep away from explosive, corrosive and flammable materials; keep out of reach of children; do not drill into water pipes or cables



Install where the inverter is sheltered from sunlight, rain and snow (top row), not where it is exposed to them (bottom row)

Installation Angle Requirements

- Install the inverter upright (vertical), or tilted back by no more than 15 degrees.
- Do not install the inverter upside down, tilted forward, or lying flat.



Allowed: tilted back no more than 15° (angle to the floor more than 75°) or upright at 90°. Not allowed: lying flat, tilted forward (more than 90°), or upside down

Installation Tool Requirements

The following tools are recommended for installing the equipment. Use other tools on site if needed.



Recommended installation tools (see the table below for the correct tool names)

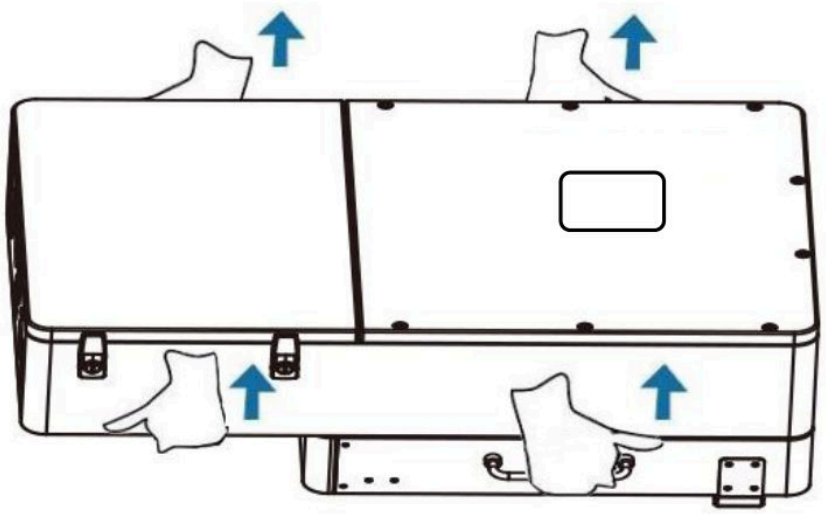
Picture row	Tools, left to right
1	Goggles · Safety shoes · Safety gloves · Dust mask · Ferrule crimping pliers
2	Diagonal pliers · Wire stripper · Hammer drill · Heat gun · Hydraulic crimping pliers
3	Marker · Level · Heat shrink tube · Rubber hammer · Vacuum cleaner
4	Multimeter · Cable tie · Torque screwdriver (with M4/M5/M6 bits) · Hex key (Allen wrench) · 10/13/22 mm socket or wrench

Three tiles in the picture carry the wrong name: "Tube type bundle terminal crimping pliers" are ferrule crimping pliers, "Torque wrench" is a torque screwdriver, and "L-type Hexagon wrench" is a hex key (Allen wrench).

5.2 Inverter Installation

5.2.1 Moving the Inverter

CAUTION The unit is heavy. Do not lift it alone. While lifting, make sure the unit is held firmly so it cannot tip over or drop. Parts used to support or hold the unit in place must be designed and made to keep the risk of injury, and of fixings coming loose, as low as possible. Make sure the lifting method cannot let the unit slip out of chains or slings, turn over, or slide off the lifting devices.



Moving the inverter: lift it from below at both ends

- Transport must be done by trained people (truck operators, hook-up personnel) who wear the needed protective equipment: overalls, safety shoes, protective gloves, helmets and goggles.
- Do not walk or stand under or near the load. Avoid sudden movements and jolts when you unload and position the unit. Handle the unit with care when you move it on site.
- Do not use any part of the unit as a lever. If the unit is not balanced, add ballast. Do not hold up any part that sticks out by hand. Install the inverter so that the operating panel is easy to reach and the electrical power connection point is easy to access.
- Keep the inverter accessible for maintenance and repair work.

- Check the load capacity and hardness of the supporting surface. Per IEC 62109-1, the load rating of the mounting bracket must be at least four times the weight of the device. Wear, corrosion, material fatigue and ageing weaken the support over time. Work out the support from the design data of the supporting material, and consult a structural engineer.

5.2.2 Installing Steps

NOTICE

- Use screws or anchors that suit the support surface and can carry the weight of the inverter.
- When you drill holes, stay clear of water pipes and cables buried in the wall.
- Wear goggles and a dust mask when you drill, so you do not breathe in the dust or get it in your eyes.
- Make sure the inverter is firmly installed so it cannot fall.

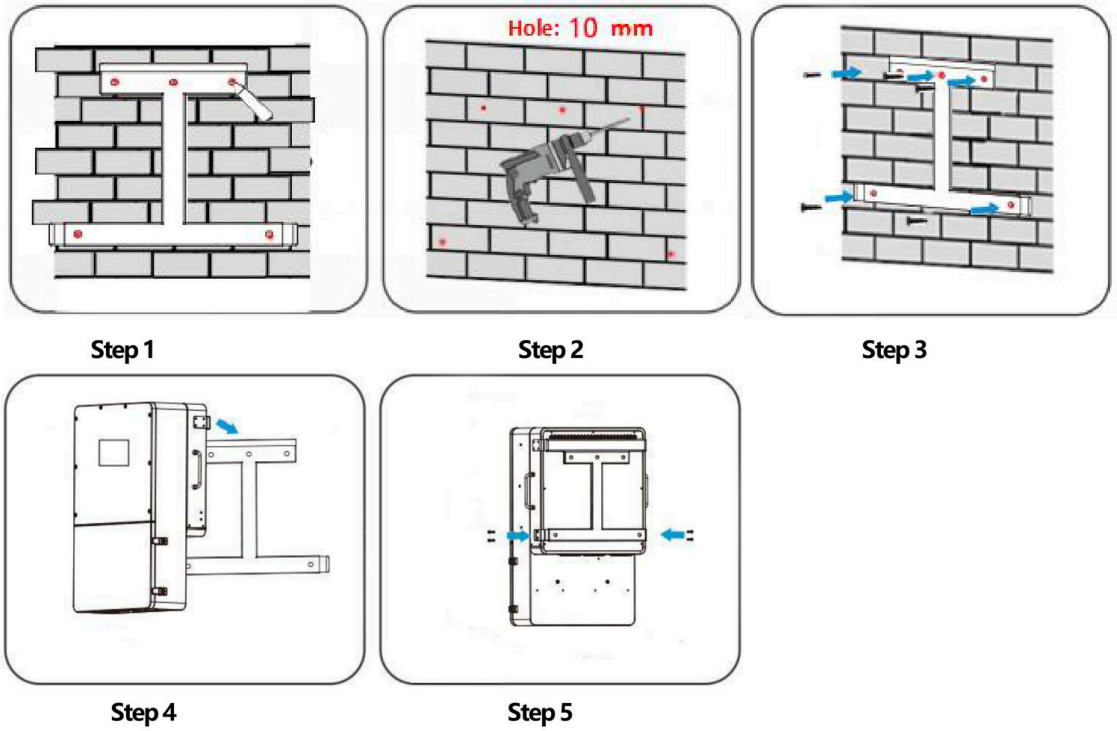
Step 1: Hold the mounting bracket flush against the wall and mark the positions of the holes.

Step 2: Drill 10 mm holes at the marked positions.

Step 3: Fasten the mounting bracket to the wall through the drilled holes and tighten the screws.

Step 4: Hang the inverter on the mounting bracket.

Step 5: After the inverter is hung, lock the screws at both ends, two at each side, as the step 5 drawing shows.



Installing steps 1 to 5 (step 2: hole diameter 10 mm)

6. Electrical Connection

6.1 Safety Precautions

DANGER

- All work, cables and parts used for the electrical connection must comply with local laws and regulations.
- Before you make any electrical connection, turn off the DC switch and the AC output switch to power off the inverter. Then wait at least 5 minutes so the capacitors can discharge. Do not work with the power on. Otherwise, electric shock may occur.
- Tie cables of the same type together, and keep them apart from cables of other types. Do not let cables tangle or cross.
- A cable under too much tension may make a poor connection. Leave some slack in each cable before you connect it to the inverter cable port.
- When you crimp a terminal, make sure the bare conductor is in full contact with the terminal. Do not crimp the cable jacket into the terminal. Otherwise, the inverter may not operate, or the loose connection may heat up during operation and damage the terminal block.

NOTICE

- Wear personal protective equipment, such as safety shoes, safety gloves and insulating gloves, during electrical connections.
- All electrical connections must be made by qualified professionals.
- Cable colors in this manual are for reference only. Cable specifications must meet local laws and regulations.

6.2 Connection Requirements

6.2.1 AC / DC Connection Requirements

Port	Terminal / breaker rating	Terminal wire size range (min.–max.)
GRID	200 Aac	2/0–4/0 AWG
LOAD	200 Aac	2/0–4/0 AWG
GEN	63 Aac	6–4 AWG
MPPT	26 Adc (per MPPT)	12–10 AWG
Battery port	200 Adc × 2	2/0–4/0 AWG
I-LOAD (intelligent load)	90 Aac	6–4 AWG

The GRID and LOAD ratings are the same (200 A) on all four models, KR-L8K, KR-L10K, KR-L12K and KR-L15K.

Wire colors

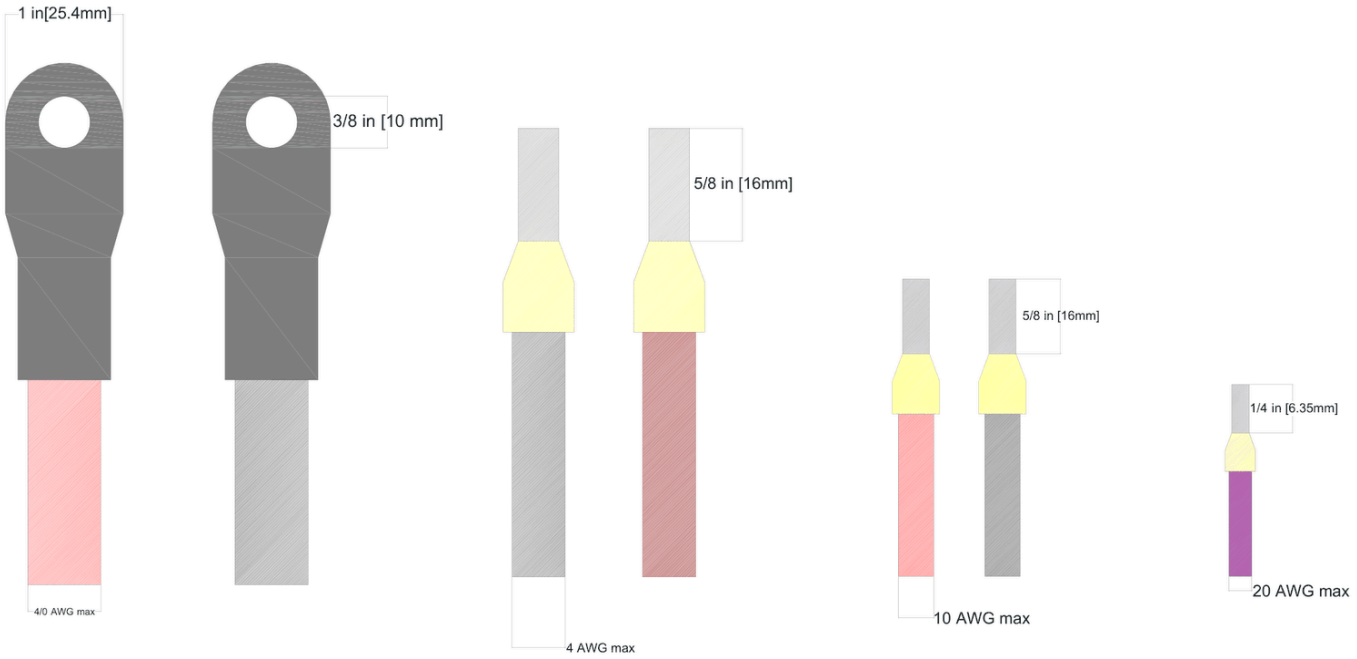
Every wiring drawing in this manual uses the standard US wire colors. Use the same colors on the job so anyone who opens the system later can read it.

Wire	Color
L1	Black
L2	Red
L3 (120/208 V only)	Blue
Neutral (N)	White (drawn grey in the drawings so it shows on white paper)
Ground	Green, or bare copper
DC positive (+)	Red — battery and PV
DC negative (−)	Black — battery and PV

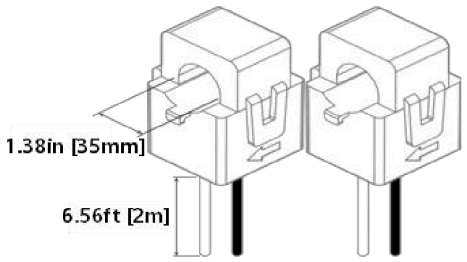
6.2.2 Sensors and Communications Requirements

Component	Wire size range	Maximum distance
CT sensor	16–20 AWG	0–13 ft (4 m): 16 AWG 13–20 ft (6 m): CAT6, extendable
Communications	23–24 AWG	0–100 ft (30 m): 24 AWG 100–400 ft (120 m): 23 AWG
RJ45 parallel communication	CAT5e or better	0–7 ft (2.1 m): included 7–20 ft (6 m): extendable

The CT sensors in the box come with 6.56 ft (2 m) leads.



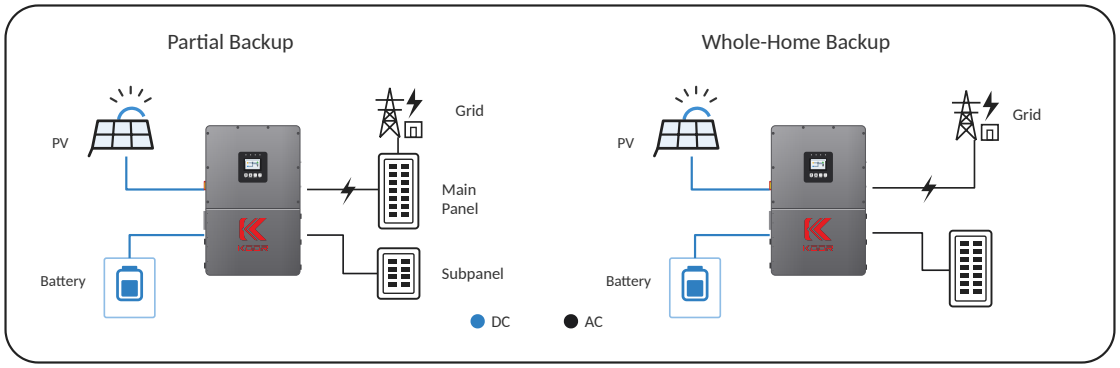
Terminal and ferrule sizes: ring terminal for up to 4/0 AWG (1 in [25.4 mm] and 3/8 in [10 mm]); ferrules for up to 4 AWG (5/8 in [16 mm]), up to 10 AWG (5/8 in [16 mm]) and up to 20 AWG (1/4 in [6.35 mm])



CT sensors (included): 1.38 in (35 mm) opening, 6.56 ft (2 m) leads

6.3 System Integration

The system connection diagrams are shown below (US version).



System connection: Partial Backup (left) and Whole-Home Backup (right). Blue lines are DC; black lines are AC.

Single system installation guide

Partial Backup (Essential Loads Only)

- Connect the output of the back-feed breaker or line-side tap (depending on your interconnection method) to the "GRID" terminal of the inverter.
- Install an external disconnect switch between the point of interconnection and the inverter. Size the disconnect according to local electrical code.
- Connect the "LOAD" terminal of the inverter to the essential loads panel. Choose the wire gauge according to electrical code.

Whole-Home Backup (Entire Load)

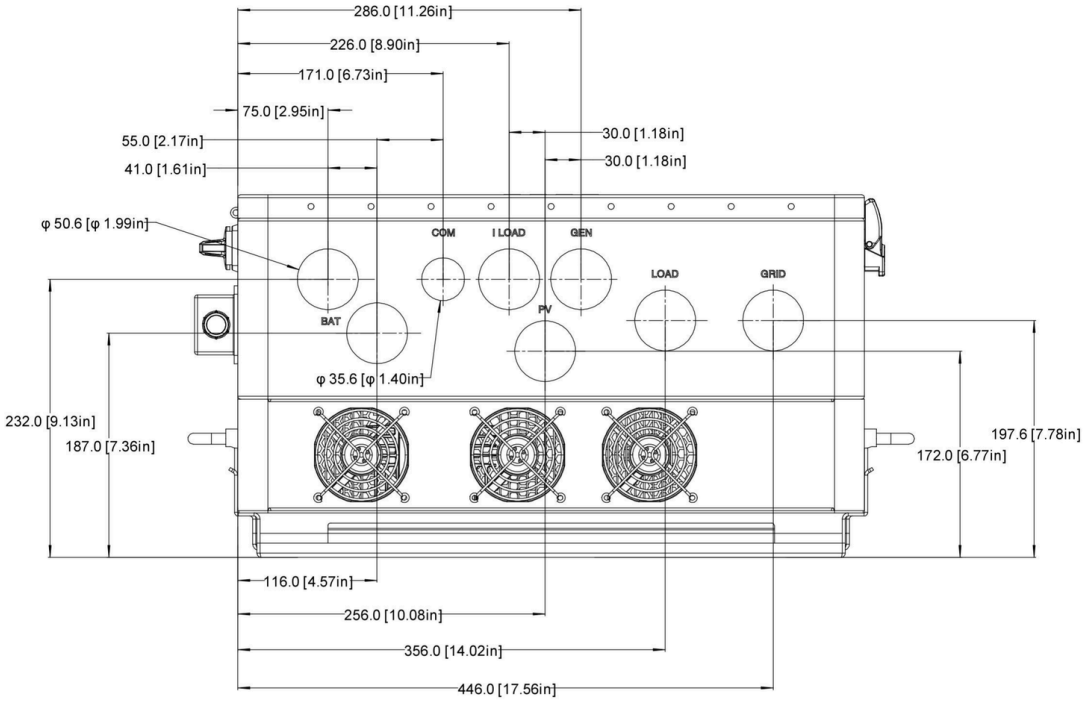
- Connect the utility grid directly to the inverter's "GRID" terminal.
- Install an external disconnect switch between the grid and the inverter. Size the disconnect according to code requirements.
- Connect the "LOAD" terminal to the main service panel. Size the wire according to code.

AC Generator or AC-Coupled Source (Optional)

- A generator or an AC-coupled power source (for example, a string inverter or microinverters) can be connected to the "GEN" terminal.
- The maximum input for the GEN terminal is 63 A (about 15 kW). Use the GEN terminal only for a 120/240 V generator under 15 kW (Fantastech rule, 6.8.1). An AC-coupled source is limited to 6 kW.
- Only one AC source may be connected to the GEN terminal at a time.

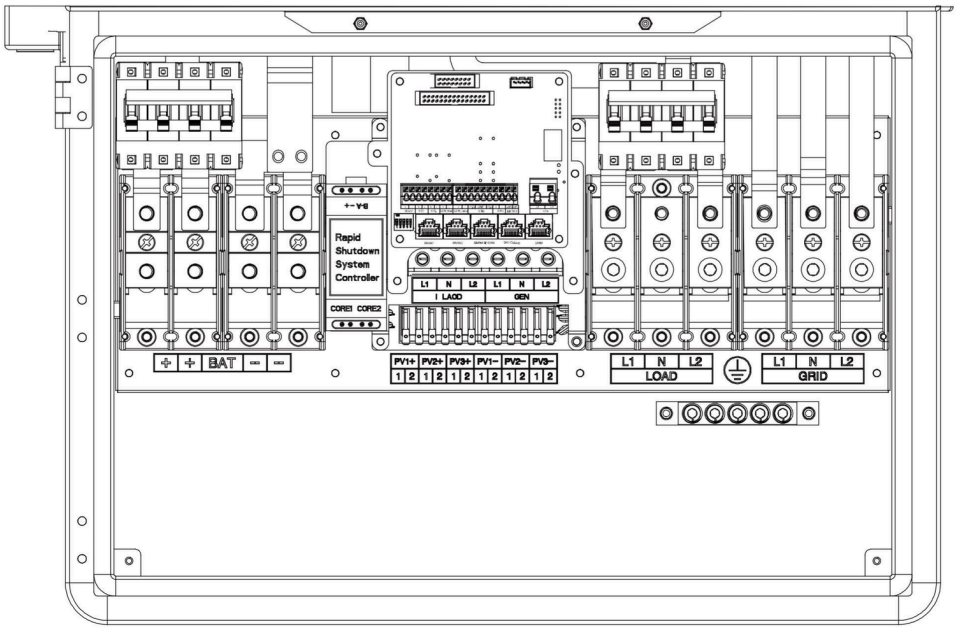
- Which port a generator goes on depends on its voltage and size. A 120/240 V split-phase generator under 15 kW goes on the GEN port. A generator of 15 kW or more, and every 120/208 V generator, goes on the GRID port. Never connect a 120/208 V or three-phase generator to the GEN port. See section 6.8.1.

Bottom view of cable box knockouts



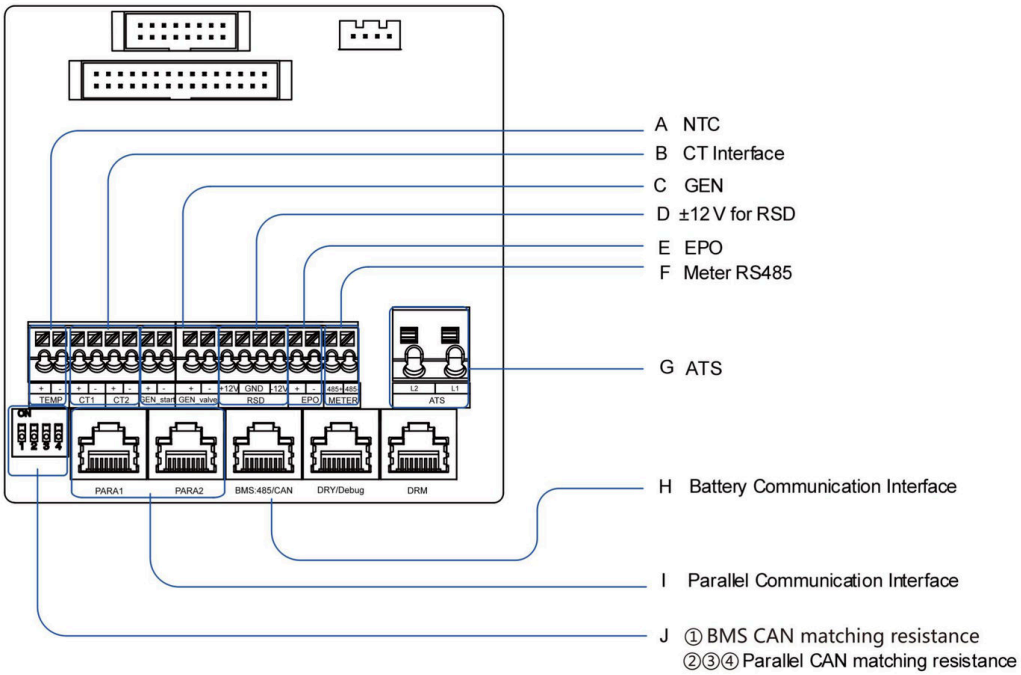
Cable box knockouts, bottom view (dimensions in mm [in]): BAT, COM, I-LOAD, GEN, PV, LOAD and GRID

Overview of cable box



Cable box overview. The terminal labeled "I LAOD" in the drawing is the I-LOAD terminal.

Communication terminals



Communication board terminals A to J

A. **NTC:** Connection for the lead-acid battery temperature sensor.

B. **CT interface:** See section 6.7.4 for the pin definition.

C. **GEN (NO):** Connection for the generator auto-start function. This is a dry-contact signal to start the generator. When the "GEN signal" is active, the normally open contact (GS) closes (no voltage output).

D. **±12 V for RSD:** Power supply for the RSD PLC transmitter (max. current 500 mA).

E. **EPO:** Reserved for an external EPO (emergency power off).

F. **Meter RS485:** For meter communication.

G. **ATS:** 240 Vac output port when the inverter is on.

H. **Battery communication port (CAN & RS485):** See section 6.6.2 for the pin definition.

I. **Parallel communication port (CAN & RS485):** See section 6.9.1 for parallel system connection.

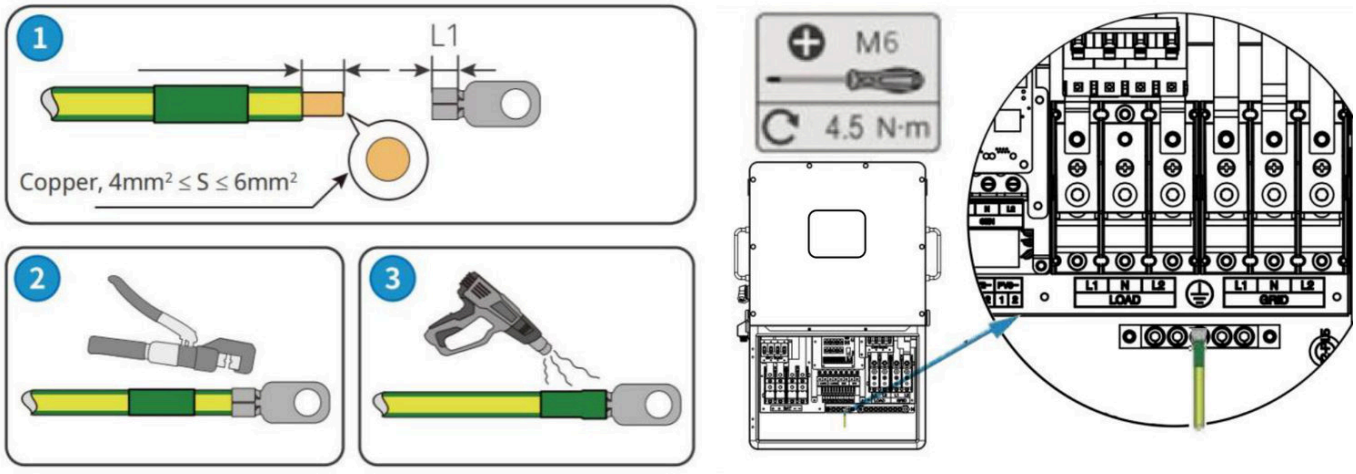
J. **DIP switches 1–4:**

- Switch 1 is the BMS CAN matching resistance. It must be ON when the battery uses CAN communication. Keep switch 1 ON on every inverter.
- Switches 2, 3 and 4 are the parallel CAN matching resistance. Set them when inverters are used in parallel: ON on the first and last inverter of the chain, OFF on every inverter in between. On a single inverter, set switches 2, 3 and 4 to OFF (see section 6.9.1).

6.4 PE Cable Connection

WARNING

- The PE cable connected to the inverter enclosure cannot replace the PE cable connected to the AC output port. Both PE cables must be securely connected.
- When there are multiple inverters, make sure all grounding points on the enclosures are bonded together so they are at the same potential (equipotential bonding).
- To improve the corrosion resistance of the terminal, KOOR recommends that you apply silicone sealant or paint to the ground terminal after you install the PE cable.
- Prepare PE cables to the recommended specification:
  - Type: outdoor single-core copper wire
  - Cross-sectional area: 6 mm² (10 AWG)



PE cable: (1) strip the cable (copper, 4 mm² ≤ S ≤ 6 mm², 12–10 AWG), (2) crimp the terminal, (3) shrink the heat-shrink tubing; then fasten it to the ground terminal (M6 screw, Phillips screwdriver, 4.5 N·m)

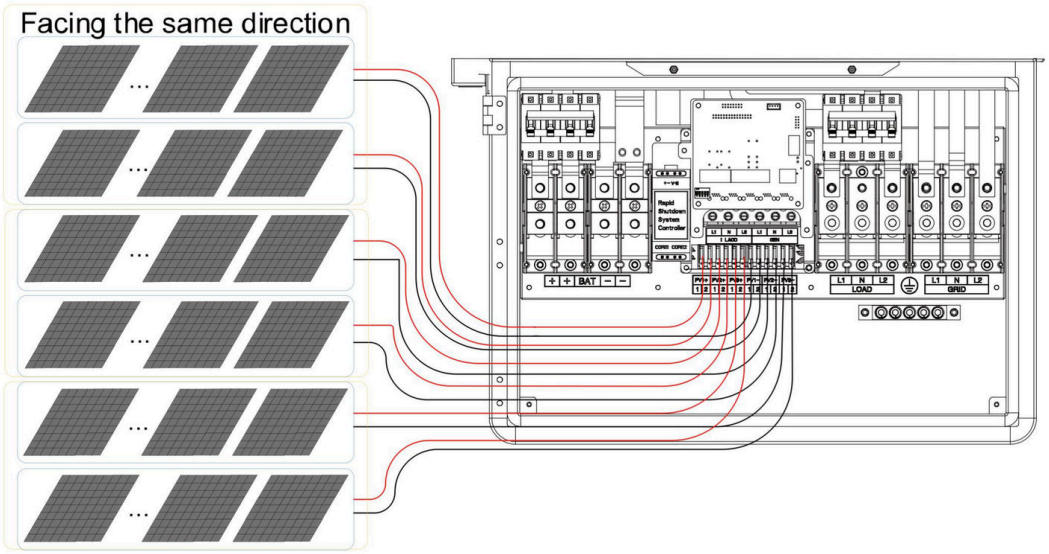
6.5 PV Connection

The PV connection of this hybrid inverter is the same as that of a traditional grid-tied solar inverter (string inverter).

WARNING Check the lowest ambient temperature at the installation site. The Voc on the solar panel nameplate is measured at 25 °C (STC). As the temperature drops, the panel Voc rises. Size every string so that its open-circuit voltage, corrected for the lowest temperature, stays at or below 525 Vdc. The absolute maximum PV input voltage of the inverter is 550 Vdc.

NOTE

- The inverter has three MPPTs. You can connect two strings of solar panels to each MPPT.
- When you connect two strings to the same MPPT, make sure each string has the same number of panels. Keep the location and orientation of the two strings as alike as possible. The inverter automatically limits the total input current of each MPPT to 26 A.
- The inverter limits the maximum solar input power to 12 kW (KR-L8K), 15 kW (KR-L10K), 18 kW (KR-L12K) or 22.5 kW (KR-L15K).
- Strings in parallel on the same MPPT must have the same design open-circuit voltage (Voc). Otherwise, the system will be limited to the lowest string voltage.
- If panels facing different directions are connected to the same MPPT, PV efficiency will be lost.
- For ground-mounted arrays, KOOR recommends an auxiliary grounding electrode near the array. It keeps the earth-to-ground resistance of the grounding system low. This auxiliary electrode must follow the NEC and local code.
- Connect the solar panel strings as shown in the configuration below.



PV string connection: six strings facing the same direction, two strings on each MPPT (PV1, PV2, PV3)

PV connection steps

a. Strip 5/8–11/16 in (16–18 mm) of insulation from the positive and negative conductors of the PV string.

b. If the PV string wires are stranded, use wire ferrules.

c. Install the conduit connector in the PV connection port.

d. Pass the PV conductors through the conduit fittings and into the inverter.

e. Check that the cable connections are correct and secure. Then make sure the conduit and conduit fittings are properly tightened and sealed into the cable hole.

6.6 Battery Connection

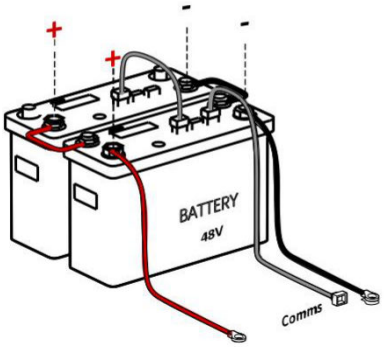
Battery connection guidelines for the KR-L8K–L15K-SPUS series

> **IMPORTANT** Make sure the inverter is turned OFF before you connect any batteries.

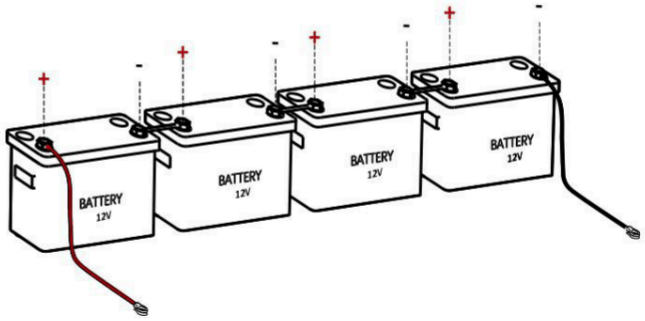
- Connect your battery bank in the configuration that fits the battery voltage (see the wiring diagrams below).
- Battery breakers must be OFF during wiring.
- If your battery bank does not have built-in breakers, take all necessary safety precautions when you handle exposed terminals or cables.
- When both sets of battery terminals are used, the maximum battery charge/discharge current is 167 A (KR-L8K), 210 A (KR-L10K), 250 A (KR-L12K) or 275 A (KR-L15K).
- If you use only one set of battery terminals, the charge/discharge current is limited accordingly.
- This inverter is designed for a 48 Vdc nominal system.
- **DO NOT** connect the inverter to a battery configuration outside the 40–60 Vdc range.
- If you use 12 V batteries, **do not** put more than 4 batteries in series (max. 48 V nominal).

The inverter supports many battery chemistries, as long as the voltage stays within 40–60 Vdc. Depending on the battery voltage, wire the battery bank in one of the configurations shown below.

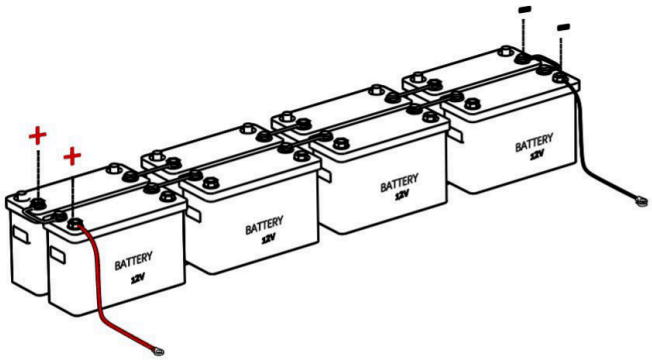
KOOR batteries: the battery set-up (communication protocol, address switches and cable) is in the "Appendix — KOOR batteries" at the front of this manual.



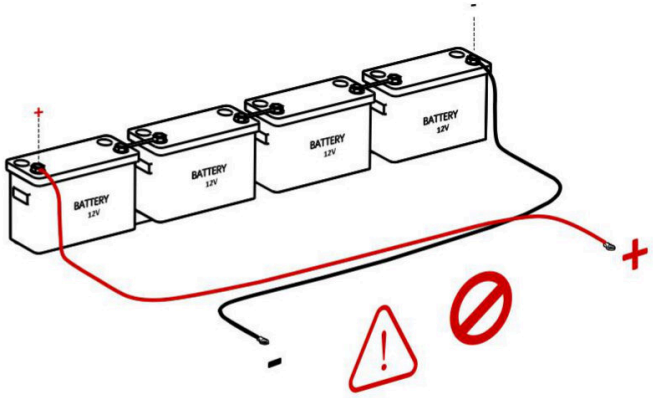
48 V batteries in parallel connection



12 V batteries in series connection



Series and parallel connections for a complete 48 V battery bank



Reversed polarity: never do this

WARNING DO NOT reverse the polarity. The system will be damaged, and the warranty will be void!

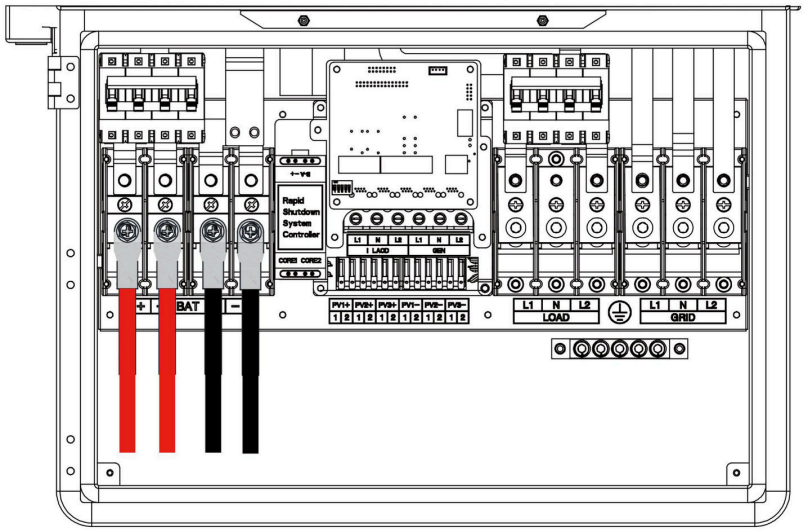
6.6.1 Battery Power Cable Connection

To connect the battery terminals:

1. Strip 1/4–5/16 in (6–8 mm) of insulation from the cable ends and crimp on the OT ring terminals.
2. Lay the battery power cables. Connect the positive cable to BAT+ and the negative cable to BAT−.
3. Use the lock nut to secure the conduit fittings to the enclosure.
4. Fasten the OT rings of the battery positive and negative cables to the terminal block, following the markings.

Multi-terminal installation

The inverter's two pairs of battery terminals are connected in parallel inside the inverter. This gives the battery banks a common connection and makes battery installation simpler. To get the full rated charge/discharge current (167 A for KR-L8K, 210 A for KR-L10K, 250 A for KR-L12K, 275 A for KR-L15K), connect the batteries to both pairs of terminals. If you use 3 or more batteries, use external busbars for the (+) and (−) connections.



Battery cables connected to both pairs of BAT terminals: two positive (red) and two negative (black)

WARNING **Multiple inverters in a parallel system**

- All parallel inverters must connect to a single battery bank. Otherwise, the system will NOT operate properly.
- DO NOT use separate battery banks in parallel systems.
- Follow all values specified by the battery manufacturer to ensure proper charging and discharging.

This shared-bank rule is for parallel systems only (one master inverter and its slaves). A single inverter that is not part of a parallel system has its own battery bank, and that bank must not be connected to any other inverter (see section 2.4).

Battery charging setpoint examples (48 V nominal)

Battery type	Absorption	Float
AGM	57.6 V	53.6 V
Gel	56.4 V	54.0 V
Lithium	54.6 V	54.3 V

Calculating battery bank amp-hours: examples (AGM)

Number of batteries	Voltage per battery	Ah per battery	Ah at 48 V (strings of four 12 V batteries in series)	Max. charge / discharge (A)
4	12 V	230 Ah	230 Ah	100 A
8	12 V	230 Ah	460 Ah	200 A
12	12 V	230 Ah	690 Ah	275 A
16	12 V	230 Ah	920 Ah	275 A

The 275 A figures are the KR-L15K limit. The inverter's maximum battery charge/discharge current is 167 A (KR-L8K), 210 A (KR-L10K), 250 A (KR-L12K) or 275 A (KR-L15K). On the smaller models, the current is limited to that model's maximum.

To make the battery settings on the inverter screen, see section 9.7 Battery Settings Menu.

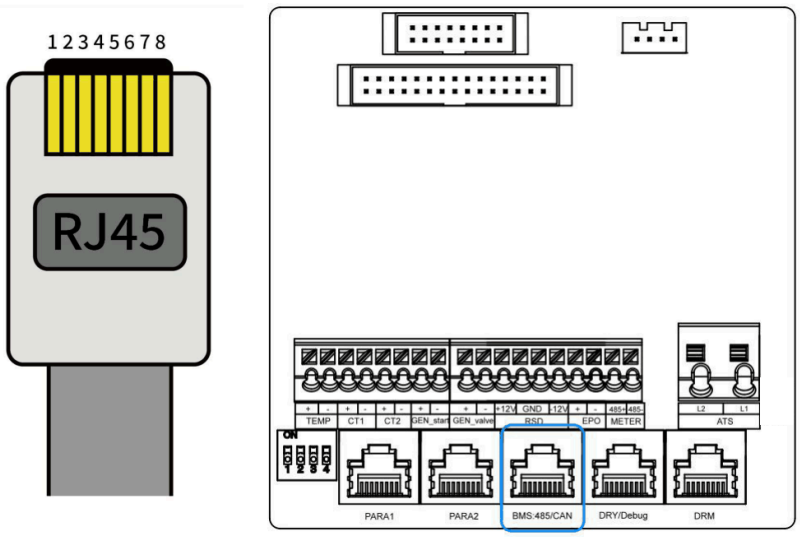
> **WARNING** **Reverse polarity will damage the inverter! Before you connect or disconnect the battery, make sure the built-in battery circuit breaker is in the "OFF" position.**

6.6.2 Battery Communication Cable Connection

When you select a lithium-ion battery type, you must use the correct battery communication cable to connect the battery to the inverter. If the lithium battery cannot communicate with the inverter, select the "lead-acid" battery type as a temporary test only. Never leave a lithium battery set up as lead-acid (see the Appendix — KOOR batteries at the front of this manual).

The inverter's battery communication port is an RJ45 socket. Its pins are listed below. Make the communication cable to match the pins of the inverter port and the battery communication port. The inverter supports both CAN and RS485 communication.

Pin	Description
1	BAT RS485 A
2	BAT RS485 B
3	NC
4	BAT CAN H
5	BAT CAN L
6	NC
7	BAT RS485 B
8	BAT RS485 A



RJ45 pin numbers (left) and the battery communication port, BMS:485/CAN (right, circled in blue)

NOTICE After the battery power cables and the communication cable are connected, set the battery on the inverter screen: Bat Protocol, Bat Type and Comm Type (section 9.7 Battery Settings Menu). After you select the right battery protocol, communication is established within 1–2 minutes. In a parallel system, also set Shared Bat, and connect the battery communication cable to the master inverter.

WARNING Damage caused by improper use of the communication protocols (CAN bus or Modbus) is not covered by the warranty. The Modbus map is available on request, for "READ" operations only. To get the Modbus map, contact Fantastech (574-218-1973, derek@fantastech.org).

6.7 Grid & EPS Load Connection

6.7.1 Grid Type and Regulation Selection

This inverter can be used on 120/240 V split-phase systems or on 120/208 V systems (two legs of a three-phase wye). If you connect the inverter to the grid, make sure the grid settings are set correctly. Select the grid standard, voltage and frequency on the inverter screen. See section 9.8 Grid Settings Menu (Grid Code, Grid Volt and Grid Freq).

For a single inverter on a 120/208 V service, select the grid standard **120/208 3P**. The inverter's input and output are then both 120/208 V: 208 V hot to hot and 120 V hot to neutral. There is no 240 V output.

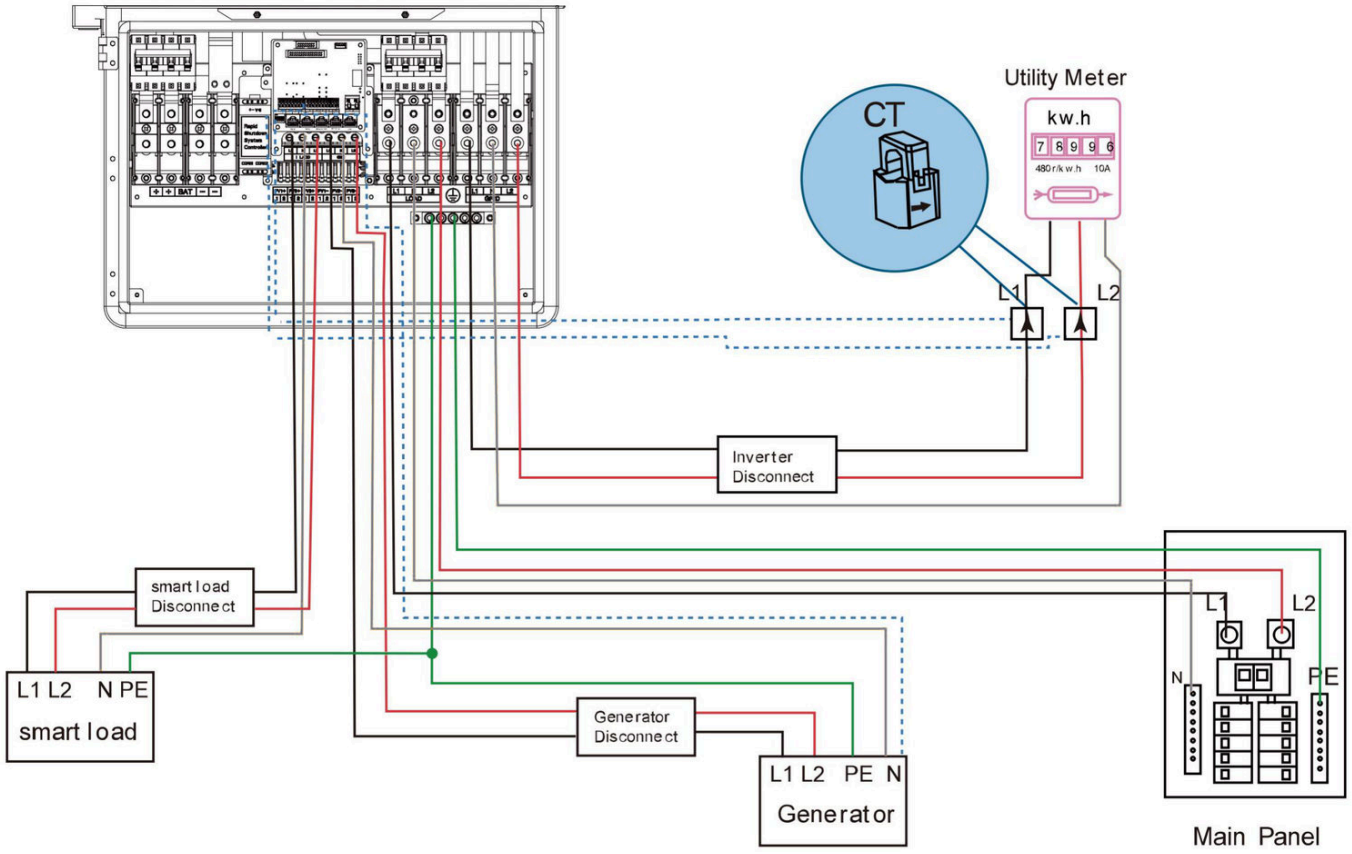
NOTICE Do not connect the internet router or the fiber modem (ONT) to the inverter's LOAD terminals. They may interfere with remote software updates. Connect these devices on the grid side so the network service is not interrupted.

6.7.2 Split-Phase Grid and EPS Load Connection

The wiring diagram below is for a 120/240 V service. The wiring for a 120/208 V service (two or three phases) is mostly the same, except for the generator (see the warning below the diagram and section 6.8.1).

When the bus rating of the main panel meets the requirements of NEC 705.12(B)(3), the inverter can be connected on the load side of the service disconnecting means. If it does not, a line-side connection can be made. This avoids a costly upgrade of the main panel.

Grid and EPS load connection for 120/240 V

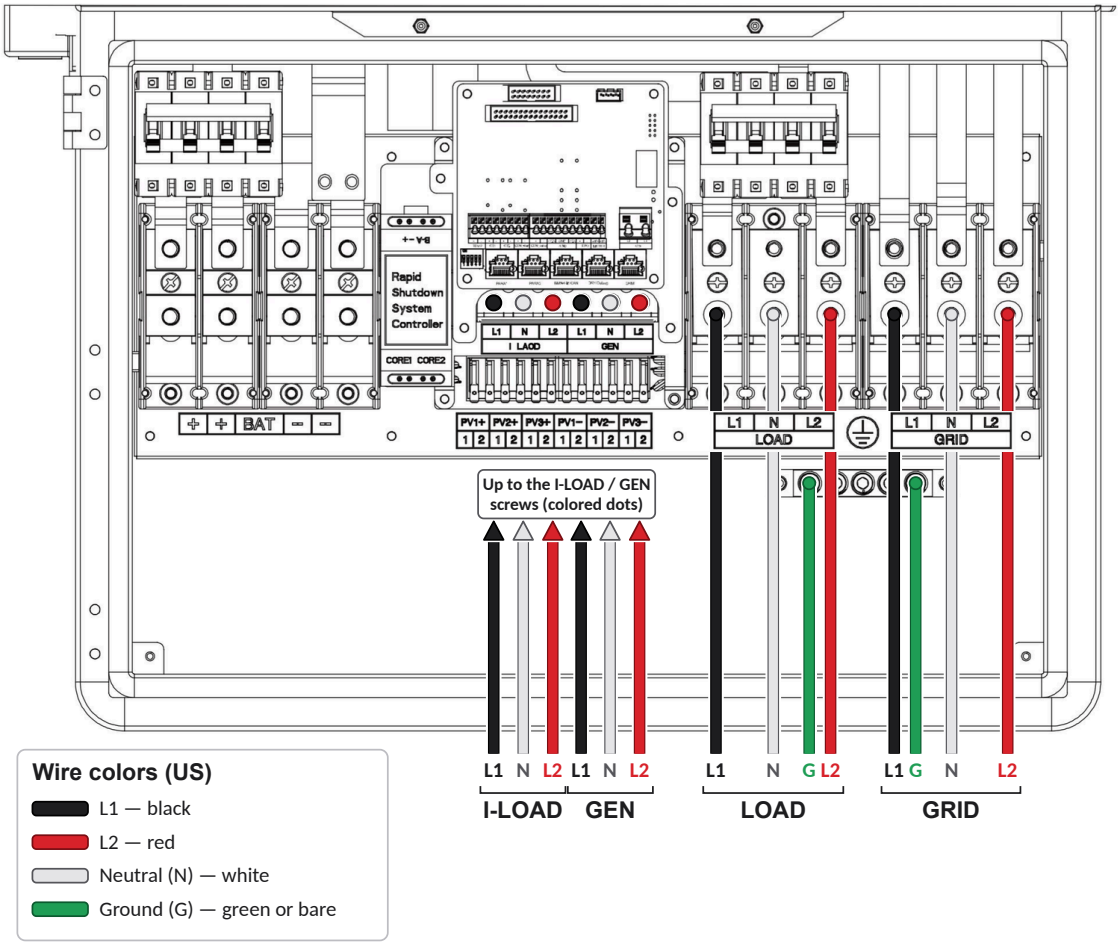


120/240 V grid and EPS load wiring, with smart load and generator. Labels in the drawing read: Utility Meter, CT, Inverter Disconnect, Main Panel, Smart Load Disconnect, Smart Load, Generator Disconnect and Generator. Wire colors: black = L1, red = L2, grey = neutral (white wire), green = ground, blue dotted = communication and CT cables.

WARNING On a 120/208 V system (two or three phases), the generator can be connected only to the GRID terminal. Connecting it to the GEN terminal is prohibited.

6.7.3 AC Cable Connection

1. Strip 5/16–3/8 in (8–10 mm) of insulation from the cable.
2. If the cable is stranded, put a wire ferrule on the end of each wire.
3. Fasten the conduit fitting to the enclosure with the top nut of the fitting.
4. Fasten the grid wires and the EPS load wires to the terminal block, as marked on the block.
5. Fasten the conduit to the conduit fitting.
6. Check that all cable connections are correct and tight. Make sure the conduit and conduit fittings are tightened and sealed into the cable entry hole.



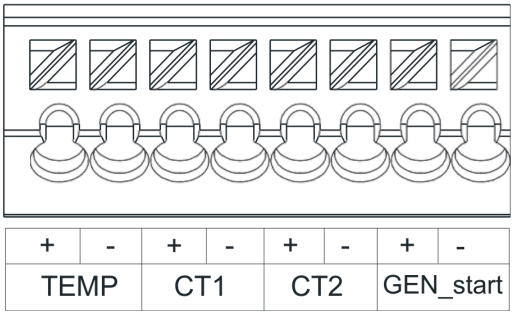
AC wiring at the I-LOAD, GEN, LOAD and GRID terminals and the ground bar. Each cable is L1 (black), N (white) and L2 (red); the ground (green) goes to the ground bar. The I-LOAD and GEN cables are drawn cut below the PV terminals so the PV labels stay readable; the colored dots show which I-LOAD and GEN terminal each wire lands on. The terminal marked "I LAOD" in the drawing is I-LOAD.

WARNING Before you supply power to the EPS load, make sure the built-in load circuit breaker is in the "ON" position.

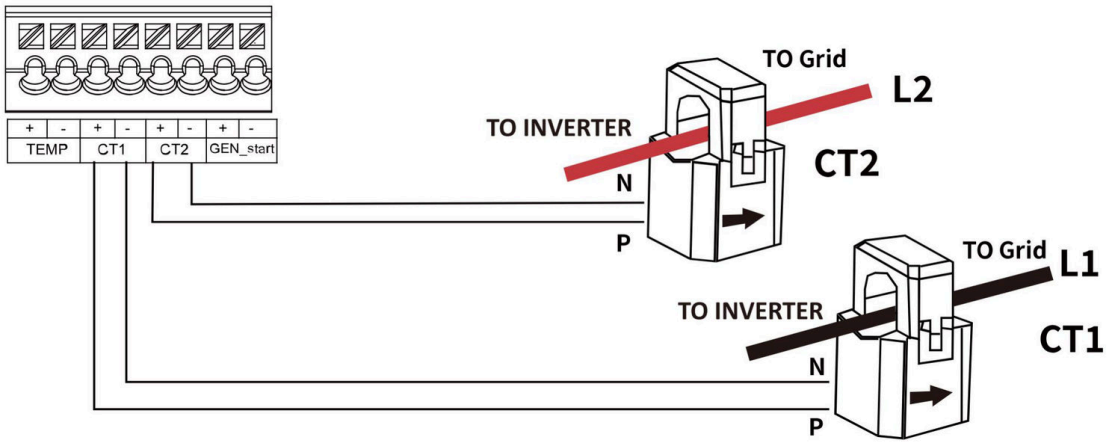
6.7.4 CT / Energy Meter Connection

To measure the power going to and coming from the grid, you must install a pair of CTs or a three-phase meter at the service entrance point. Put them inside the main service panel or near it. Two CTs come with each inverter.

CT interface pin definition



CT interface terminals: TEMP (+/−), CT1 (+/−), CT2 (+/−) and GEN_start (+/−)



CT wiring. Each CT clamps around its line wire with the arrow pointing toward the grid ("TO Grid"); "TO INVERTER" marks the inverter side. The P and N leads go to the + and − terminals of CT1 and CT2.

Use the wiring diagram to find the correct place for each CT. Clamp the two CTs onto the L1 and L2 wires at the service entrance point of the main service panel.

- **CT1** (labeled L1) goes on **L1**.
- **CT2** (labeled L2) goes on **L2**.

> **WARNING** The arrows molded into ALL of the CT sensors must point toward the grid.

If the CTs are installed wrong, the screen may show wrong readings and the inverter may not work properly.

CT clamping ratio: CT (current transformer) compatibility and setup

- The inverter supports a standard CT ratio of 2000:1.
- The CTs in the accessory package have a 2000:1 ratio and are pre-calibrated for use with the system.
- If you use a third-party CT, make sure its ratio is either 1000:1 or 2000:1.

> **CAUTION** Set the correct CT ratio for the CT you use. Make the setting on the inverter screen (section 9.8 Grid Settings Menu, CT Ratio). A wrong CT ratio setting can cause wrong power readings or make the system behave wrongly.

Temperature sensor

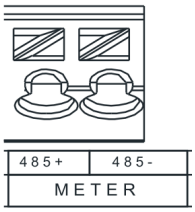
The temperature sensor connects to the **TEMP** terminals, the first pair on the terminal strip shown in the CT interface pin definition above.

- The temperature sensor terminal is required when you use lead-acid batteries.
- Place the sensor between two batteries.
- Hold it in place with tape. Keep it away from the battery terminals so it does not overheat.
- The sensor has no polarity. It lets the inverter adjust the charging voltage and the capacity calculation as the temperature changes.
- Lithium batteries do NOT need the external temperature sensor.

Energy meter connection

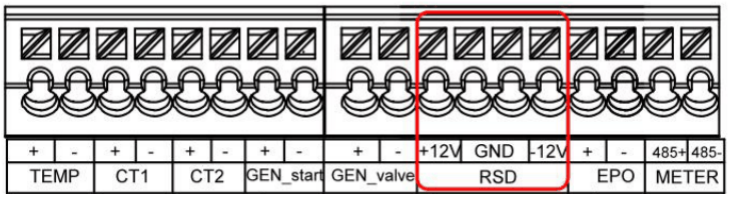
If you use an energy meter for import/export monitoring instead of CTs, connect it to the **485A** and **485B** terminals on the inverter for communication. On the terminal strip these are marked **485+** and **485−** under **METER**. These terminals use RS485.

On the inverter screen, set Grid Sensor to Meter (section 9.8 Grid Settings Menu).



METER terminals 485+ and 485−

6.7.5 Emergency Stop and Rapid Shutdown Connection



Communication terminal strip. The RSD terminals (+12V, GND, −12V) are boxed in red; the E-Stop terminals are marked EPO.

Emergency stop (E-Stop) and RSD connection guidelines

- Connect the emergency stop (E-Stop) button to its control pins on the inverter (marked EPO).
- Connect the rapid shutdown (RSD) transmitter to the RSD terminal. The RSD terminal supplies 12 Vdc or 24 Vdc power.
- The maximum current output of the RSD terminal is 500 mA. Make sure the RSD transmitter or other connected device does not go over this limit.

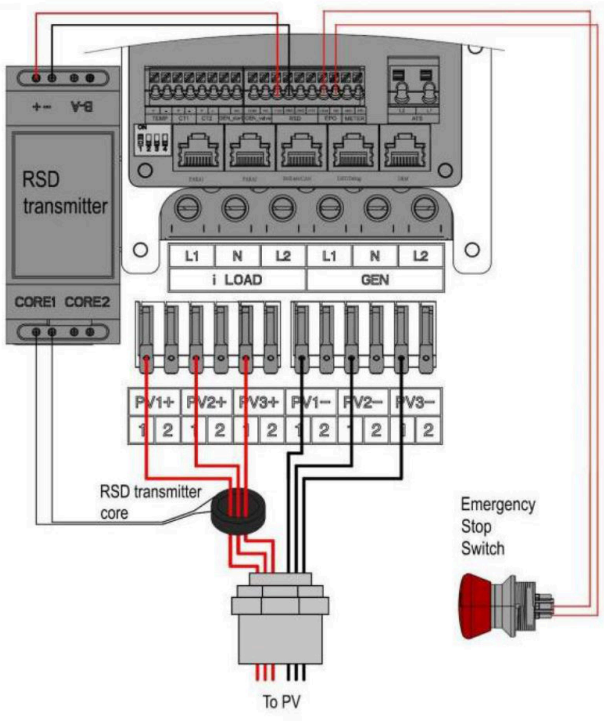
The E-Stop pins on the inverter are a normally open (NO) contact. When the circuit is closed, it triggers the rapid shutdown (RSD) function.

The inverter has two power terminals for the RSD transmitter: +12V and −12V.

- **12 V RSD transmitters:** connect the transmitter to +12V and GND.
- **24 V RSD transmitters:** connect the transmitter to +12V and −12V.

CAUTION Make sure the current draw of the RSD transmitter does not go over 500 mA. This is the maximum output of the inverter's RSD terminal.

The diagram below shows one way to wire the transmitter for rapid shutdown. Make sure the transmitter does not draw more current than the RSD terminal can supply.



Rapid shutdown wiring. The RSD transmitter takes its power from the inverter's RSD terminals (+12V and GND). The PV positive conductors pass through the RSD transmitter core on their way to the PV array ("To PV"). The Emergency Stop Switch is wired to the EPO terminals.

NOTICE The built-in 12 Vdc power supply of the inverter is rated for 500 mA. Do not exceed it! If you are not sure of the current (A) rating of the transmitter, contact Fantastech (574-218-1973, derek@fantastech.org) before you connect it.

6.8 Use a Generator

6.8.1 Generator System Connection

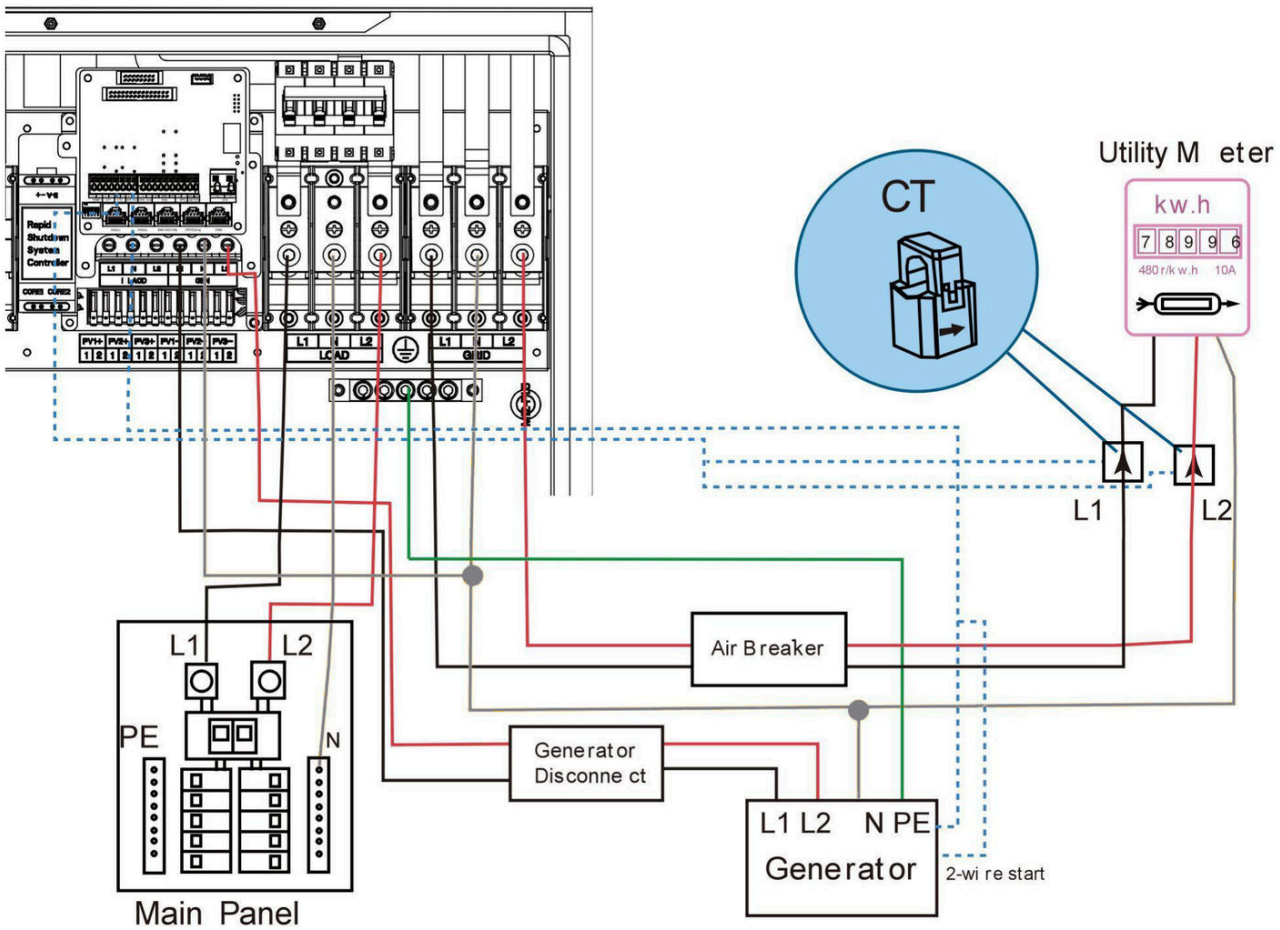
This hybrid inverter can be used with an external generator. It has a dedicated **GEN** terminal for the generator. The inverter can also take the generator on the **GRID** terminal instead. To do that, set **Gen Connect To Grid** to Yes on the inverter screen (section 9.9 Generator Port Settings Menu).

WARNING When the system is set up for 120/208 V (two- or three-phase systems), the generator can be connected only to the GRID terminal. Connecting it to the GEN terminal is prohibited. If you do, the generator will fail to start.

FANTASTECH RULES Which port the generator goes on:

- **120/240 V split-phase generator under 15 kW** → **GEN** port (63 A breaker).
- **120/240 V generator 15 kW or bigger** → **GRID** port.
- **120/208 V generator** (two legs of a three-phase generator) → **GRID** port. Never the GEN port.
- **Never run three-phase power to the GEN port.**
- **Generator on the GRID port:** set Gen Connect To Grid to Yes and Gen Port Func to No (the two settings cannot be used together). Start SOC/V and Stop SOC/V still start and stop the generator, and the generator charges the battery in Backup Power mode.
- **Generator charge time:** on Generator Setting 1/3 (section 9.9), set the Start Time 1 minute after the End Time (for example, End Time 00:00 and Start Time 00:01). The charge window then covers the whole day, so the battery charges whenever the generator runs. The factory default (Start 00:00, End 00:59) allows only 59 minutes a day.

Generator connection — whole-home backup (120/240 V)

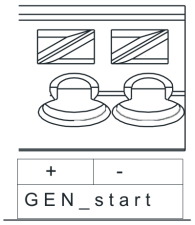


Generator connection for whole-home backup, 120/240 V. Labels in the drawing read: Utility Meter, CT, Air Breaker (circuit breaker), Main Panel, Generator Disconnect, Generator and 2-wire start. Wire colors: black = L1, red = L2, grey = neutral (white wire), green = ground, blue dotted = communication, CT and start cables. On 120/208 V the generator goes on the GRID terminal only.

When the generator starts, it powers all loads connected to the EPS and charges the battery at the same time.

The AC circuit breaker on the GEN port is rated for 63 A. When the generator is running, make sure the total load current plus charging current does not go over 63 A.

Every KOOR inverter has generator auto-start. It uses a dry-contact **GEN_start** signal that is driven by the battery SOC or voltage (see section 6.8.2). To let the inverter start the generator remotely, connect the generator's start signal (2-wire start) to the **GEN_start** terminals on the communication terminal block.



GEN_start terminals (+/−)

Generator requirements

- The generator should have a **neutral-grounding connection** and give both **240 V and 120 V output**. The generator's capacity may be larger than the capacity of the system.
- A THD (total harmonic distortion) of less than 15% is preferred.
- Make sure the generator is connected correctly and meets the voltage and capacity requirements, so it works properly with the inverter. Always follow the generator manufacturer's instructions for installation and connection.
- Do NOT use the **Grid Priority** work mode in off-grid systems. When the generator is connected to the GEN terminal, power export to the grid is not available, and Grid Priority mode must not be used. Otherwise, the generator may be damaged.

6.8.2 Generator Start and Stop Settings

These conditions depend on the **Battery (Bat) Operation Mode** settings. The system uses the **battery's state of charge (SOC)** or the **battery voltage** to decide whether the generator needs to start or stop.

Generator start conditions

The generator starts in these cases:

- **Power failure:** the grid or other power sources are lost.
- **Battery discharge cutoff:** the battery discharges to the cutoff voltage or SOC setting.
- **Forced charging:** the battery needs a forced charge.
- **Battery voltage or SOC below the generator start threshold:** the battery voltage or SOC falls below the set **generator charging start voltage/SOC**.

Generator stop conditions

The generator stops when the **battery voltage** or **SOC** is higher than the **charging endpoint voltage/SOC** setting. This means the battery is charged enough and the generator is no longer needed.

With these settings, the generator runs only when it is needed. This makes the best use of the battery and cuts down on generator run time.

Make these settings on the inverter screen, Generator Setting 1/3. For a generator on the GEN port, set Gen Port Func to Gen first. See section 9.9 Generator Port Settings Menu for how to use the screen.

Setting (Generator Setting 1/3)	What it does	Factory default
Gen Warm Time	Warm-up time. After the inverter sends the start signal, it waits this long before it accepts generator power.	60 s
Start SOC/V	Battery SOC (or voltage) at which the generator starts.	25%
Stop SOC/V	Battery SOC (or voltage) at which the generator stops.	85%
Input Rated Pwr	Maximum power output of the generator. Set it to your generator's rating.	15 (see 9.9)
Max Run Time	Longest continuous run time when the inverter controls the generator.	600 min
Cooling Time	Cool-down period after the generator stops.	10 min
Start Time / End Time	Generator charge time window. Fantastech rule: End Time 00:00, Start Time 00:01 (see section 6.8.1).	00:00 / 00:59

6.8.3 AC Coupling Installation and Connection

The inverter supports adding AC-coupled solar. You can raise the maximum solar input power by coupling microinverters or string inverters into the "GEN" or "LOAD" terminals. A fully AC-coupled solar system is not recommended, because power control and monitoring are limited. Using DC-coupled modules, or a mix of DC-coupled and AC-coupled solar panels, is always preferred.

Inverters used for AC coupling should meet the relevant industry specifications. This certification confirms that the inverter can disconnect from the grid based on frequency. It also makes sure the KOOR inverter can safely shift the frequency to control the AC-coupled production.

AC coupling to "GEN"

- CAN produce solar power during grid outages.
- CAN produce solar power for off-grid systems.
- CAN monitor solar production.

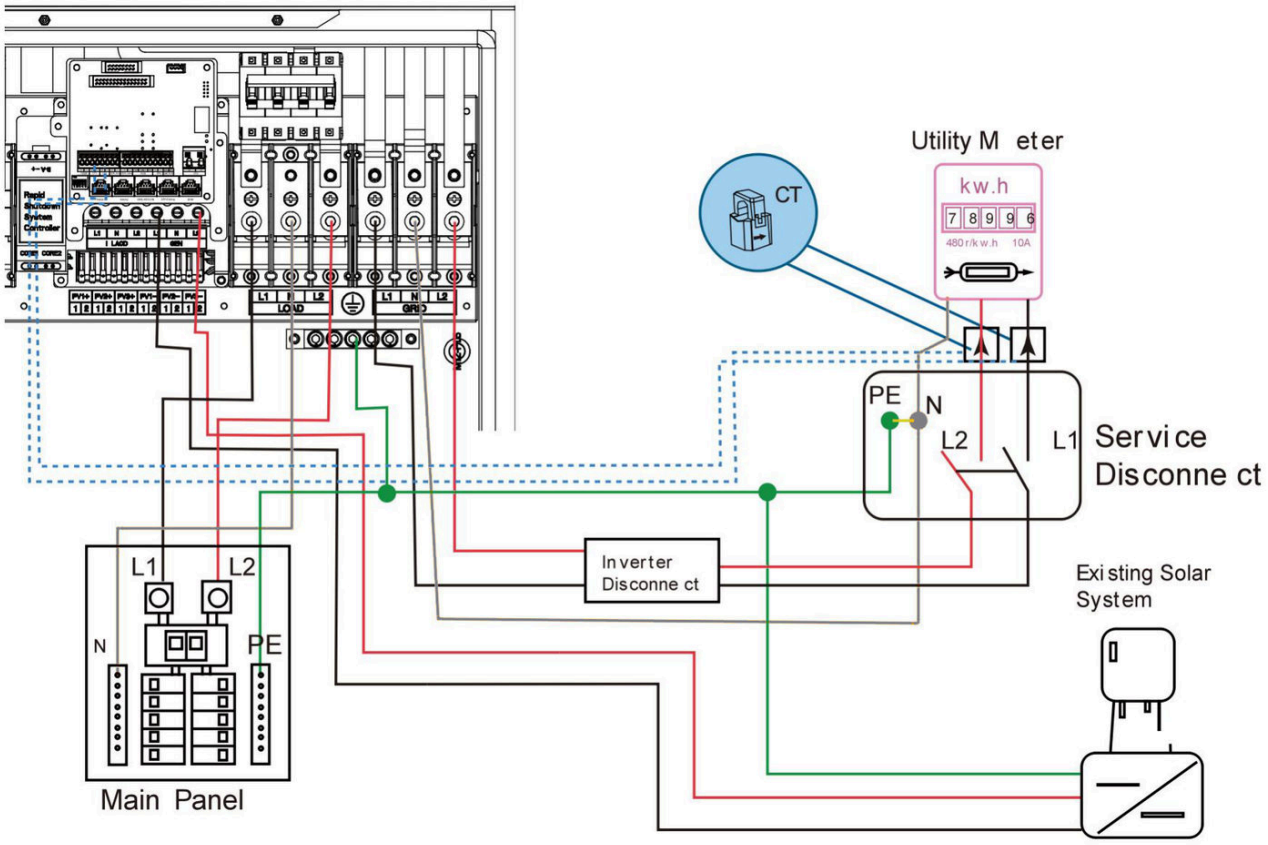
AC coupling to "LOAD"

- CAN produce solar power during grid outages.
- CAN produce solar power for off-grid systems.
- CANNOT monitor solar production.
- The "GEN" input CANNOT be used.
- The backup transfer time is extended to 2 seconds.

Size limits: the GEN terminal and the I-LOAD (intelligent load) terminal must each carry less than 15 kW. AC-coupled solar must be less than 6 kW.

For AC coupling on the GEN terminal, set Gen Port Func to AC Couple on the inverter screen (section 9.9 Generator Port Settings Menu). See also section 7.5 AC Coupling Function.

Connection for an AC-coupled solar system



AC-coupled solar system connection. Labels in the drawing read: Utility Meter, CT, Service Disconnect, Inverter Disconnect, Main Panel and Existing Solar System. Wire colors: black = L1, red = L2, grey = neutral (white wire), green = ground, blue dotted = communication and CT cables.

When the **"Grid"** is **ON**, the **GEN** terminal is connected to the inverter's internal **"Grid"** terminal. In this case, the hybrid inverter passes the power of the grid-interactive (AC-coupled) inverter through (bypass) to the grid and the EPS (Emergency Power Supply) system.

When the grid is disconnected, the **GEN** terminal is connected to the inverter's internal **EPS** terminal. The loads are then powered by solar energy first. If the solar power is more than the loads use, the extra energy is stored in the battery.

When the solar power is more than the load power plus the battery's maximum charging power (for example, when the battery is almost full), the inverter signals the grid-interactive inverter to lower its output. It does this with a **frequency-shifting power reduction** method. This keeps power production and use in balance within the microgrid, so the system runs efficiently and does not overproduce.

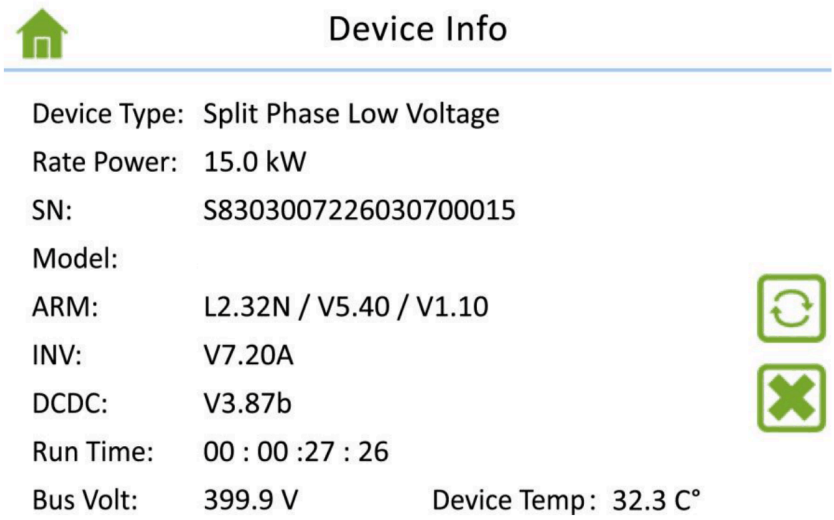
6.9 System Parallel Operation

The inverter can run in parallel with other units in two ways:

- **Split-phase 4-wire parallel** (120/240 V, section 6.9.1). This lets you add power and energy capacity to suit the job. Up to 6 units can be connected in parallel, for a total of up to 90 kW (six KR-L15K-SPUS units).

- **Three-phase 5-wire parallel** (section 6.9.2). The units are cross-connected to make a three-phase output (phases A, B and C). Up to 3 units can be connected this way.

NOTICE All inverters in a parallel group must be the same hardware model and must have the same firmware versions (ARM, DSP1, DSP2). On the Device Info screen of every inverter, check the ARM, INV and DCDC lines (section 9.12) before you start the system.



Device Info screen with the firmware versions (ARM, INV, DCDC). The model and serial number shown vary from unit to unit.

6.9.1 Parallel System Connection for Split-Phase 4-Wire System

NOTICE

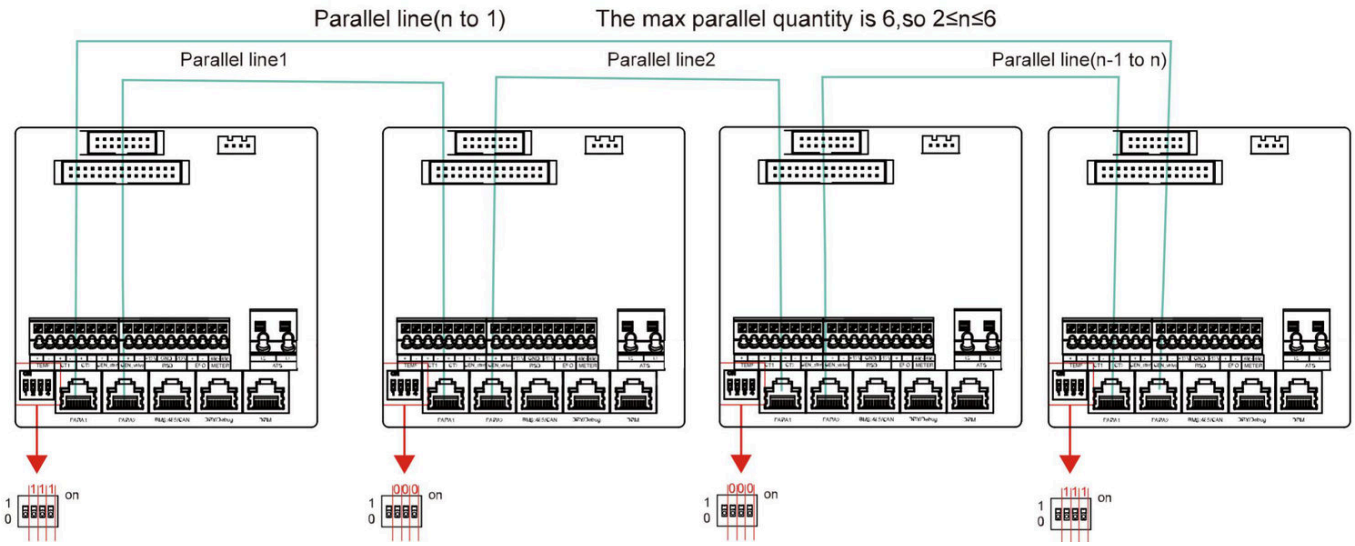
- A split-phase parallel system MUST use one shared (joint) battery bank. All the inverters in the group connect to the same bank. (The rule in section 2.4, "never connect one battery bank to more than one inverter", is for a single-inverter system.)
- All inputs and outputs must be shared by ALL the parallel inverters. The only exception is the PV (DC solar) inputs.

CAN termination for the daisy chain

The parallel cables link the inverters in a daisy chain, from the PARA ports of one inverter to the next. Set the CAN termination switches (DIP switches 2, 3 and 4, see below) to ON on the first and last inverters in the chain. Switch 1 stays ON on every inverter.

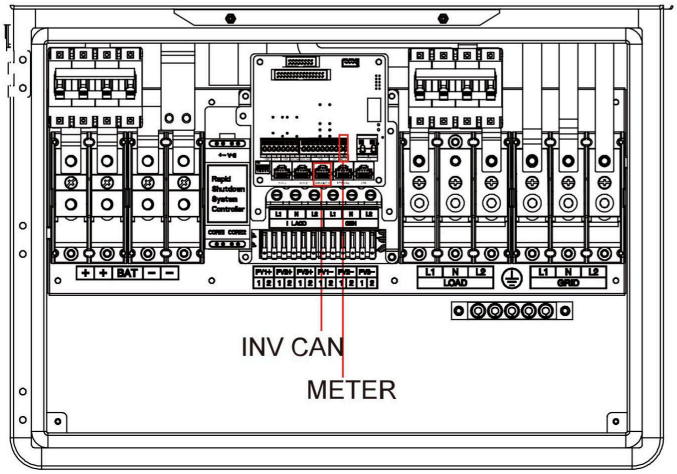
Steps:

1. **First inverter:** Set the CAN termination switches to **ON**. This enables communication and sets the starting point of the CAN bus.
2. **Last inverter:** Set the CAN termination switches to **ON** as well. This terminates the communication line properly at the end of the bus.
3. **Inverters in between:** Leave the termination switches **OFF** on every inverter between the first and last units in the chain. These units act as repeaters and do not need termination.



Parallel cables between the PARA1 and PARA2 ports (2 to 6 units). The "n to 1" cable from the last unit back to the first closes the chain into a ring and is installed. DIP switches 2, 3 and 4: ON (1) on the first and last units, OFF (0) on the units in between. Switch 1 stays ON on every unit.

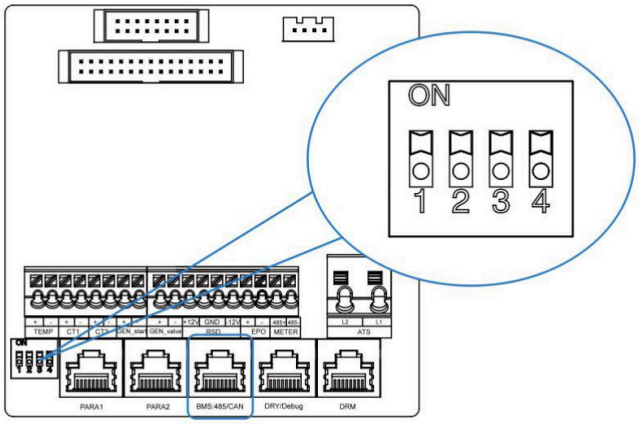
If the **parallel cables** are too short or not available, you can make a **direct pin-to-pin cable** to extend the connections between the inverters.



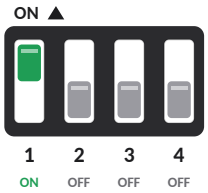
Wiring area with the INV CAN and METER connection points marked

DIP switch settings for parallel systems

The DIP switch is on the communication board, next to the PARA1 port.



Communication board with the 4-position DIP switch shown enlarged



Slider pushed UP, toward ON
= that switch is ON

Slider pushed DOWN
= that switch is OFF

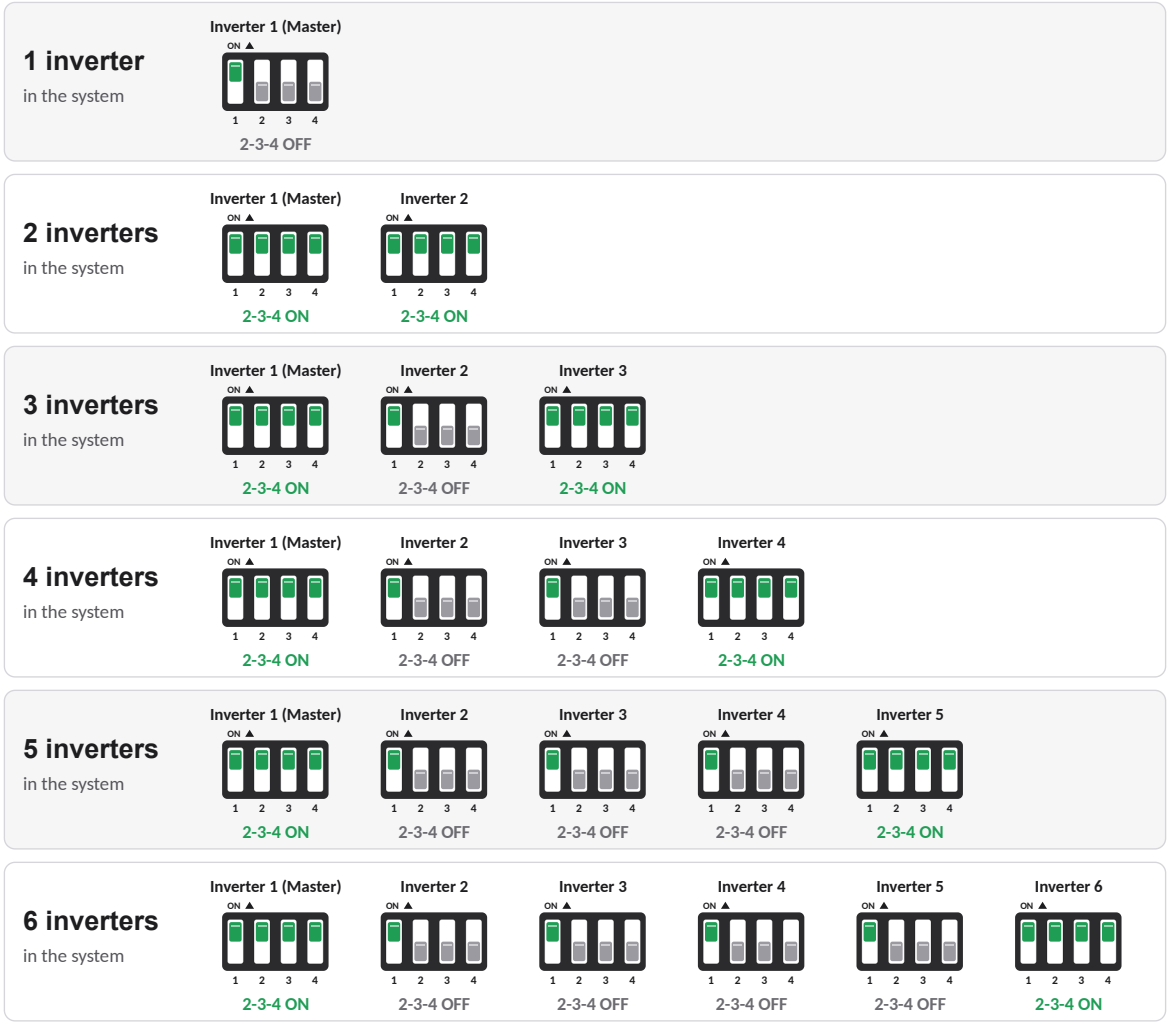
Switches are numbered 1, 2, 3, 4 from the left.
Use a small screwdriver or a pen tip.
This one reads: 1 ON, 2 OFF, 3 OFF, 4 OFF.

How to read the switch: green slider pushed up toward ON = ON. Grey slider pushed down = OFF.

This switch tells each inverter whether it sits at an **end** of the parallel cable chain. **Switch 1 is for communication. It always stays up (ON) on every inverter.** Switches 2, 3 and 4 are the CAN termination: they are ON on the two inverters at the ends of the chain and OFF on every inverter in between. A single inverter on its own has switches 2, 3 and 4 OFF.

Set your inverters like this

Find the row for the number of inverters in your system. Set each inverter's switch like its picture. Switch 1 is always up. **2-3-4 ON** = push switches 2, 3 and 4 up toward ON. **2-3-4 OFF** = push switches 2, 3 and 4 down. Inverter 1 is the master.



Parallel DIP switch settings for 1 to 6 inverters. Switch 1 is always ON. Switches 2, 3 and 4 are ON on the inverters at both ends of the chain and OFF on the ones in between.

The same settings as a table (switches 2, 3 and 4 — switch 1 is always ON):

Units in system	Inverter 1 (Master)	Inverter 2	Inverter 3	Inverter 4	Inverter 5	Inverter 6
1	OFF	—	—	—	—	—
2	ON	ON	—	—	—	—
3	ON	OFF	ON	—	—	—
4	ON	OFF	OFF	ON	—	—
5	ON	OFF	OFF	OFF	ON	—
6	ON	OFF	OFF	OFF	OFF	ON

Parallel settings on the inverter screen

Make the parallel settings on each inverter's screen, on Advanced Function page 2/2 (section 9.10). Example for a three-inverter split-phase system:

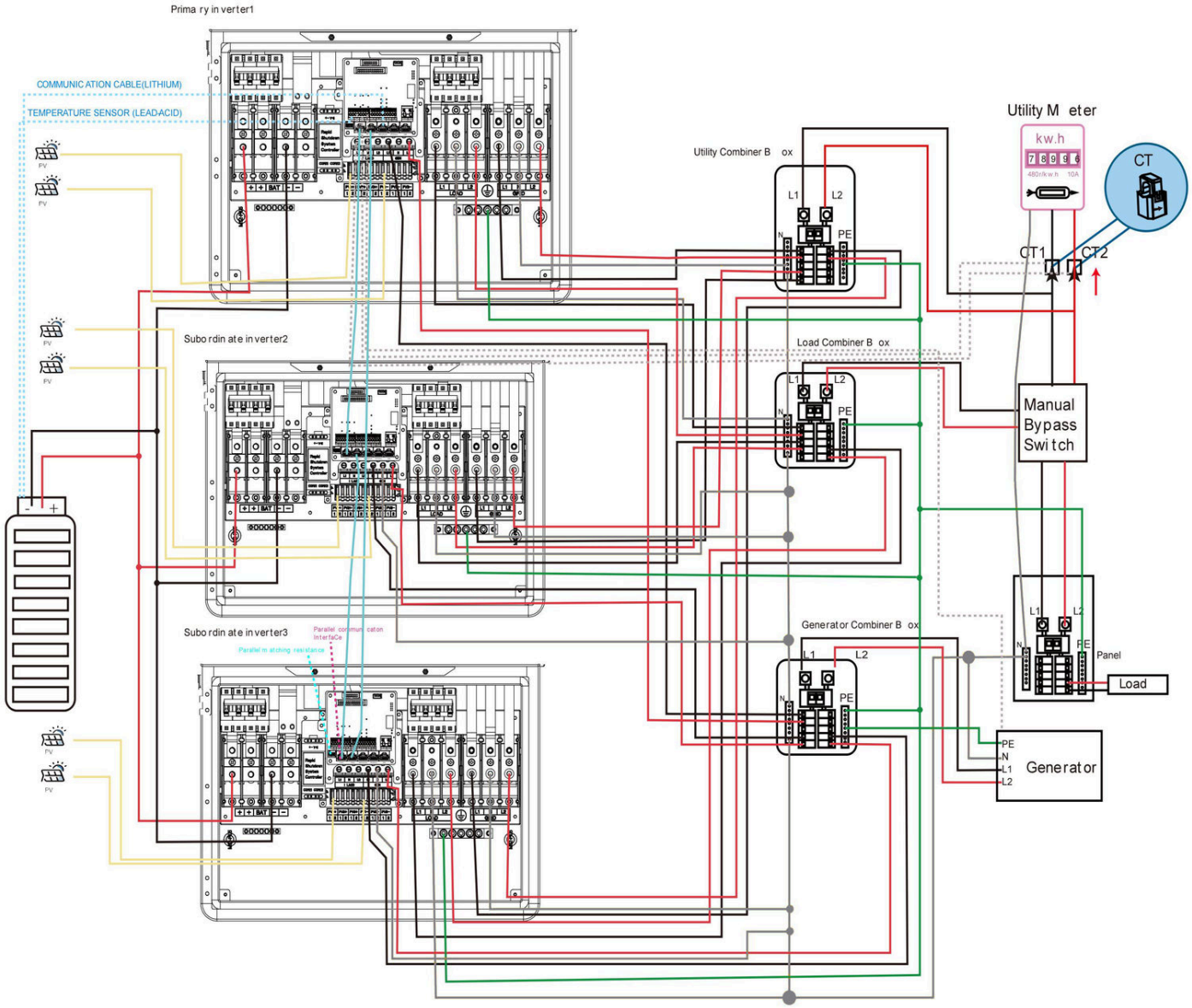
Setting	Master	Slave 1	Slave 2
Parallel Type	Master	Slave	Slave
Parallel Num (total number of inverters)	3	3	3
Parallel Addr	1	2	3
Three Phase	No	No	No

The battery settings for a battery bank shared by parallel inverters are in section 9.7 (Shared Bat). In a parallel system, the battery's BMS communication cable connects to the master inverter. For KOOR batteries, see the Appendix — KOOR batteries at the front of this manual.

Parallel wiring diagrams

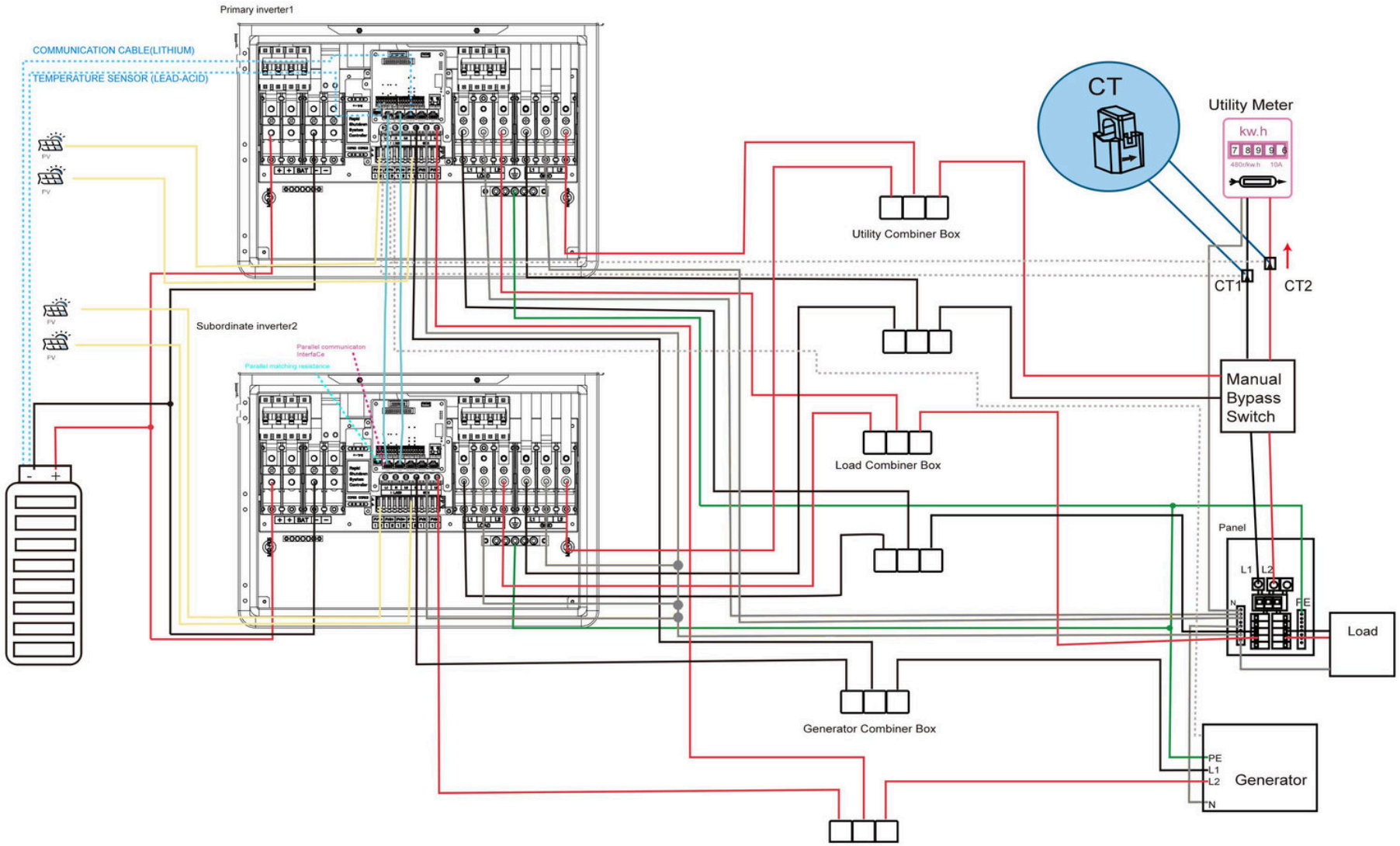
The wiring diagrams for split-phase parallel connection are shown below. In each diagram, "Parallel matching resistance" points to the CAN termination DIP switch, and "Parallel communication interface" points to the parallel cable ports. The diagrams are printed sideways so they can be as large as possible. Turn the manual a quarter turn to the right to read them.

A. Three 15 kW units in parallel, 120/240 V



Primary inverter 1 and subordinate inverters 2 and 3 on one battery bank, with utility, load and generator combiner boxes, manual bypass switch, utility meter with CT1 and CT2, panel, load and generator. Wire colors: black = L1 and battery –, red = L2 and battery +, grey = neutral (white wire), green = ground, yellow = PV, dotted = communication.

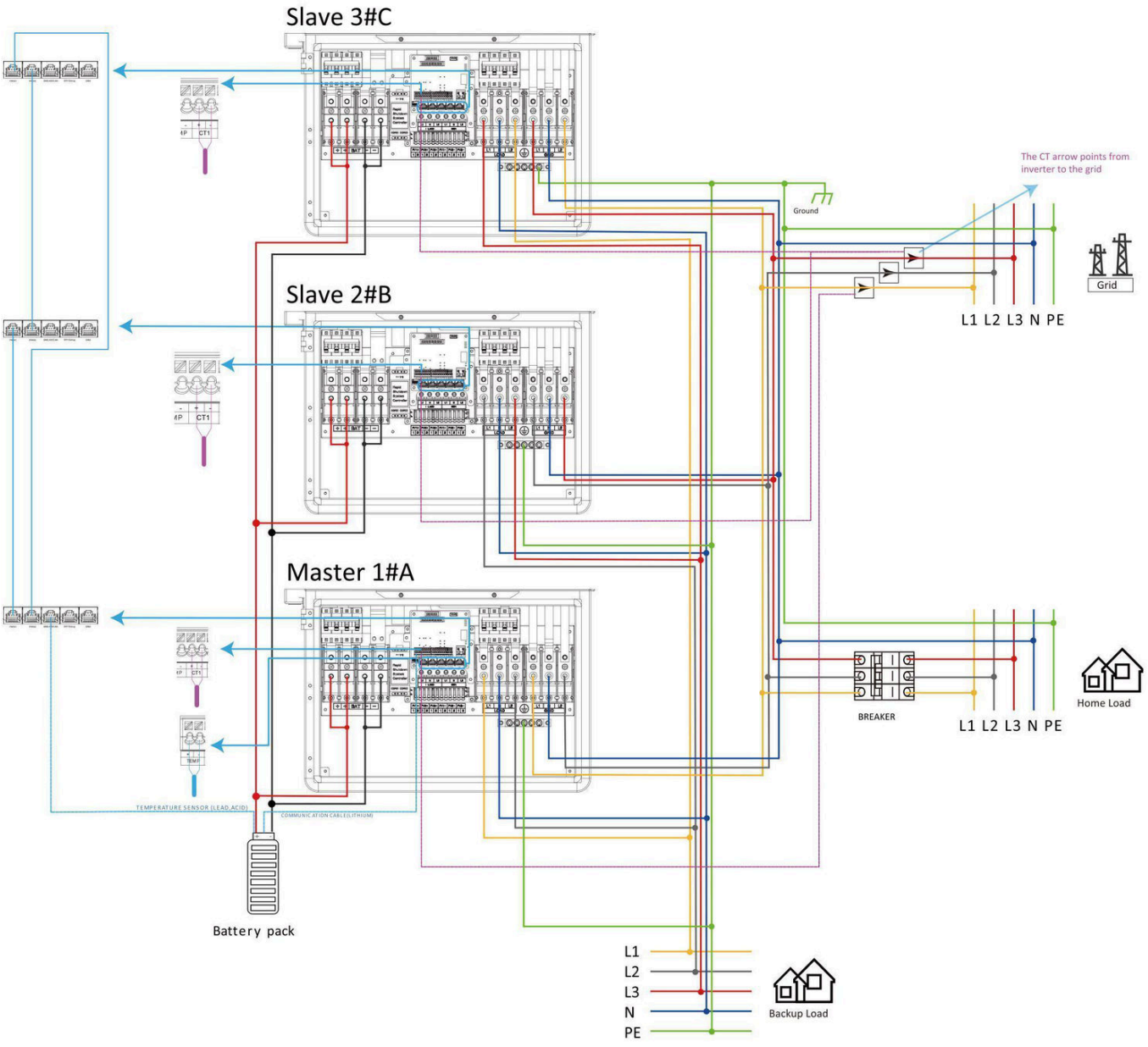
B. Two 15 kW units in parallel, 120/240 V



Primary inverter 1 and subordinate inverter 2 on one battery bank, with utility, load and generator combiner boxes, manual bypass switch, utility meter with CT1 and CT2, panel, load and generator. Wire colors: black = L1 and battery −, red = L2 and battery +, grey = neutral (white wire), green = ground, yellow = PV, dotted = communication.

6.9.2 Parallel System Connection for Three-Phase 5-Wire System

Three inverters can be cross-connected to make a three-phase output. Each inverter's L1 and L2 connect to two different phases. Unit 1 is the master (Master 1#); units 2 and 3 are slaves (Slave 2# and Slave 3#). The wiring diagram is shown below.



Three inverters in three-phase parallel: Master 1#A, Slave 2#B and Slave 3#C, with one battery pack, the CTs on the grid phases, the home-load breaker and the backup load. This drawing uses international wire colors: yellow = L1, grey = L2, red = L3, blue = neutral (N), green = PE (ground). In the US, wire L1 black, L2 red, L3 blue, neutral white (grey) and ground green or bare. Battery: red = +, black = –. The CT arrow points from the inverter to the grid.

FANTASTECH RULE Never run three-phase power to the GEN port. A generator in a three-phase or 120/208 V system goes on the GRID side (section 6.8.1).

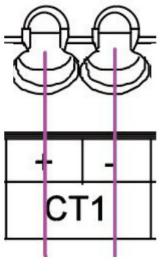
Step 1: Wire the system

Wire the system as shown above. Install a 200 A 3-phase 4-wire AC switch at the point where the loads are combined. The cables from each inverter to the grid combining point must all be the same length. The cables from each inverter to the load combining point must all be the same length.

NOTICE

1. **Grid phases and CTs.** Connect the grid terminals as in the table below. Mount one CT on each phase at its combining point, and connect that CT's sampling wires to the CT1 terminal of the inverter named in the table.
2. **Grid neutral (N):** Connect the Grid N terminals of Master 1#, Slave 2# and Slave 3# together, then to the grid N.
3. **Protective earth (PE):** Connect the PE terminals of Master 1#, Slave 2# and Slave 3# together, then to the grid PE.
4. **Load phases:** Connect the load (EPS) terminals as in the table below.
5. **Load neutral (N):** Connect the load N terminals of Master 1#, Slave 2# and Slave 3# together, then to the load N.
6. **Load PE:** Connect the load PE to the incoming grid PE.
7. **PV:** Follow the single-unit wiring rules (section 6.5). Independent MPPT mode is supported. Up to 9 PV strings can be connected in total for the three inverters.
8. **Battery:** Use one shared battery bank, with the BMS communication cable to Master 1#.

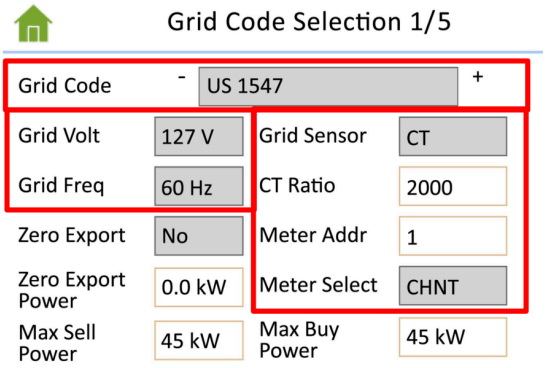
Phase	Grid terminals joined	CT on this phase goes to	Load (EPS) terminals joined
Phase A	Grid L1 of Master 1# + Grid L2 of Slave 3#	CT1 terminal of Master 1#	EPS L1 of Master 1# + EPS L2 of Slave 3#
Phase B	Grid L2 of Master 1# + Grid L1 of Slave 2#	CT1 terminal of Slave 2#	EPS L2 of Master 1# + EPS L1 of Slave 2#
Phase C	Grid L2 of Slave 2# + Grid L1 of Slave 3#	CT1 terminal of Slave 3#	EPS L2 of Slave 2# + EPS L1 of Slave 3#



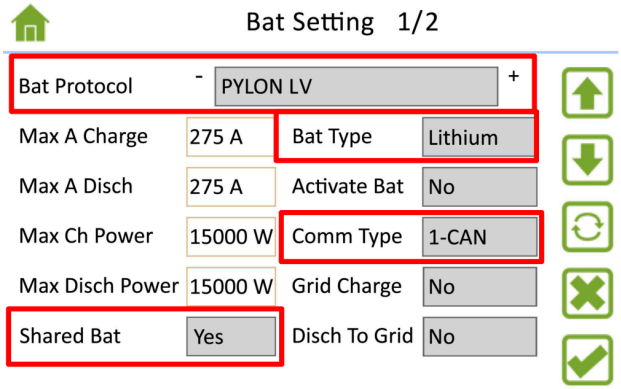
CT sampling wires on the CT1 terminal (+ and −)

Step 2: Set the parameters on the screen

Set the grid, battery and PV settings on each inverter. The settings marked in red on the example screens are the ones to check. See sections 9.7, 9.8 and 9.10. For KOOR batteries, see the Appendix — KOOR batteries at the front of this manual.

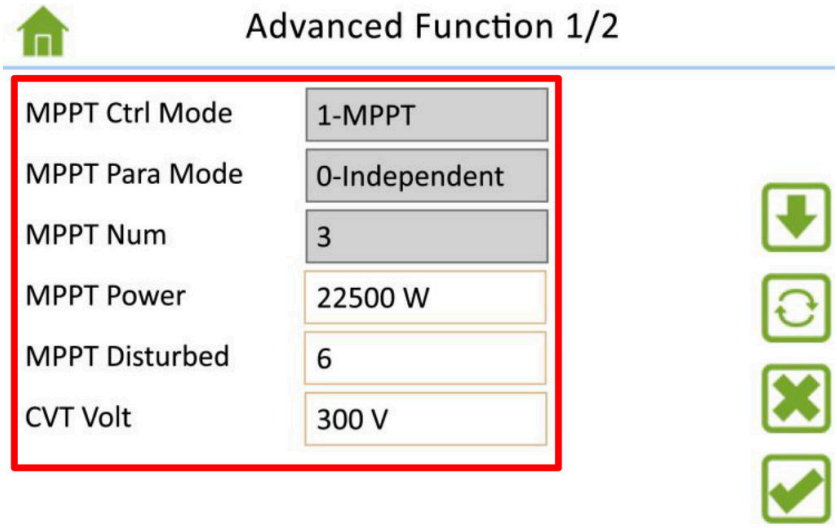


Grid Code Selection 1/5: Grid Code, Grid Volt, Grid Freq, Grid Sensor, CT Ratio, Meter Addr and Meter Select



Bat Setting 1/2: Bat Protocol, Bat Type, Comm Type and Shared Bat (Yes for a shared battery bank)

US SETTING The example screens show Grid Volt 127 V (a 220/127 V grid). On a US 120/208 V three-phase service, set **Grid Volt** to **120 V** (line to neutral, the same as the single-unit default in section 9.8) and keep **Grid Freq** at **60 Hz**.



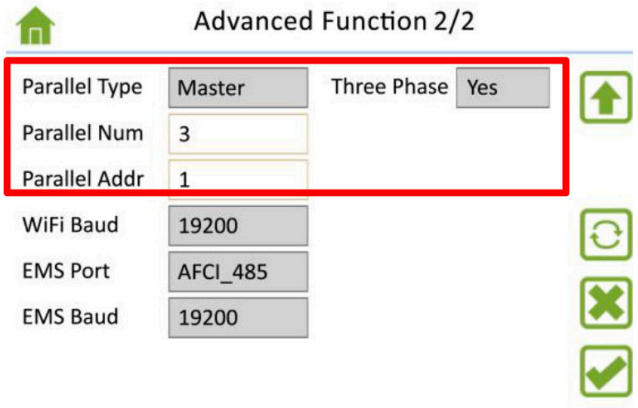
Advanced Function 1/2: MPPT Ctrl Mode 1-MPPT, MPPT Para Mode 0-Independent, MPPT Num 3, MPPT Power, MPPT Disturbed and CVT Volt

The example screens show the KR-L15K-SPUS values (275 A, 15000 W, 22500 W). Use the values for your model (specifications, section 3.3.4). The maximum battery charge/discharge current is 167 / 210 / 250 / 275 A for the KR-L8K / L10K / L12K / L15K-SPUS.

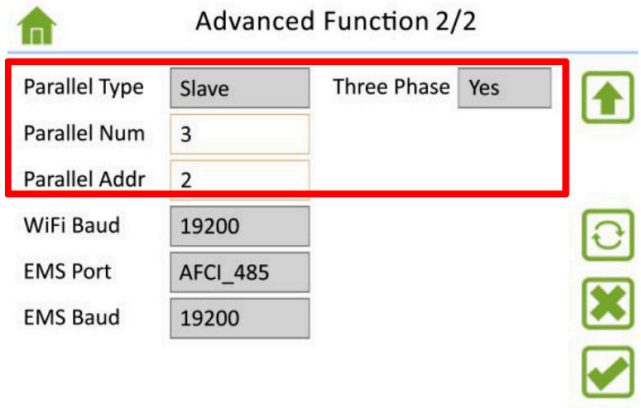
Step 3: Set the parallel parameters

On Advanced Function page 2/2 (section 9.10) of each inverter, set Parallel Type, Parallel Num and Parallel Addr, and set **Three Phase** to **Yes**:

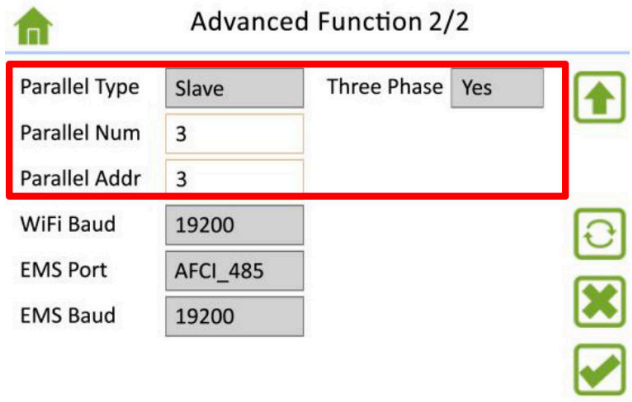
Setting	Master 1#	Slave 2#	Slave 3#
Parallel Type	Master	Slave	Slave
Parallel Num	3	3	3
Parallel Addr	1	2	3
Three Phase	Yes	Yes	Yes



Advanced Function 2/2 on Master 1#



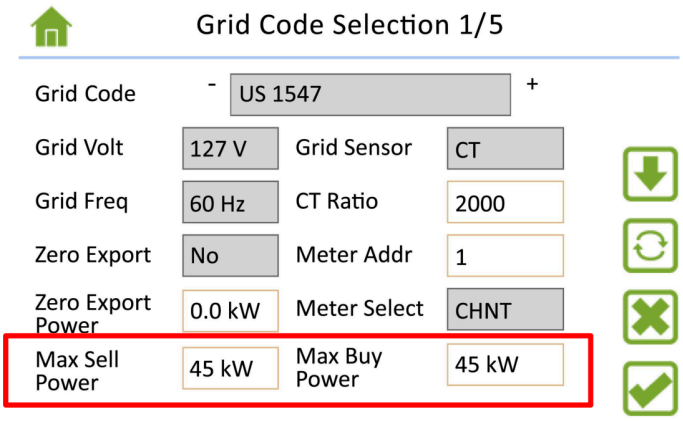
Advanced Function 2/2 on Slave 2#



Advanced Function 2/2 on Slave 3#

Step 4: Set the grid parameters

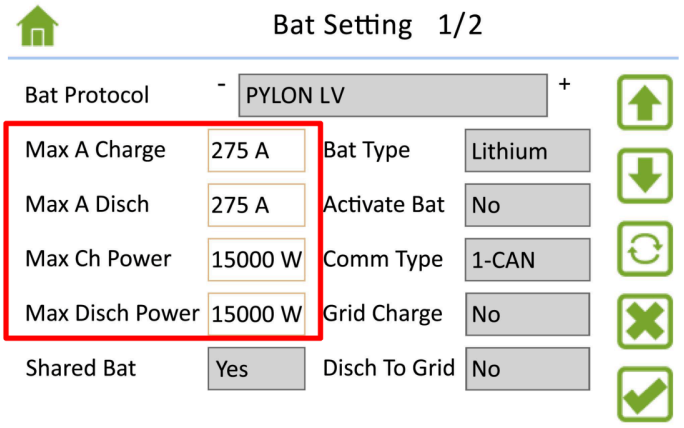
The maximum grid input and output power of the three-inverter system is 45 kW (three KR-L15K-SPUS units). Set **Max Sell Power** and **Max Buy Power** on Grid Code Selection page 1/5 (section 9.8).



Grid Code Selection 1/5: Max Sell Power and Max Buy Power set to 45 kW

Step 5: Set the battery parameters

Set **Max A Charge**, **Max A Disch**, **Max Ch Power** and **Max Disch Power** on Bat Setting page 1/2 (section 9.7).



Bat Setting 1/2: the KR-L15K-SPUS example (275 A, 15000 W)

Step 6: Check the parallel DIP switches

On Master 1# and Slave 3#, DIP switches 1, 2, 3 and 4 must all be ON. On Slave 2#, only DIP switch 1 is ON; switches 2, 3 and 4 are OFF. (This is the 3-inverter row of the DIP switch table in section 6.9.1.)

Steps 7 to 9: Start the system

7. Check that the parallel communication link is working and that no fault alarms are showing.
8. If there are no faults: disconnect the grid supply first, and power on the system from the batteries only.
9. After the system runs steadily off-grid on the batteries, power down all the units completely. Close the grid-side and PV circuit breakers, then power the system back on.

7. Operating Modes and Functions

7.1 Operate the Inverter and Run a Test

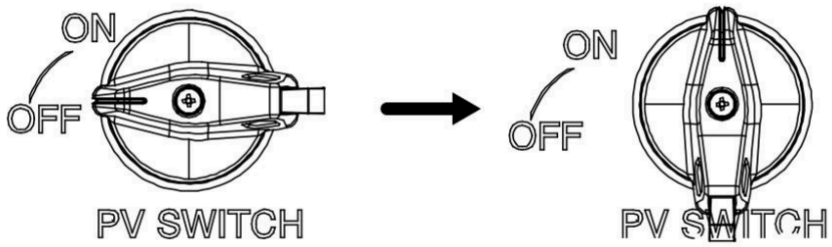
The inverter needs at least one energy source to run: battery, PV or grid.

7.1.1 Check the Voltage of the Battery Bank

- The battery voltage must be 40–60 Vdc.
- If the batteries have their own internal switches, turn them ON. Measure the voltage of each battery.
- Check that the battery bank voltage at the inverter's battery terminals is correct.
- DO NOT reverse the polarity. DO NOT turn OFF the battery disconnect while any current is flowing into or out of the battery.

7.1.2 Check the Voltage of Each PV Input Circuit

- The PV input voltage must never go above 550 Vdc. Size every string so its cold-corrected open-circuit voltage (Voc) stays at or below 525 Vdc.
- The input voltage must be above the start-up voltage of 120 Vdc.
- Do not ground PV+ or PV−.
- Check the polarity of each PV string. If a string is connected backward, the inverter reads 0 Vdc on that input, and the reversed polarity will cause long-term damage.
- PV power alone only turns on the LCD screen. The inverter needs the grid, the batteries or both to operate.
- The PV DC disconnect switch on the side of the inverter turns the PV ON or OFF.



PV switch: turn it from OFF (left) to ON (right)

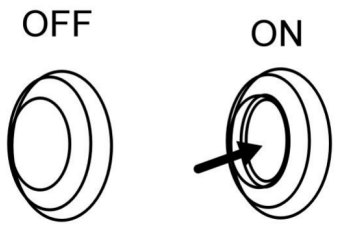
7.1.3 Check the Grid Input Voltage

- Use a multimeter to measure the AC voltage at the "GRID" terminals.
- Measure line (L) to neutral (N) on the "GRID" terminals. Make sure it reads 120 Vac on both L1 and L2.
- Measure line (L) to line (L) on the "GRID" terminals. Make sure it reads 240 Vac. If it reads close to 208 Vac, the site has a 120/208 V service (two legs of a three-phase service). Then set the grid standard for 120/208 V (section 6.7.1) before you go on.
- Check that the voltage between neutral and ground is 0 Vac.
- Check that the voltage between "GRID" L1 and "LOAD" L1 is 0 Vac. Do the same for L2.

7.1.4 Power ON the Inverter

This is the same start-up order as section 11.1.

1. Turn ON the battery breaker(s).
2. Press the power button down to the ON position. Wait for the LED indicator to turn on. This may take a few minutes.
3. Turn ON the PV DC disconnect switch (turn the PV SWITCH to ON).
4. Turn ON the external "GRID" disconnect (the AC breaker between the inverter and the grid).
5. Turn ON the internal "LOAD" breaker (the AC breaker between the inverter's load port and the backup loads) and the external "GEN" breaker.
6. The first time you power on the inverter, send the start command by hand: on the inverter screen, set **Power On/Off** to ON (Work Mode page 2/4, section 9.6). For safety, the inverter can be set to start by itself after this first start (**Auto Power On**, same page).
7. The inverter should now start running.



Power button: OFF (left) and pressed in to ON (right)

7.1.5 Power Cycle Sequence

Steps 1 to 4 are the same shut-down order as section 11.2.

1. Turn OFF all AC breakers and disconnects ("GRID", "GEN" and "LOAD").
2. Turn OFF the built-in PV DC disconnect switch on the side of the inverter.
3. Press the power button and make sure it is in the OFF position. An "OFF" message appears after the LED turns off.
4. Turn OFF the battery breakers.
5. Wait at least 5 minutes so the inverter is completely de-energized (the label on the unit says 5 minutes).
6. Make sure the inverter is properly connected to the batteries, solar panels, "GRID", "GEN" and "LOAD".
7. To turn the inverter back ON, do these steps in reverse order (section 7.1.4).

7.2 Operating Modes and Functions

The inverter has several operating modes and functions to meet different customer needs. They are described below.

You choose the work mode on the inverter screen with the **Work Mode** setting (Work Mode page 2/4, section 9.6). The choices are Self Use, TOU ECO, Backup Power and Grid Priority.

7.2.1 Self Use Mode (Default)

In **Self Use** mode, the system uses this order of priority:

- **Power for the loads comes from:** Solar → Battery → Grid
- **Solar energy goes to:** Load → Battery → Grid

When solar power is available, it powers the local loads first. Extra solar energy is stored in the battery. Anything left over is exported to the grid. When solar power is not enough, the system uses battery energy to help power the loads. The system draws power from the grid only when solar and battery together are not enough.

Self Use is the default operating mode. It is designed to **use as much of your own solar energy as possible**. This lowers your need for grid power and cuts your electricity costs.

When to use it

Self Use mode is a good fit for homes and businesses that want lower energy bills and more energy independence by making full use of their own solar power.

Battery stop settings

These settings are on the inverter screen, on Work Mode page 2/4 (section 9.6).

- **On-Grid Bat Stop SOC** (10–100%, default 10%): While the inverter is on-grid, the battery can discharge only while its SOC is higher than this setting. If the SOC drops below this setting minus 5%, the battery goes into charge mode and charges back up to this setting.
- **Off-Grid Bat Stop SOC** (10–100%, default 10%): While the inverter is off-grid, the battery can discharge only while its SOC is higher than this setting.
- **Charge Max SOC** (default 95%): Charging stops when the battery reaches this SOC.

7.2.2 TOU ECO Mode

TOU ECO (time-of-use economy) mode is designed to save the most money. It changes how the system uses and stores energy to follow time-of-use (TOU) electricity prices. You divide the day into Discharge, Self-Use and Charge periods:

- **Discharge periods — priority: Load → Grid → Battery.** Earn the most by selling solar energy to the grid while the price is high.
- **Self Use periods — priority: Load → Battery → Grid.** Use solar power mainly for your own loads and store the extra in the battery.
- **Charge periods — priority: Battery → Load → Grid.** Charge the battery from the grid while electricity is cheap, and use the stored energy during expensive periods.

You can set the Discharge, Self-Use and Charge periods to match your local electricity rates and your own usage. Set them on the inverter screen or through the monitoring app.

Time setting

On the inverter screen, the time periods are set on Work Mode page 3/4, TOU Mode Settings (section 9.6). Each time slot has a start time, an end time and a type (Discharge, Self-Use or Charge). Tap "+" to add a time slot (up to 6). The Work Mode must be set to TOU ECO for the time slots to take effect.

When to use it

TOU ECO mode is a good fit where electricity prices change a lot from one time of day to another. It lowers electricity costs and earns the most by charging when electricity is cheap and discharging or selling when it is expensive.

Make this setting on the inverter screen, on Work Mode page 2/4 (section 9.6).

7.2.3 Backup Power Mode

Backup Power mode is designed to charge the battery as fully as possible and keep that energy in reserve for backup during grid outages or an unstable grid.

In this mode, during normal operation the system stores solar energy in the battery first, instead of sending it to the loads or exporting it to the grid. The battery **discharges only when the grid fails or is not available**.

PV power priority: Battery → Load → Grid

When to use it

Backup Power mode is a good fit for areas with frequent power outages or an unstable grid, or where critical loads (such as medical equipment, servers or refrigeration) must keep running at all times. It puts available energy and system reliability ahead of cost savings or feed-in income.

Make this setting on the inverter screen, on Work Mode page 2/4 (section 9.6).

7.2.4 Grid Priority Mode

Grid Priority mode is designed to export power to the grid whenever there is enough, to earn the most feed-in income.

In this mode, the system supplies power in this order:

Power supply priority: Load → Grid → Battery

Solar energy first powers the local loads. Any extra energy is then exported to the grid for income. The battery is charged only if there is still surplus after the loads and the grid export are met.

When to use it

Grid Priority mode is a good fit where there are **good feed-in tariffs (FiTs)** or net metering rules. It suits users who want to **earn the most from their solar power**, more than they want to use their own energy or keep backup power.

Make this setting on the inverter screen, on Work Mode page 2/4 (section 9.6).

7.3 Generator Function

The GEN port can be set up as a generator input or as Intelligent Load 1 (Smart Load, section 7.4). The generator function lets the inverter run a generator connected to the GEN port. To use it, set **Gen Port Func** to **Gen** on the inverter screen (Generator Setting page 1/3, section 9.9).

The inverter starts the generator automatically through the GEN_start dry contact (wiring in section 6.8.1). The generator starts when the battery SOC (or battery voltage) falls to the **Start SOC/V** setting. It stops when the battery reaches the **Stop SOC/V** setting. While the generator runs, its power is used to charge the battery.

> **NOTE** Automatic generator control works only if the inverter can control the generator. If it cannot, automatic control is not possible and the generator must be started and stopped by hand.

Settings

- **Gen Warm Time:** Generator warm-up time. After the inverter sends the start signal, it waits this long before it accepts generator power.
- **Start SOC/V and Stop SOC/V:** The levels at which the generator starts and stops automatically. With lithium batteries they are battery SOC levels. With non-lithium batteries they are battery voltages.
- **Input Rated Pwr:** The maximum power output of the generator.
- **Max Run Time:** The longest time the generator may run without stopping. This applies when the inverter controls the generator automatically.

- **Start Time and End Time:** The generator charge window. The generator charges the battery only inside this window (Fantastech rule: set the Start Time 1 minute after the End Time, see below and section 6.8.1).
- **Cooling Time:** The generator cool-down period after it stops. This applies when the inverter controls the generator.
- **Gen Force:** Force-starts the generator. Turn it on the first time you connect a generator, so the inverter recognizes and starts it.

> **FANTASTECH RULES**
>
> - **Generator charge time:** set the Start Time 1 minute after the End Time (for example End Time 00:00, Start Time 00:01). The window then covers the whole day, so the battery always charges whenever the generator runs. The factory default (Start 00:00, End 00:59) allows only 59 minutes a day.
> - **Off-grid:** keep the generator **Start SOC/V** above the **Off-Grid Bat Stop SOC** (section 7.2.1), so the generator starts before the battery stops.
> - Which port the generator goes on: section 6.8.1.

Example

- Generator Start SOC/V = 20%
- Generator Stop SOC/V = 90%

How it works

- When the battery SOC falls below 20%, the inverter starts the generator. The generator's power charges the battery.
- When the battery SOC reaches 90%, the inverter stops the generator.

When to use it

Use this function when a generator backs up the battery, for example in an off-grid system or during a long grid outage. The battery is recharged without anyone having to start the generator by hand.

7.4 Smart Load Function

The **Smart Load** function lets a power port work as a load output. When Smart Load is on, the inverter powers the load on that port when the **battery state of charge (SOC)** is above the levels you set. There are two smart load ports:

- **Intelligent Load 1** shares the GEN port. To use it, set **Gen Port Func** to **Smart Load** on the inverter screen (section 9.9).
- **Intelligent Load 2** is the I-LOAD terminal (iLoad, section 7.6).

> **IMPORTANT** **Do not connect a generator** when the Smart Load or AC Coupling function is on. This can **damage the equipment**.

Settings

- **Stop Output SOC:** When the battery SOC drops to this level, the smart load output turns off automatically. (Lithium batteries only.)
- **Stop Output Volt:** If voltage-based control is selected, the smart load output turns off when the battery voltage reaches this value.
- **Control Start Time:** The start of the time window in which the SOC or voltage conditions apply.

- **Control End Time:** The end of the time window in which the SOC or voltage conditions apply.

Two more settings on the Smart Load page — **On Grid Always On** and **Off Grid Immediately Off** — see section 9.9.

> **NOTICE** Outside the time window you set, the SOC and voltage limits do not apply to the smart load output. It then follows the same discharge behavior as the EPS output.

Example

- Smart Load Stop Output SOC = 50% (lithium batteries only)
- Smart Load Stop Output Volt = 50 V (non-lithium batteries only)
- Control Start Time = 08:00
- Control End Time = 16:00

How it works

- From 08:00 to 16:00, the relay for the smart load port opens automatically when the lithium battery SOC falls below 50% (or when the voltage of a non-lithium battery falls below 50 V).

When to use it

Use this function when you want solar energy to power non-critical loads only when the battery is well charged and solar production is above a certain level. This way, only extra power goes to those loads.

> **NOTICE** Smart load output is not supported in a parallel system. Even if the port is set to Smart Load, the output stays off.

7.5 AC Coupling Function

The **AC Coupling** function lets an existing PV (grid-tie) inverter connect to the GEN port. When the conditions are met, power from that PV inverter can flow into the unit. To use it, set **Gen Port Func** to **AC Couple** on the inverter screen, then set the AC Coupling settings (AC Coupling Function, section 9.9). Wiring and size limits for AC coupling are in section 6.8.3.

> **IMPORTANT** **Do not connect a generator** when the AC Coupling function is on. This can **damage the equipment**.

Settings

SOC values are used with lithium batteries. Voltage values are used with non-lithium batteries.

- **Start SOC/V:** The battery SOC or voltage level that turns AC coupling on.
- **Stop SOC/V:** The battery SOC or voltage level that turns AC coupling off.
- **Backlash SOC/V:** The hysteresis (dead band) for the Start and Stop levels.

Example (lithium battery)

- AC Coupling Start SOC/V = 10%
- AC Coupling Stop SOC/V = 90%
- AC Coupling Backlash SOC/V = 5%

How it works

- When the battery SOC is above 15% and below 85%, the GEN port relay closes.
- When the battery SOC is below 10% or above 90%, the GEN port relay opens.

When to use it

Use this function to add an existing PV inverter to the system and make flexible use of its power. When the conditions are met, the AC power from the PV inverter flows into the inverter automatically to charge the battery, power the loads and more. This makes good use of extra PV power and manages the system's energy automatically.

7.6 iLoad Function

The **iLoad** function controls the I-LOAD terminal (Intelligent Load 2, section 7.4) as a load output. When iLoad is on, the inverter powers the load on that terminal when the **battery state of charge (SOC)** is above the levels you set. Make these settings on the inverter screen.

Settings

- **Stop Output SOC:** When the battery SOC drops to this level, the iLoad output turns off automatically. (Lithium batteries only.)
- **Stop Output Volt:** If voltage-based control is selected, the iLoad output turns off when the battery voltage reaches this value.
- **Control Start Time:** The start of the time window in which the SOC or voltage conditions apply.
- **Control End Time:** The end of the time window in which the SOC or voltage conditions apply.

Example

- iLoad Stop Output SOC = 50% (lithium batteries only)
- iLoad Stop Output Volt = 50 V (non-lithium batteries only)
- Control Start Time = 08:00
- Control End Time = 16:00

How it works

From 08:00 to 16:00, the relay for the iLoad port opens automatically when the lithium battery SOC falls below 50% (or when the voltage of a non-lithium battery falls below 50 V). Outside this time window, the relay stays open (off) by default.

When to use it

Use this function when you want solar energy to power non-critical loads only when the battery is well charged and solar production is above a certain level. This way, only extra power goes to those loads.

7.7 Peak Shaving Function

The **Peak Shaving (PS)** function keeps the power drawn from the grid below a limit you set, during a time window you set, while the battery is able to discharge. It works on a single inverter and on parallel systems. Make the settings on the inverter screen, on Grid Settings page 2/5 (section 9.8).

Settings

- **PS Enable:** Turns the Peak Shaving function on or off.

- **PS Power:** The maximum power allowed into the inverter from the grid port.
- **PS Start Time:** The start of the peak shaving time window.
- **PS End Time:** The end of the peak shaving time window.
- **PS Overdrive:** When it is on, grid power may go above the PS limit to cover the loads when PV and battery power are not enough. When it is off, the unit shuts down automatically after 1 minute of continuous overload at 110% of the PS limit. If Auto Power On is turned on, it locks after 3 failed restart attempts. The retry counter clears after 5 minutes of normal running.

Example

- PS Enable = ON
- PS Power = 2.0 kW
- PS Start Time = 09:00
- PS End Time = 18:00
- PS Overdrive = OFF

How it works

From 09:00 to 18:00, while the battery is able to discharge, the inverter keeps the power drawn at the grid port below 2.0 kW. If PV and battery together cannot cover the total load and PS Overdrive is off, the inverter shuts down automatically after the grid power stays at 110% of the 2.0 kW limit for 1 minute. If Auto Power On is turned on, it locks after three restart attempts in a row fail, and the inverter stops restarting by itself. The retry counter resets only after 5 minutes of steady normal operation.

How it works in each work mode

When PS Enable is on, the system checks whether peak shaving applies in the current work mode. Within the time window, it keeps grid power below PS Power, following these rules:

1. **Self Use mode:** Peak shaving starts when the battery SOC (the lowest SOC of the units in a parallel system) drops below the On-Grid Bat Stop SOC. The grid powers the loads first. PV and battery discharge cover any load above the PS Power limit.
2. **Backup Power mode:** Peak shaving is on by default. The grid charges the battery first. The battery discharges to cover load power above the PS limit.
3. **Grid Priority mode:** If the battery SOC (the lowest SOC of the units in a parallel system) is higher than the On-Grid Bat Stop SOC, the system runs normal Grid Priority without a peak shaving limit. If the battery SOC falls below the On-Grid Bat Stop SOC, peak shaving starts automatically and the system switches to Self Use mode.

Battery recharge

When peak shaving is active and the battery SOC (the lowest SOC of the units in a parallel system) is lower than the Off-Grid Bat Stop SOC minus 5%, the system starts a recharge. It charges the battery until the SOC reaches the On-Grid Bat Stop SOC or higher.

7.8 Off-Grid Installation Tips

- **Limit sensors (CTs)** are **not needed** in a fully off-grid system, **unless** "Grid Priority" is turned on with a generator connected to the **GRID** terminal.
- A generator can go on the **GRID** terminal (which port to use depends on the generator: see the Fantastech rules box below and section 6.8.1). This makes it easier to work with a service panel that is set up as a "GEN" input. With

the generator on the GRID terminal, set **Gen Connect To Grid** to **Yes** and **Gen Port Func** to **No** (section 9.9).

- A **transfer switch is not needed** in an off-grid system. The **LOAD** output can be connected directly to the main panel.
- **Do not use "Grid Priority" mode** in an off-grid system.
- When you use a **generator in an off-grid system**, the recommended **Work Mode** is **"Self Use"** (Work Mode page 2/4, section 9.6).
- The **Auto Generator Start (AGS)** function turns on when the battery voltage or SOC drops below the **Start SOC/V** level you set (section 7.3). Keep the generator Start SOC/V above the Off-Grid Bat Stop SOC so the generator starts first.
- If you add a **wind turbine**, you must install a **48 V wind charge controller with a dump load** to keep the battery from overcharging. The wind charge controller must be **connected directly to the battery bank**.
- Make sure the **battery capacity** is set correctly and the charge and discharge current limits are set within safe ranges (Battery Settings, section 9.7).

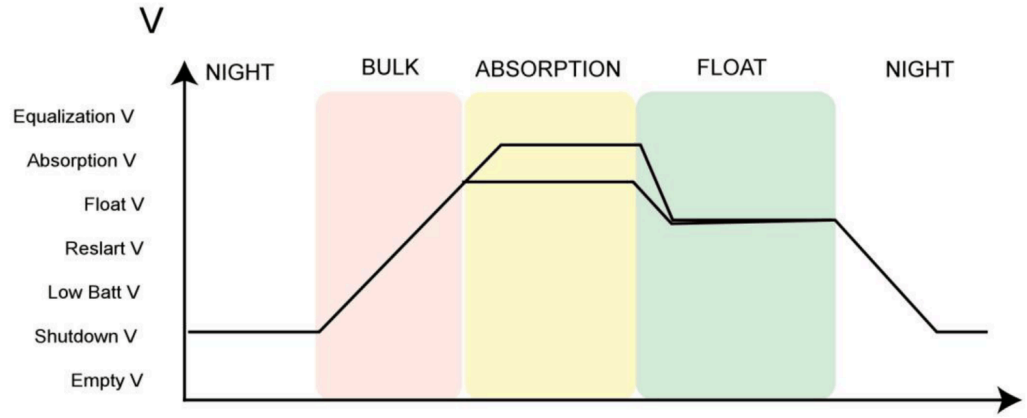
FANTASTECH RULES Which port the generator goes on:

- **120/240 V split-phase generator under 15 kW** → **GEN** port (63 A breaker).
- **120/240 V generator 15 kW or bigger** → **GRID** port.
- **120/208 V generator** → **GRID** port. Never the GEN port.
- **Never run three-phase power to the GEN port.**
- A generator on the GRID port still starts and stops on the **Start SOC/V** and **Stop SOC/V** settings (section 9.9), and it charges the battery in Backup Power mode.

7.9 Battery Charge Controller

Three-stage charging: Bulk, Absorption and Float

The inverter charges the battery in 3 stages so charging is fast, efficient and safe. There is no equalization stage. The figure below shows the order of the stages.



Charge controller curve: battery voltage through the Night, Bulk, Absorption, Float and Night periods. The label "Reslart V" means Restart V. There is no equalization stage, so the "Equalization V" level on the chart is not used.

Bulk stage

In the Bulk stage, the battery is below 100% state of charge and has not yet reached the Absorption voltage setpoint. The controller sends 100% of the available solar power to recharge the battery.

Absorption stage

When the battery reaches the Absorption voltage setpoint, the inverter holds the battery voltage at that setpoint (constant-voltage regulation). This prevents overheating and too much battery gassing. The battery is allowed to reach full charge at the Absorption voltage setpoint. Absorption lasts until the battery charge current (A) drops to 2% of the programmed battery capacity (Ah).

Float stage

After the Absorption stage has fully charged the battery, the MPPT lowers the battery voltage to the Float voltage setpoint. When a battery is 100% charged, no more chemical reactions can take place, and all charging current turns into heat and gassing. The Float stage gives a small maintenance charge while it reduces the heating and gassing of the fully charged battery. The purpose of the Float stage is to protect the battery from long-term overcharge.

8. Monitoring (SOLARMAN Smart)

Your KOOR inverter can send its data to your phone over the internet. It uses the free SOLARMAN Smart app and the Wi-Fi logger stick that comes in the box with the inverter.

The inverter works with two kinds of logger stick:

- **LSW-5 (standard, in the box):** connects over Wi-Fi and Bluetooth (BLE). Wi-Fi band: 2.4 GHz.
- **LS4G-6-G (optional):** connects over Wi-Fi and Bluetooth (BLE), or over the 4G cellular network with a SIM card. Wi-Fi band: 2.4 GHz. Cellular band: 4G.

Read the parts of this chapter that match the stick you have. The install drawings show the LSW-5. The 4G stick installs the same way.

8.1 What You Need

- The Wi-Fi logger stick from the inverter box.
- A smartphone (iPhone or Android).
- The name and password of the Wi-Fi network at the site. The Wi-Fi signal must reach the inverter.
- The site Wi-Fi must use the **2.4 GHz** band. The stick does not work on 5 GHz Wi-Fi.

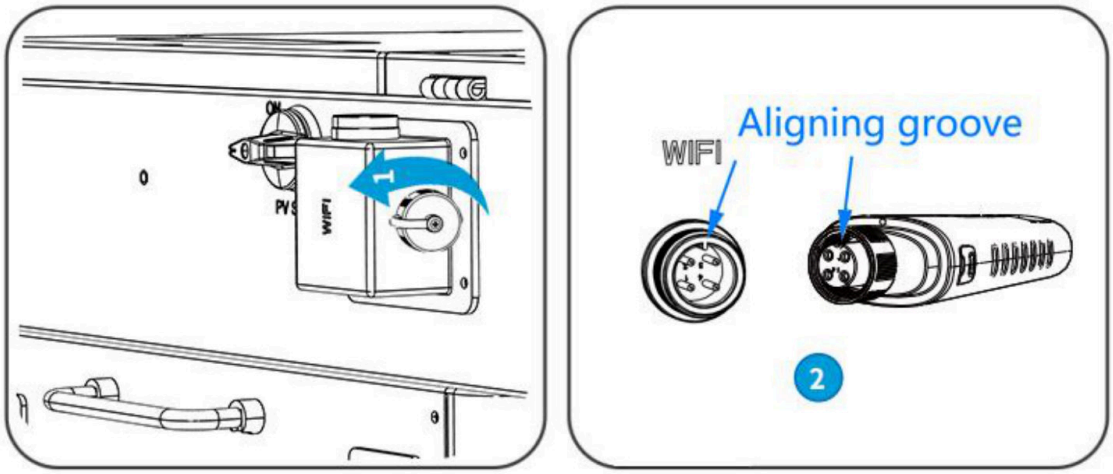
> NOTE No Wi-Fi at the site? A 4G LTE hotspot can be used instead, or the optional LS4G-6-G 4G stick with a SIM card (section 8.2.1). Call Fantastech at 574-218-1973.

> NOTICE Do not power the internet router or the fiber modem (ONT) from the inverter's LOAD terminals. This can interfere with remote software updates. Power them from the grid side instead, so the network service is not interrupted.

8.2 Install the Logger Stick

The stick plugs into the Wi-Fi port on the inverter (item H in section 3.3.2). Follow these steps:

1. Remove the waterproof cover from the Wi-Fi port.
2. Line up the groove on the stick's connector with the key in the Wi-Fi port.
3. Push the stick straight into the port.
4. Turn the stick's locking ring to lock the stick in place.



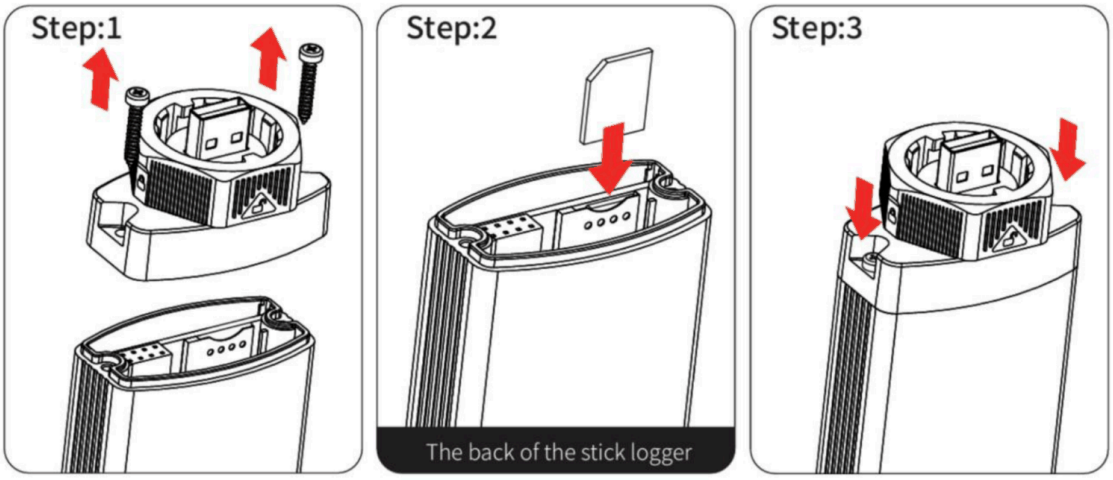
Step 1: remove the waterproof cover. Step 2: line up the aligning groove with the Wi-Fi port (steps 3 and 4 are not pictured).

WARNING When you install or remove the logger stick, turn the locking ring only. Do not hold the stick body and turn it.

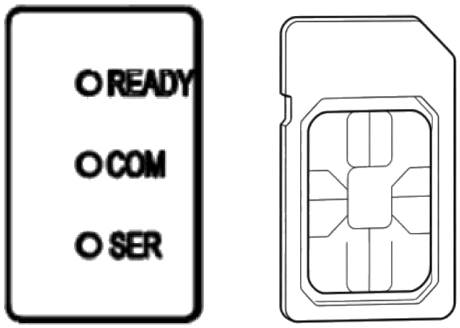
8.2.1 SIM Card (4G Stick Only)

The optional LS4G-6-G stick can use Wi-Fi or the 4G cellular network. For a better and more stable signal, buy a local 4G SIM card. Skip these steps if you do not need to put in or replace a SIM card.

1. Hold the stick with the side **without** the lights facing you. Take out the screws on both sides of the end cover and remove the end cover.
2. Push the SIM card in as shown until it clicks and locks in place.
3. Put the end cover back on and tighten the screws.



SIM card steps 1 to 3. Step 2 shows the back of the stick.

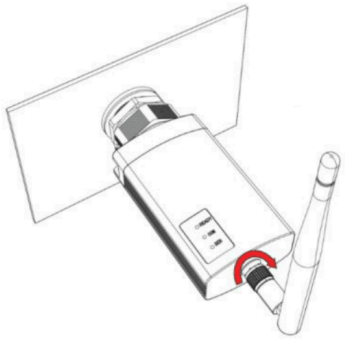


The chip side faces the same way as the lights

NOTICE The notch of the SIM card faces up. The chip side of the card must be on the same side as the lights (READY, COM, SER) on the stick. Do NOT put the card in backwards. To take the card out, press it in until it clicks, then pull it out.

8.2.2 Antenna (4G Stick Only)

Line up the threaded connector of the antenna with the antenna port on the stick. Turn it clockwise until it is tight.



Antenna on the 4G stick

8.3 Get the App

1. On an iPhone, open the App Store. On an Android phone, open Google Play.
2. Search for **SOLARMAN Smart** and install it.
3. Open the app and create an account with your e-mail address.

Installers and service companies can use the SOLARMAN Business app instead.

8.4 Add Your System

1. In the app, create a new plant. Enter a name, the site address and the size of the solar array.
2. Add the logger to the plant. Scan the barcode on the logger stick label, or type the logger serial number (SN) printed on the label. The SN is also printed on the stick's box.
3. When the app asks for the PWD, type the PWD from the stick label in **lowercase letters**.

IMPORTANT The PWD is printed on the label in CAPITAL letters, but the app rejects it in capitals. Type the same letters and numbers in lowercase. Example: a label that reads FEE32734 is entered as fee32734.

8.5 Connect the Stick to the Site Wi-Fi

The LSW-5 stick needs a Wi-Fi connection. The LS4G-6-G stick uses the 4G cellular network by default: you only need to add it to your plant (section 8.4), and Wi-Fi is optional.

1. With the inverter on, the logger stick sends out its own Wi-Fi network. On your phone, open the Wi-Fi settings and join the stick's network.
2. Open the phone's web browser and go to **10.10.100.254**.
3. Pick the site's Wi-Fi network from the list, type its password, and save. The stick restarts and joins the site Wi-Fi.

Another way, in the app: right after you add the logger, the app offers **Go to Configure**. Turn on the phone's Bluetooth and Wi-Fi, tap Go to Configure, pick the site Wi-Fi, type its password and start the configuration. Wait a few minutes, then tap **Done**. The app does not accept a 5 GHz network.

Your own SIM card in the 4G stick: the installer sets the SIM card's APN settings with the SOLARMAN Business app.

1. Tap **APN Configuration**. Scan the logger's code or type its SN.
2. Enter the SIM card's APN settings. If the SIM card has a user name and password, enter them too. Tap **Settings**. The setup is done when the app shows "Set Successfully".
3. Tap **Query** and check that the APN settings read back match what you entered.

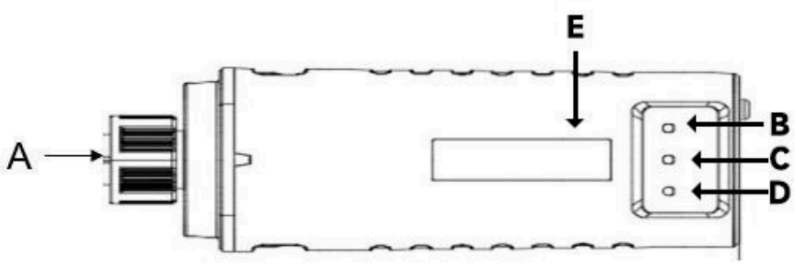
8.6 After It Connects

- The stick uploads new data every 5 minutes.
- A newly added logger shows **Offline** until its first upload. Wait at least 10 minutes before you check again.
- If it stays Offline, make sure the site Wi-Fi reaches the inverter and join the stick to the Wi-Fi again (section 8.5). Then check the stick lights (sections 8.7 and 8.8). If it still will not connect, call Fantastech at 574-218-1973.

8.7 Logger Stick Lights

8.7.1 LSW-5 Wi-Fi Stick

- **A:** round connector. It plugs into the inverter and carries the communication.
- **B: NET** (green light): communication with the router and the server.
- **C: COM** (green light): communication with the inverter.
- **D: READY** (green light): the operating status of the stick.
- **E:** label with the stick's SN and password.



LSW-5 Wi-Fi logger stick

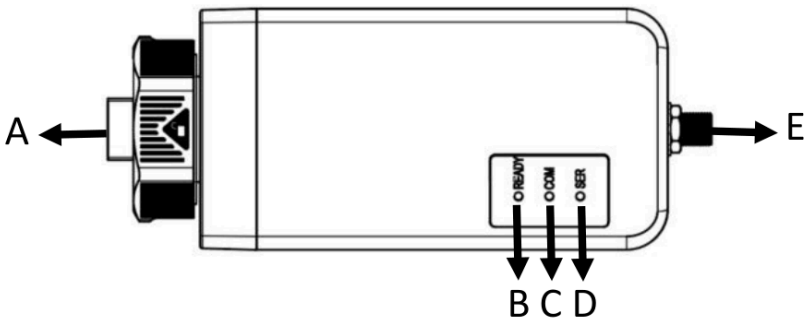
Light	Shows	What the light means
NET	Communication with the router	On steady: connected to the server. Off: the connection to the router failed. Slow flash (1 s on, 1 s off): connected to the router. Fast flash (100 ms on, 100 ms off): network setup in progress.
COM	Communication with the inverter	On steady: the stick is connected to the inverter. Off: the connection to the inverter failed. Slow flash (1 s on, 1 s off): communicating with the inverter.
READY	Stick operating status	Off: the stick is not running normally. Slow flash (1 s on, 1 s off): running normally. Fast flash (100 ms on, 100 ms off): restoring factory settings.

Normal operation (the router is connected to the internet):

1. NET stays on after the stick powers up: connected to the server.
2. READY flashes: the stick is running normally.
3. COM stays on: connected to the inverter.

8.7.2 LS4G-6-G 4G Stick (Optional)

- **A:** round connector. It plugs into the inverter and carries the communication.
- **B: READY** (green light): the operating status of the stick.
- **C: COM** (green light): communication with the inverter.
- **D: SER** (blue-green light): communication with the router or the cell tower, and the server.
- **E:** antenna port for the 4G and Wi-Fi signal.



LS4G-6-G 4G logger stick

Light	Shows	What the light means
READY	Stick operating status	Off or on steady: the stick is not running normally. Slow flash (1 s on, 1 s off): running normally.
COM	Communication with the inverter	On steady: the stick is connected to the inverter. Off: the connection to the inverter failed. Slow flash (1 s on, 1 s off): communicating with the inverter.
SER	Communication with the router or the cell tower	Green: 4G mode. Blue: Wi-Fi mode. On steady: connected to the server. Slow flash (1 s on, 1 s off): connected to the router or the cell tower. Fast flash (500 ms on, 5 s off): the connection to the router or the cell tower failed.

8.8 Logger Stick Troubleshooting

If the data in the app looks wrong while the stick is running, use the stick lights and the tables below to find the problem. If that does not fix it, or the lights do not match any row, contact Fantastech (574-218-1973, derek@fantastech.org).

> **NOTICE** Check the lights only after the stick has been powered on for at least 2 minutes.

Reset button on the stick: press and hold it for 5 seconds to restart the stick. Press and hold it for 10 seconds to restore the stick's factory settings.

LSW-5 Wi-Fi stick

NET	COM	READY	Problem	Cause	What to do
Any	Off	Slow flash	The stick cannot talk to the inverter.	1. The stick is loose in the inverter port. 2. The inverter's communication rate does not match the stick.	1. Check the connection. Remove the stick and install it again. 2. Check that the inverter's communication rate matches the stick (WiFi Baud, section 9.10). 3. Press and hold the Reset button for 5 s to restart the stick.
Off	On	Slow flash	The stick cannot reach the router.	1. The stick has no network set up. 2. The router's Wi-Fi signal is weak.	1. Check that the Wi-Fi network is set up (section 8.5). 2. Make the router's Wi-Fi signal stronger at the inverter.
Slow flash	On	Slow flash	The stick reaches the router but not the remote server.	1. The router's internet connection is not working. 2. The stick's server address was changed. 3. The network blocks the server.	1. Check that the router has internet access. 2. Check the router settings for a blocked or limited connection. 3. Contact Fantastech.
Off	Off	Off	No power to the stick.	1. The stick is loose or badly connected. 2. The inverter is not supplying enough power. 3. The stick is faulty.	1. Check the connection. Remove the stick and install it again. 2. Check the inverter output power. 3. Contact Fantastech.
Fast flash	Any	Any	Network setup mode (normal).	Normal.	1. It ends by itself after 2 minutes. 2. Hold the Reset button for 5 s to restart the stick. 3. Hold the Reset button for 10 s to restore factory settings.
Any	Any	Fast flash	Restoring factory settings (normal).	Normal.	1. It ends by itself after 1 minute. 2. Hold the Reset button for 5 s to restart the stick. 3. Hold the Reset button for 10 s to restore factory settings.

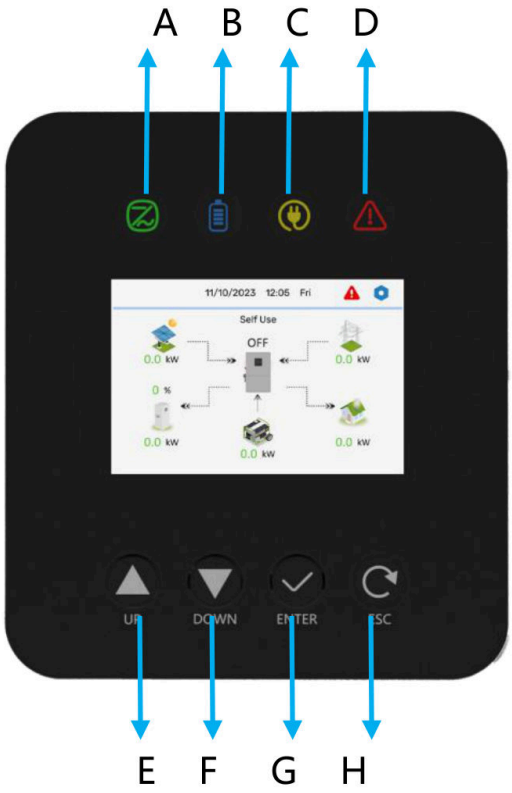
LS4G-6-G 4G stick

SER	COM	READY	Problem	Cause	What to do
Any	Off	Any	The stick cannot talk to the inverter.	1. The stick is loose in the inverter port. 2. The inverter's communication rate does not match the stick.	1. Check the connection. Remove the stick and install it again. 2. Check that the inverter's communication rate matches the stick (WiFi Baud, section 9.10).
Fast flash or slow flash	Fast flash or on	Any	The stick cannot connect to the cell tower.	1. The SIM card is out of credit. 2. The 4G signal is weak.	1. Check that the SIM card account has enough balance. 2. The cell signal is poor: replace the antenna with a suction-cup antenna.
Off	Off	Off	No power to the stick.	1. The stick is loose or badly connected. 2. The inverter is not supplying enough power. 3. The stick is faulty.	1. Check the connection. Remove the stick and install it again. 2. Check the inverter output power. 3. Contact Fantastech.

9. Operation

9.1 Indicator Panel

The operation panel is on the front of the inverter. It has four indicator LEDs (A to D), an LCD touchscreen and four buttons (E to H).



Operation panel: indicator LEDs A–D and buttons E–H



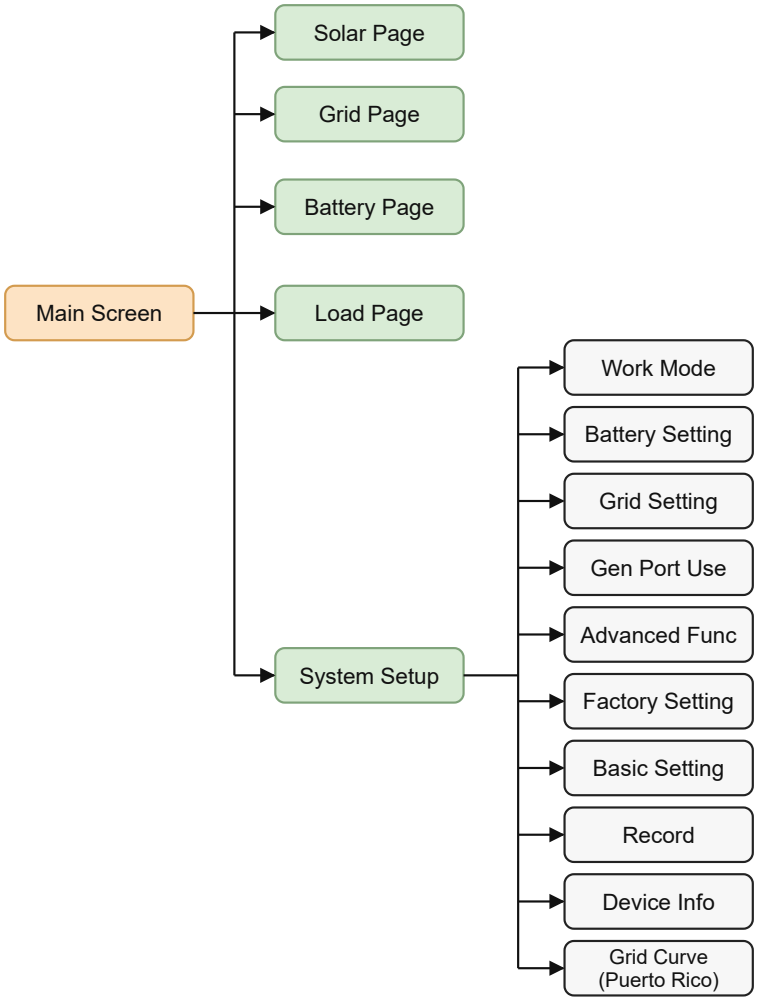
Location of the operation panel on the inverter

Each LED is marked with a symbol in its own color. The table below gives the symbol, the color and the meaning of each LED.

Name	Item	Symbol on panel	Function
Indicator LED	A	Inverter symbol	Green. On: the inverter is running. Flashing: the inverter is in standby.
	B	Battery symbol	Blue. On: communication with the BMS is normal.
	C	Plug symbol	Yellow. On: the inverter is in EPS mode.
	D	Warning triangle	Red. The inverter is in fault mode.
Button	E	UP	Go to the previous selection.
	F	DOWN	Go to the next selection.
	G	ENTER	Confirm the selection.
	H	ESC	Exit setting mode.

9.2 LCD Operation Flow Chart

From the main screen you can open the Solar, Grid, Battery and Load pages and the System Setup menu. The chart below shows the menus under System Setup.



LCD menu structure

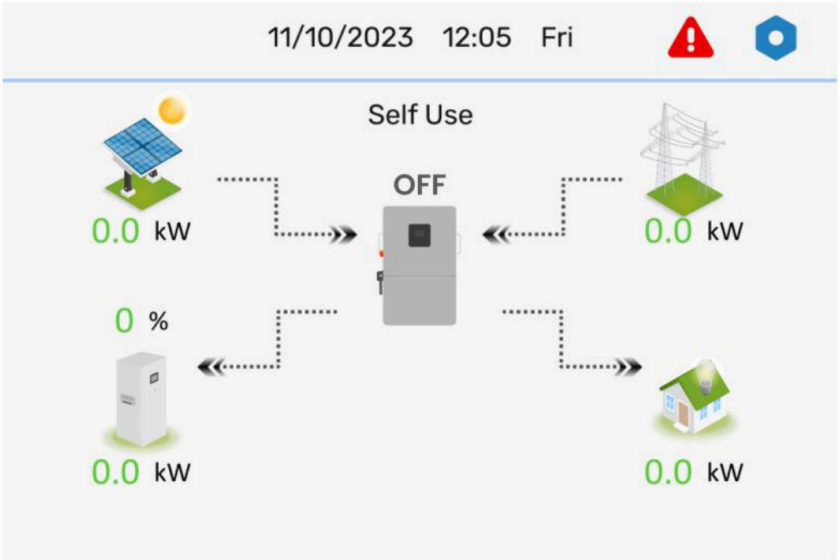
Where each menu is described in this manual:

- Solar, Grid, Battery and Load pages: sections 9.3.2 to 9.3.5.
- Work Mode: section 9.6.
- Battery Setting: section 9.7.
- Grid Setting: section 9.8.
- Gen Port Use: section 9.9.
- Advanced Func: section 9.10.
- Factory Setting: section 9.13.
- Basic Setting: section 9.5.
- Record: section 9.11.
- Device Info: section 9.12.
- Grid Curve (Puerto Rico): section 9.14.

9.3 LCD Display Icons

9.3.1 Main Screen

The LCD is a touchscreen. The main screen shows the overall information of the inverter.



LCD main screen

1. System information display

- **1.1 Date and time.** The current system date and time are shown at the top of the screen.
- **1.2 Warning icon (red "!").** A red "!" means there is a system fault or an abnormal condition. Tap the icon to open the fault detail page. There you can check the alarm messages and fault codes. Follow the on-screen guidance or chapter 10 (Troubleshooting) to fix the problem.
- **1.3 Settings icon (gear).** Tap it to open the system settings menu. It includes settings such as operating mode, battery parameters and grid settings.

2. Inverter status and operating mode

- **2.1 Inverter power status.** The icon in the center is the inverter. It shows whether the inverter is ON or OFF.
- **2.2 Current operating mode.** The active system mode is shown above the inverter icon. Possible modes include Self Use, TOU ECO, Backup Power and others (see section 7.2). This helps you see how the system is working right now.

3. Real-time energy device overview

The four device icons show the status and power of the main energy parts. You can tap each icon:

- **3.1 Top left: PV.** Shows the real-time solar power. Tap it to see the detailed PV status (section 9.3.2).
- **3.2 Top right: Grid.** Shows the grid import power. Tap it to see the grid connection information (section 9.3.3).
- **3.3 Bottom left: Battery.** Shows the battery state of charge (SOC) and the charge or discharge power. Tap it to see the battery details (section 9.3.4).
- **3.4 Bottom right: Load.** Shows the real-time load use. Tap it to see the detailed load information (section 9.3.5).

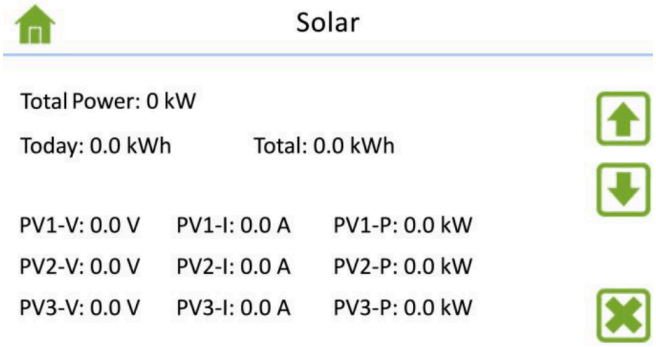
4. Energy flow and connection indicators

- **4.1 Arrows.** The arrows show the real-time power flow paths. The arrow direction is the direction of power flow (for example, from PV to the load).
- **4.2 Red "X" icons.** A red "X" means a path or device is not connected or not active. Examples: the battery is not installed, or the grid is offline.

How to read the power values (plus and minus signs):

- PV power and load power are always positive.
- Grid power: negative means power is taken from the grid. Positive means power is sold to the grid.
- Battery power: negative means the battery is charging. Positive means the battery is discharging.

9.3.2 PV Information Page Overview

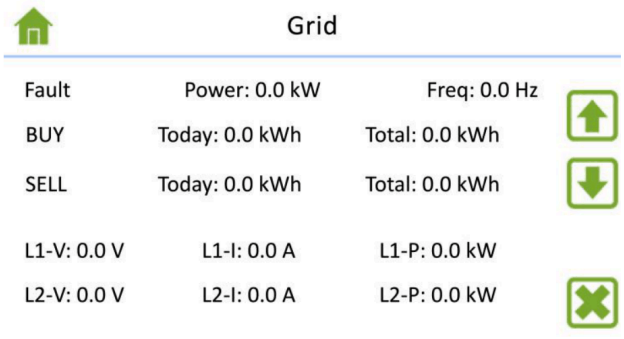


PV (Solar) information page

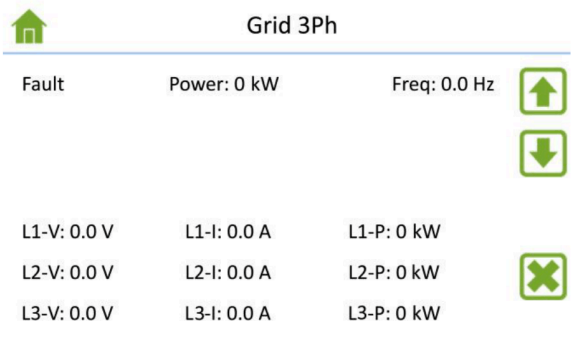
This page may vary depending on the system configuration. Tap the PV icon on the main screen to open the PV information page. PV1 to PV3 show the voltage (V), current (I) and power (P) of each of the three PV input channels.

The page also shows the total PV power right now (Total Power), the energy generated today (Today) and the total energy generated to date (Total). Use the arrow buttons on the side of the screen to scroll to more data pages.

9.3.3 Grid Information Page Overview



Grid information page



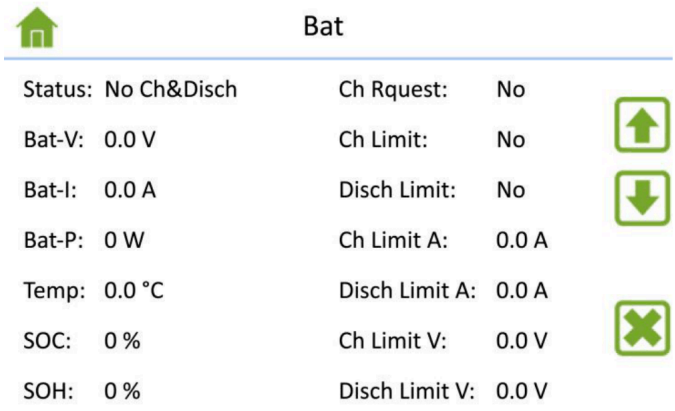
Grid 3Ph page (three-phase inverter groups only)

This page may vary depending on the system configuration.

- The top line shows the grid state (the example screen shows **Fault**), the total grid power (**Power**) and the present grid frequency (**Freq**).

- **BUY** shows the energy bought from the grid today (Today) and in total (Total). **SELL** shows the same for energy sold to the grid.
- **L1 and L2** rows show the voltage (V), current (I) and power (P) of each line.
- The **Grid 3Ph** page applies only to three-phase inverter groups. It adds a third row, L3.

9.3.4 Battery Information Page Overview

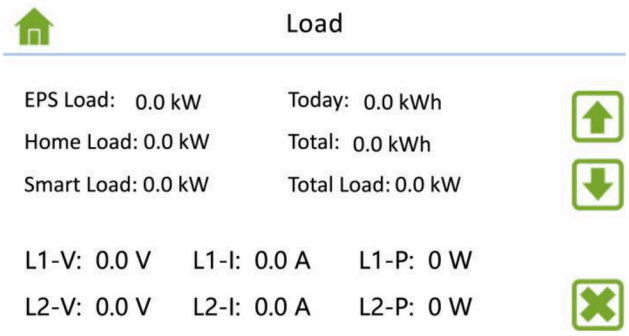


Battery information page (the screen prints "Ch Rquest" for Ch Request)

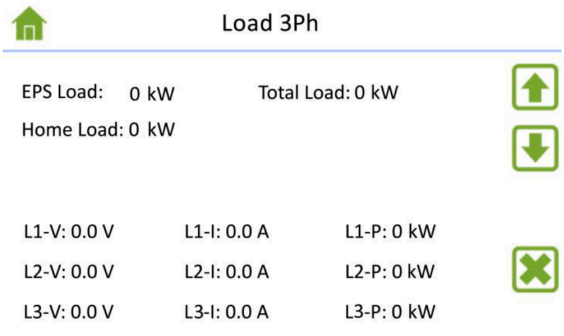
This page shows the operating information of the battery. It may vary depending on the system configuration.

- The left column shows the battery status (the example screen shows **No Ch&Disch**, no charging or discharging), voltage (Bat-V), current (Bat-I), power (Bat-P), temperature (Temp), state of charge (SOC) and state of health (SOH).
- The right column shows the charge request (Ch Request), the charge and discharge limit flags (Ch Limit, Disch Limit), the charge and discharge current limits (Ch Limit A, Disch Limit A) and the charge and discharge voltage limits (Ch Limit V, Disch Limit V).

9.3.5 Load Information Page Overview



Load information page



Load 3Ph page (three-phase inverter groups only)

This page may vary depending on the system configuration. It shows the EPS load power (EPS Load), the household load power (Home Load), the smart load power (Smart Load) and the total load power (Total Load). Today and Total show the load energy for today and in total.

Below these values, the page shows the voltage (V), current (I) and power (P) of the two split-phase lines, L1 and L2.

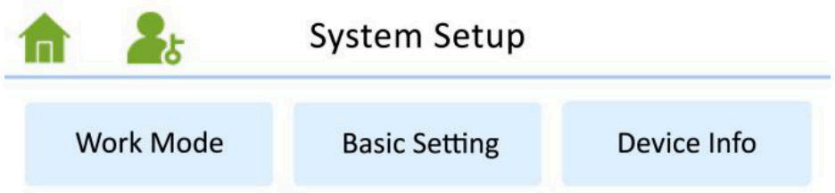
The **Load 3Ph** page applies only to three-phase inverter groups. It shows EPS Load, Home Load and Total Load, and the voltage, current and power of L1, L2 and L3.

9.4 System Setup Menu

The System Setup menu has two access levels: standard user access and advanced user access (password 7777).

Standard user access

To enter basic settings mode, tap the settings icon (gear) in the upper-right corner of the main screen. The System Setup page opens with three menus: **Work Mode**, **Basic Setting** and **Device Info**.

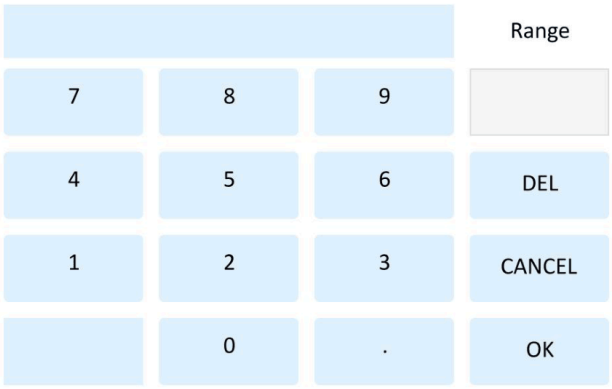


System Setup page (standard user access)

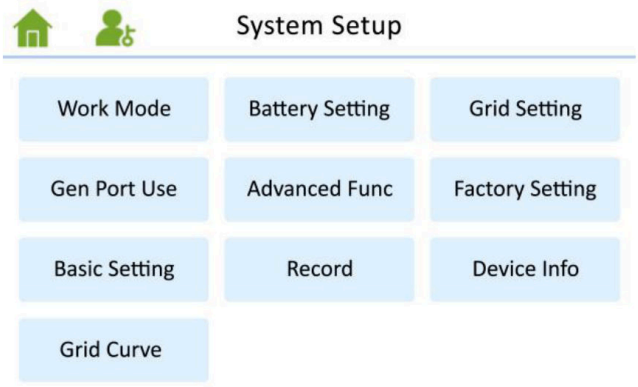
Advanced user access

To enter advanced settings mode, tap the user icon (a person with a key) in the upper-left corner of the System Setup page. Then enter the password **7777** on the keypad and tap OK.

When the password is accepted, the System Setup page opens with more menus: Work Mode, Battery Setting, Grid Setting, Gen Port Use, Advanced Func, Factory Setting, Basic Setting, Record, Device Info and Grid Curve. From there you can reach settings that are not available with standard access.

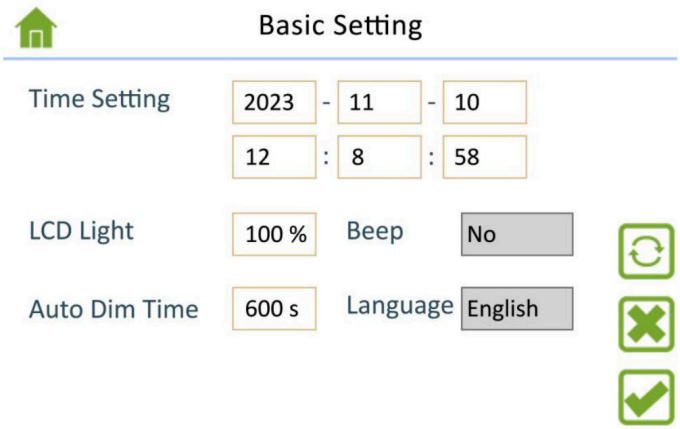


Password keypad



System Setup page (advanced user access)

9.5 Basic Settings Menu



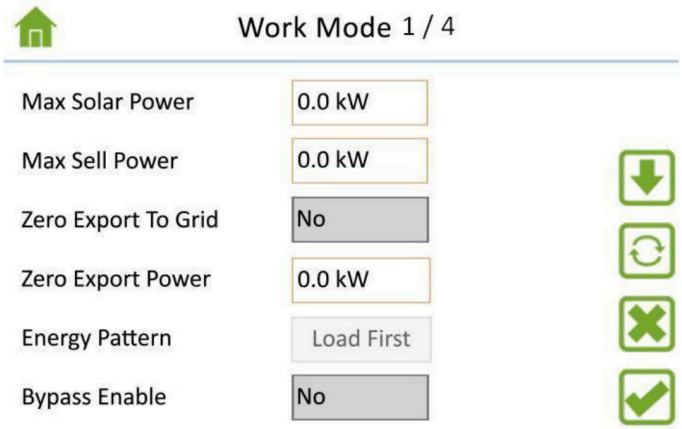
Basic Setting screen (default values)

Setting	Default	What it does
Time Setting	Set on site	Sets the system date and time: year, month, day, hour and minute.
LCD Light	100%	Sets the screen brightness level.
Beep	No (see note)	Turns the buzzer on or off. When it is on, the buzzer sounds each time you touch the screen or press a key.
Auto Dim Time	600 s	Sets how long the screen waits before it dims or turns off by itself.
Language	English	Selects the language used on the screen.

The default screen shows Beep = No (buzzer off).

9.6 Work Mode Settings Menu

Work Mode Settings — Page 1/4

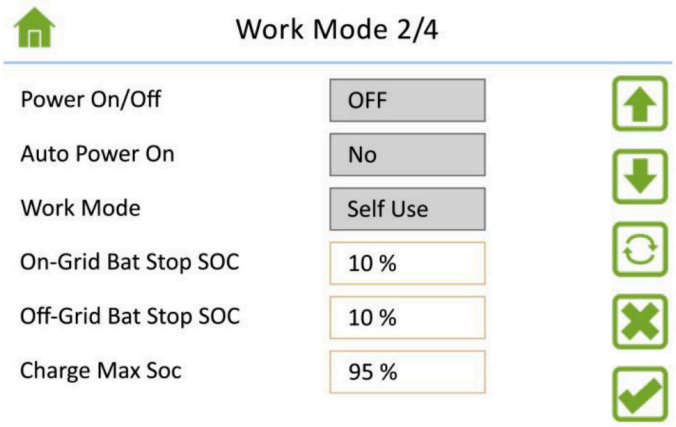


Work Mode 1/4 screen (default values)

Setting	Default	What it does
Max Solar Power	0.0 kW	Sets the maximum PV input power.
Max Sell Power	0.0 kW	Sets the maximum power that can be exported to the grid.
Zero Export To Grid	No	When enabled, the inverter limits the power it exports to the grid. The actual export limit is set by **Zero Export Power**.
Zero Export Power	0.0 kW	Sets the inverter's export power limit to the grid. This setting works only when **Zero Export To Grid** is enabled.
Energy Pattern	Load First	Sets the system's energy priority. At this time only **Load First** is supported. **Load First:** PV power supplies the load first. Any extra PV energy then charges the battery. If PV power is not enough, the grid supplies the load.
Bypass Enable	No	Turns the grid bypass mode on or off.

Max Sell Power also appears on Grid Code Selection 1/5 (section 9.8) with a default of 15 kW. Check both screens.

Work Mode Settings — Page 2/4



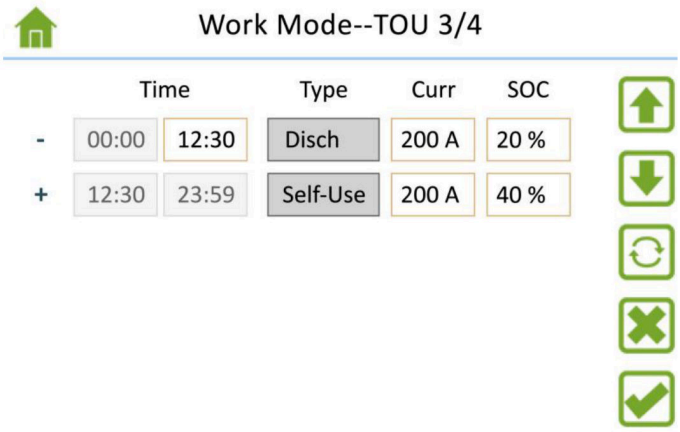
Work Mode 2/4 screen (default values; the screen prints "Soc" for SOC)

Setting	Default	What it does
Power On/Off	OFF	Turns the inverter ON or OFF by hand.
Auto Power On	No	Turns automatic startup and shutdown on or off. When it is on, the inverter powers on by itself.
Work Mode	Self Use	Sets the system's working mode. The choices are **Self Use**, **TOU ECO**, **Backup Power** and **Grid Priority**. The modes are explained in section 7.2.
On-Grid Bat Stop SOC	10%	Used when the inverter works on-grid. The battery can discharge only while its SOC is higher than this setting. If the battery SOC drops below this setting minus 5% (Stop SOC − 5%), the battery goes into charge mode and charges back up to this setting.
Off-Grid Bat Stop SOC	10%	Used when the inverter works off-grid (no utility). The battery can discharge only while its SOC is higher than this setting.
Charge Max SOC	95%	Sets the highest battery SOC for charging. When the battery reaches this SOC, charging stops by itself.

FANTASTECH RULE Before you turn the inverter **OFF** from the app, set **Auto Power On** to **No** first. If it is left on, the inverter powers itself back on after about 10 seconds. In a parallel system, set all the inverters together.

FANTASTECH RULE On an off-grid system with a generator, keep the generator **Start SOC/V** (section 9.9) higher than the **Off-Grid Bat Stop SOC**, so the generator starts before the battery stops.

Work Mode Settings — Page 3/4 (TOU Mode Settings)



Work Mode TOU 3/4 screen with two example time slots

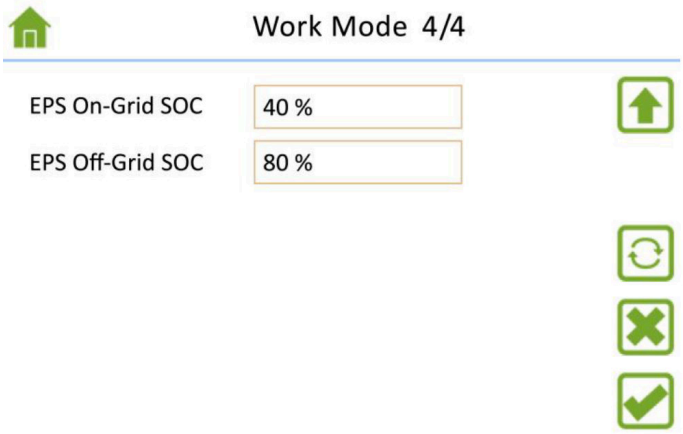
> **NOTE** TOU scheduling works only when **Work Mode** on Work Mode page 2/4 (section 9.6) is set to **TOU ECO**. Only then do the time-based settings take effect.

- **Time:** sets the time slots used for energy management. The whole 24-hour day must be divided into back-to-back periods that do not overlap. Each time slot has a start time and an end time. You can add up to 6 time slots by tapping the "+" icon.
- **Type:** sets what the system does during the time slot.
  - **Discharge** (Disch): the battery discharges according to the settings for that slot.
  - **Charge:** the battery charges from PV or grid power.
  - **Self-Use:** PV power is used for your own loads first (self-consumption).
- **Curr:** sets the maximum charging or discharging current of the battery during the slot. The inverter uses the lowest of three limits: this TOU Curr setting, **Max A Charge / Max A Disch** and **Max Ch Power / Max Disch Power** (section 9.7).
- **SOC:** the target battery SOC for the time slot.

A different example with three time slots (not the screen above)

Time period	Priority type	Zero Export disabled	Zero Export enabled
00:00–12:30	Discharge	The battery discharges if SOC > 20%. For the maximum battery current, see the specification table (section 3.3.4). Extra PV is exported to the grid.	Same discharge behavior. Extra PV charges the battery so it is not wasted.
12:30–16:30	Self-Use	The battery discharges if SOC > 40% when PV cannot meet the load. The battery charges if SOC < 36%, using PV or the grid. Remaining PV is exported.	Same behavior. No export allowed: extra PV that is not used for the load or for charging is curtailed.
16:30–23:59	Charge	The battery charges if SOC < 90%, using PV + grid (up to 30 A). For the maximum battery current, see the specification table (section 3.3.4). Extra PV is exported.	Same charging behavior. No export allowed: any extra PV after charging is curtailed.

Work Mode Settings — Page 4/4



Work Mode 4/4 screen (default values)

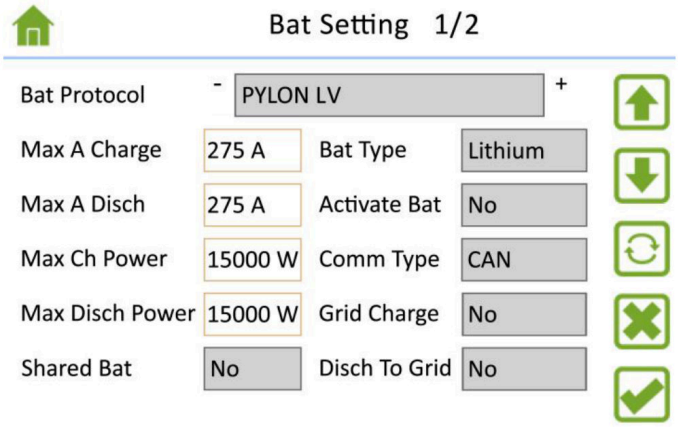
These two settings are used when the active off-grid function is turned on (**Off-grid** in the Factory Setting menu, section 9.13). Then the inverter switches between on-grid and off-grid by itself, based on the battery SOC, so that no power flows back to the grid. Battery charging from the grid still follows the **Grid Charge** setting (section 9.7).

Setting	Default	What it does
EPS On-Grid SOC	40%	The SOC point for switching from off-grid to on-grid. When the battery SOC falls below this value, the inverter connects to the grid again to supply the loads and charge the battery. It also turns on charging from the grid during the TOU ECO Self-Use or Charge time slots.
EPS Off-Grid SOC	80%	The target SOC for switching back to off-grid. The real switch point is the higher of this setting and EPS On-Grid SOC + 6%, so the relay does not switch too often. When the SOC rises above that point, the inverter disconnects from the grid and powers the local loads by itself.

These two settings are also limited by the TOU SOC setting and by the On-Grid / Off-Grid Bat Stop SOC settings. The inverter uses the highest of them.

9.7 Battery Settings Menu

Battery Settings — Page 1/2

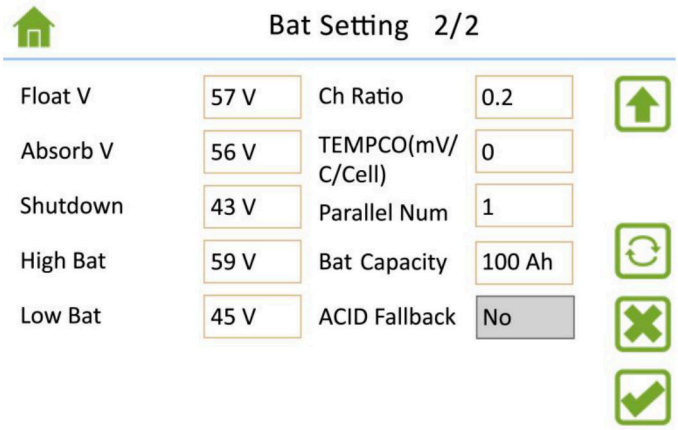


Bat Setting 1/2 screen (default values, KR-L15K-SPUS)

Setting	Default	What it does
Bat Protocol	PYLON LV	Selects the battery maker. The system then picks the correct communication protocol for that battery by itself. For KOOR batteries, see the Appendix — KOOR batteries at the front of this manual.
Max A Charge / Max A Disch	275 A 275 A	Set the **maximum battery charging current** and the **maximum battery discharging current**.
Max Ch Power / Max Disch Power	15000 W 15000 W	Set the **maximum charging power** and the **maximum discharging power** for the battery.
Shared Bat	No	Lets several inverters in a parallel system share and run on one battery bank. In a parallel system, connect the BMS communication cable to the master inverter.
Bat Type	Lithium	Selects the battery type. The choices are **Lithium**, **Lead-acid**, **DC Source** and **No Battery**.
Activate Bat	No	Starts battery activation (wake-up) by hand.
Comm Type	CAN	Selects the battery communication protocol: **CAN** or **RS485**.
Grid Charge	No	Turns battery charging from the utility grid on or off. A generator wired to the GRID port does charge the battery in Backup Power mode (Fantastech rule; see section 9.9).
Disch To Grid	No	Turns battery discharging to the utility grid on or off. Available only when the system works in TOU mode.

The screen shows 275 A and 15000 W, the KR-L15K-SPUS values. The maximum battery charge/discharge current is 167 / 210 / 250 / 275 A for the KR-L8K / L10K / L12K / L15K-SPUS. The full current needs both battery terminal pairs connected.

Battery Settings — Page 2/2



Bat Setting 2/2 screen (default values)

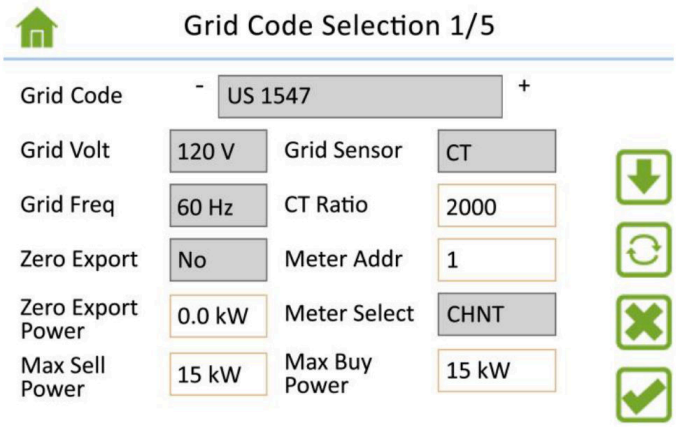
Setting	Default	What it does
Float V	57 V (see note)	Sets the **battery float charging voltage**.
Absorb V	56 V	Sets the **absorb charging voltage**.
Shutdown	43 V	Sets the **battery discharge cutoff voltage** that protects the battery.
High Bat / Low Bat	59 V 45 V	Set the **battery port over-voltage** and **under-voltage protection points**.
Ch Ratio	0.2	Sets the **charging C-rate** (charging ratio). Used **only for lead-acid batteries**.
TEMPCO(mV/C/Cell)	0	Battery **temperature compensation coefficient**, in mV per °C per cell. Used **only for lead-acid batteries**.
Parallel Num	1	Sets the **number of battery groups connected in parallel**. This setting is needed **only for lead-acid batteries** used in more than one parallel cluster.
Bat Capacity	100 Ah	Sets the **battery capacity**. Used **only for lead-acid batteries**.
ACID Fallback	No	When enabled, the inverter can keep its output on for 30 minutes after battery communication is lost.

> **FANTASTECH RULE** Set **ACID Fallback** to **Yes** on off-grid jobs. If the battery stops talking to the inverter, the inverter keeps its output on for 30 minutes instead of dropping the loads. In the field this stopped generator charging from cutting out.

Note: the factory default screen shows Float V 57 V, higher than Absorb V 56 V. Float is normally the lower voltage (section 7.9). Confirm these two values with Fantastech before you rely on them.

9.8 Grid Settings Menu

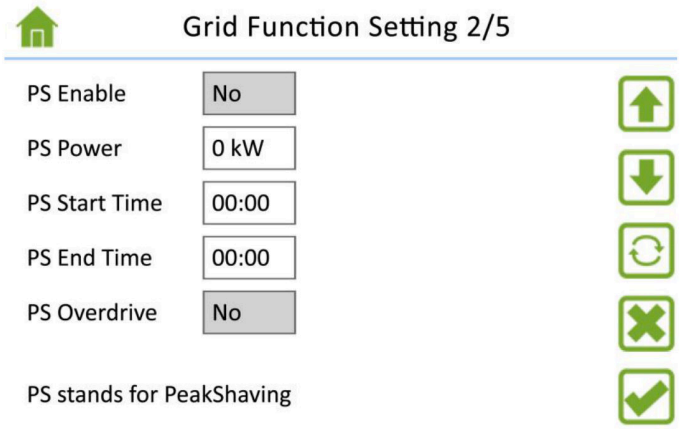
Grid Settings — Page 1/5



Grid Code Selection 1/5 screen (default values)

Setting	Default	What it does
Grid Code	US 1547	Selects the national grid standard for the region where the inverter is installed. Select **US 1547**. For a single inverter on a 120/208 V grid, see section 6.7.1 for the 120/208 3P grid standard.
Grid Volt	120 V	Sets the nominal grid voltage required by the local standard.
Grid Freq	60 Hz	Sets the nominal grid frequency of the local grid.
Zero Export	No	When enabled, the inverter limits the power it exports to the grid. The limit is set by **Zero Export Power**.
Zero Export Power	0.0 kW	Sets the maximum export power of the inverter. It works only when **Zero Export** is enabled.
Max Sell Power	15 kW	Sets the maximum output power allowed to be exported to the grid.
Grid Sensor	CT	Sets how the inverter measures grid-side current. The choices are: **None:** the inverter measures the current inside, at its grid port.
		CT: an external current transformer (CT) measures the current.
		Meter: an external smart meter, connected by RS485, supplies the grid data.
CT Ratio	2000	Sets the CT transformation ratio.
Meter Addr	1	Sets the RS485 address of the external energy meter.
Meter Select	CHNT	Selects the meter manufacturer. Make sure the model you select matches the meter that is installed.
Max Buy Power	15 kW	Sets the maximum input power the inverter is allowed to draw from the grid.

Grid Settings — Page 2/5 (Peak Shaving)

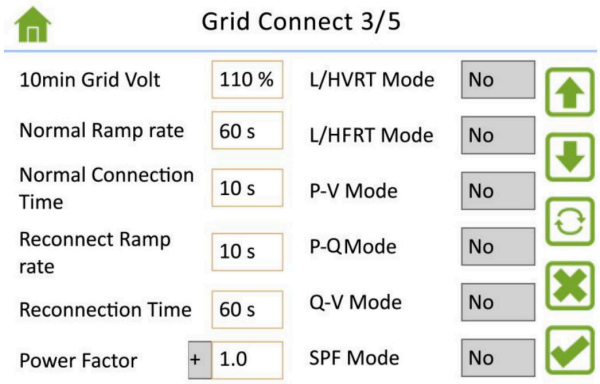


Grid Function Setting 2/5 screen (default values)

Peak Shaving (PS) lets the battery discharge during high-load times, to keep the power drawn from the grid under a limit you set. During off-peak hours the battery can be charged with low-cost grid power or extra PV power. This cuts demand charges and protects the household wiring from overload. Section 7.7 explains the function.

Setting	Default	What it does
PS Enable	No	Turns the Peak Shaving function on or off.
PS Power	0 kW	Sets the maximum input power allowed from the grid port.
PS Start Time	00:00	The start of the time window when peak shaving is active.
PS End Time	00:00	The end of the time window when peak shaving is active.
PS Overdrive	No	When enabled, the grid power can go above the PS Power limit to cover the load when PV and battery power are not enough. When disabled, the inverter shuts down by itself after 1 minute of continuous overload at 110% of the PS limit. If **Auto Power On** is on, it locks after 3 failed restarts. The restart counter clears after 5 minutes of normal running.

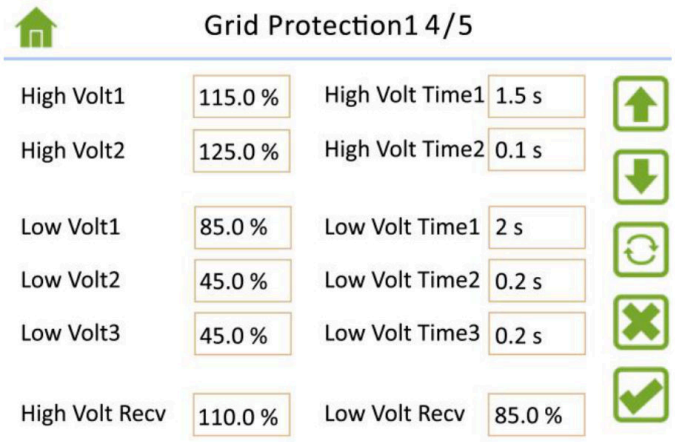
Grid Settings — Page 3/5



Grid Connect 3/5 screen (default values; the screen prints "P-QMode" for P-Q Mode)

Setting	Default	What it does
10min Grid Volt	110%	Sets the 10-minute grid over-voltage protection point. The inverter shuts down for protection when the 10-minute average RMS grid voltage goes above this percentage.
Normal Ramp rate	60 s	Sets the power ramp-up time after the inverter starts.
Normal Connection Time	10 s	Sets the time delay after startup before the inverter sends active power to the grid.
Reconnect Ramp rate	10 s	Sets the ramp-up time for active power after the inverter recovers from a fault.
Reconnection Time	60 s	Sets the reconnection delay after a fault is cleared.
Power Factor	+1.0	Sets the power factor of the inverter's grid-tied output. Keeping a proper power factor helps keep the voltage steady at the point of common coupling (PCC).
L/HVRT Mode	No	Low/high voltage ride-through. During a short grid voltage dip (LVRT) or swell (HVRT), the inverter stays connected to the grid instead of disconnecting at once, and supplies or absorbs reactive power to support the grid. It goes back to normal operation smoothly after the fault clears. This meets grid interconnection codes and keeps many inverters from tripping one after another.
L/HFRT Mode	No	Low/high frequency ride-through. During a short grid under-frequency (LFRT) or over-frequency (HFRT) event, the inverter stays connected to the grid instead of disconnecting at once. It brings its output back smoothly after the frequency returns to the normal range. This meets grid interconnection codes and keeps many inverters from tripping at the same time.
P-V Mode	No	Adjusts the active power output by itself according to the grid voltage, to keep the voltage at the point of common coupling (PCC) within limits.
P-Q Mode	No	Controls the active power and the reactive power output separately, following the commands it is given.
Q-V Mode	No	Adjusts the reactive power by itself according to the grid voltage, to support the voltage at the point of common coupling (PCC).
SPF Mode	No	Adjusts the target power factor by itself with the active power output, to meet the grid's reactive power requirements.

Grid Settings — Page 4/5



Grid Protection1 4/5 screen (default values)

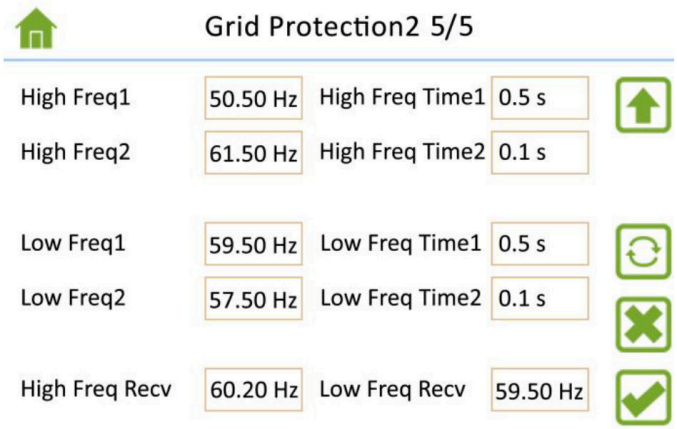
Setting	Default	What it does
High Volt1 / High Volt2	115.0% 125.0%	Level 1 and Level 2 over-voltage protection points, as a percentage of the nominal voltage.
High Volt Time1 / High Volt Time2	1.5 s 0.1 s	Time delays before Level 1 and Level 2 over-voltage protection trips.
Low Volt1 / Low Volt2 / Low Volt3	85.0% 45.0% 45.0%	Level 1, Level 2 and Level 3 under-voltage protection points, also as a percentage of the nominal voltage.
Low Volt Time1 / Low Volt Time2 / Low Volt Time3	2 s 0.2 s 0.2 s	Time delays before Level 1, Level 2 and Level 3 under-voltage protection trips.
High Volt Recv	110.0%	Grid over-voltage recovery point.
Low Volt Recv	85.0%	Grid under-voltage recovery point.

Example: to change the grid voltage range the inverter accepts for grid connection, you only need to set **High Volt1** and **Low Volt1**.

Nominal voltage	High Volt1 at 120%	Low Volt1 at 80%
240 Vac (120/240 V split phase)	Over-voltage point 288 Vac	Under-voltage point 192 Vac
208 Vac (120/208 V)	Over-voltage point 249.6 Vac	Under-voltage point 166.4 Vac

The percentages work the same way on any nominal voltage.

Grid Settings — Page 5/5



Grid Protection2 5/5 screen (default values)

Setting	Default	What it does
High Freq1 / High Freq2	50.50 Hz (see note) 61.50 Hz	First- and second-level over-frequency protection points (in Hz). These values are based on the system's nominal frequency (60 Hz in the US).
High Freq Time1 / High Freq Time2	0.5 s 0.1 s	The delay time before protection acts when the grid frequency goes above the matching point.
Low Freq1 / Low Freq2	59.50 Hz 57.50 Hz	First- and second-level under-frequency protection points. If the frequency falls below these values, protection trips.
Low Freq Time1 / Low Freq Time2	0.5 s 0.1 s	The delay time before under-frequency protection acts once the frequency drops below the set limits.
High Freq Recv	60.20 Hz	The recovery point for over-frequency protection. When the frequency comes back down below this value, the inverter reconnects to the grid or resumes operation.
Low Freq Recv	59.50 Hz	The recovery point for under-frequency protection. The inverter reconnects or resumes operation when the grid frequency rises above this value.

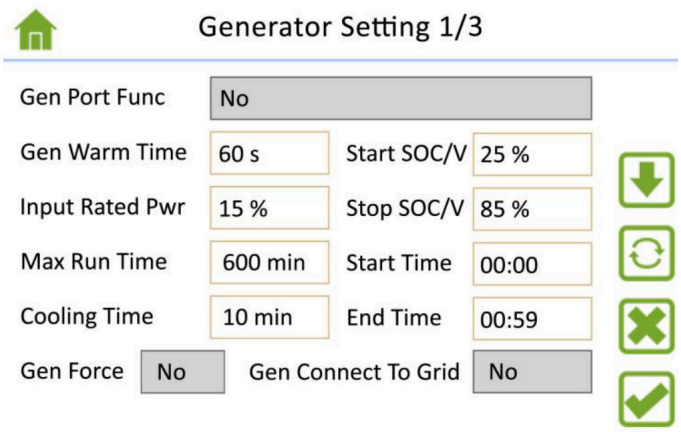
> **NOTE** The factory default screen shows High Freq1 = 50.50 Hz, which is below the 60 Hz grid frequency. Confirm this value with Fantastech before you rely on it.

Example: the system nominal frequency is 60 Hz, with these settings:

- **High Freq1 = 62 Hz, Low Freq1 = 57 Hz:** the system trips first-level protection when the frequency goes above 62 Hz or below 57 Hz, after the set time delays.
- **High Freq2 = 63 Hz, Low Freq2 = 56 Hz:** larger deviations trip second-level protection.

9.9 Generator Port Settings Menu

Gen Port Function



Generator Setting 1/3 screen (default values)

Gen Port Func selects what the generator port is used for. The choices are:

- **No:** no function is assigned. This is the default setting.
- **Gen:** the port is an input for generator power. Once the generator starts, its power is used to charge the battery.
- **Smart Load:** the port is a dedicated load output, with working conditions you set for smart-load control (see Smart Load Configuration below and section 7.4).
- **AC Couple:** an external PV inverter is connected to the generator terminal. When this function is on, the power from the external PV inverter can be used (see AC Coupling Function below and section 7.5).

When **Gen Port Func** is set to **Gen**, these settings take effect:

Setting	Default	What it does
Gen Warm Time	60 s	Generator warm-up time. After the inverter sends the start signal, it waits this long before it accepts generator power.
Start SOC/V / Stop SOC/V	25% 85%	The battery SOC points at which the generator starts and stops automatically. SOC is used for lithium batteries and voltage for other batteries (section 7.3).
Input Rated Pwr	15	The maximum power output of the generator. The default screen shows "15%"; this setting is a generator power value.
Max Run Time	600 min	The longest time the generator runs without stopping. Used when the inverter controls the generator automatically.
Cooling Time	10 min	The generator cool-down period after it stops. Used when the inverter controls the generator.
Start Time / End Time	00:00 00:59	The time window when the generator may run and charge the battery (section 7.3). See the Fantastech rule below.
Gen Force	No	Force-starts the generator. Turn it on the first time a generator is connected, so the inverter recognizes and activates the generator.
Gen Connect To Grid	No	Set it to **Yes** when the generator is wired to the GRID terminal (see below).

NOTE Every KOOR inverter has generator auto-start. The inverter starts and stops the generator through the GEN_start dry contact (wiring in section 6.8.1). The generator starts when the battery SOC falls to the **Start SOC/V** setting and stops when it reaches the **Stop SOC/V** setting. Section 6.8.2 lists all start and stop conditions. The GEN breaker is rated 63 A: the generator input must not exceed 63 A.

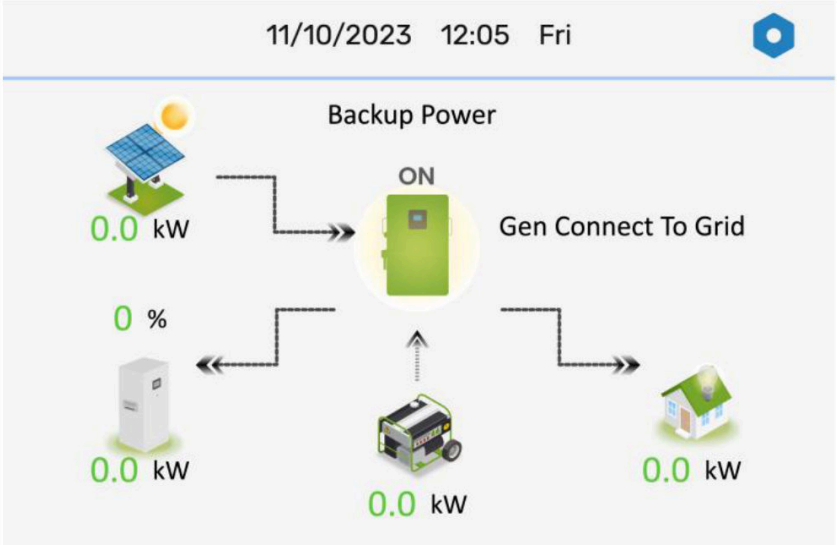
FANTASTECH RULE **Generator charge time:** the factory default window (Start 00:00, End 00:59) lets the generator charge for only 59 minutes a day. Set the **Start Time** 1 minute after the **End Time** (End Time 00:00, Start Time 00:01), so the battery charges whenever the generator runs. Keep the generator **Start SOC/V** higher than the **Off-Grid Bat Stop SOC** (section 9.6). Which port the generator goes on: section 6.8.1.

Gen Connect To Grid

This setting lets you connect the generator to the GRID terminal when the utility grid is not available. The GEN terminal has a maximum input power of 15 kW, which is lower than the GRID terminal. The generator power is limited by the **Input Rated Pwr** setting. A generator on the GRID terminal gives better use of its power. Its data is shown on the Generator/AC Couple page.

- This function cannot be used together with the Gen or AC Couple port function. Set **Gen Port Func** to **No**.
- If the unused GEN terminal is wired to a smart load, set **Gen Port Func** to **Smart Load**.
- When this function is on, the main screen changes: it shows the generator and "Gen Connect To Grid".

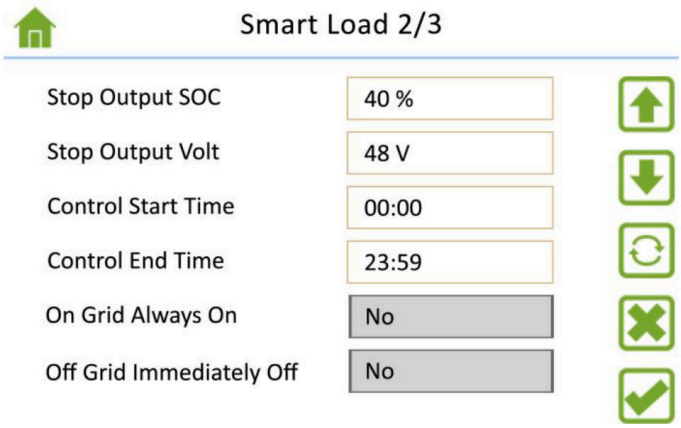
FANTASTECH RULE **Generator on the GRID port:** set **Gen Connect To Grid** to **Yes** and **Gen Port Func** to **No**. The **Start SOC/V** and **Stop SOC/V** settings still start and stop the generator. A generator on the GRID port does charge the battery in Backup Power mode.



Main screen with Gen Connect To Grid turned on

Smart Load Configuration

This page takes effect only when **Gen Port Func** is set to **Smart Load**. In this mode the port is a dedicated load output, with working conditions you set for smart-load control.



Smart Load 2/3 screen (default values)

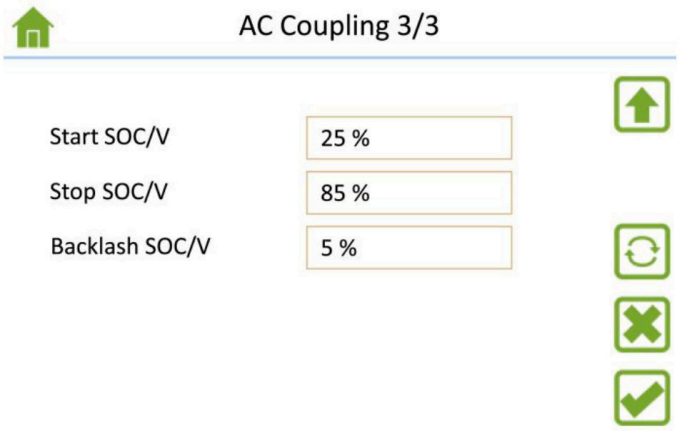
Setting	Default	What it does
Stop Output SOC	40%	When the battery SOC drops to this point, the smart load output turns off by itself.
Stop Output Volt	48 V	If voltage control is selected, the smart load output turns off when the battery voltage reaches this value.
Control Start Time	00:00	The start of the time window in which the SOC or voltage control conditions apply.
Control End Time	23:59	The end of the time window in which the SOC or voltage control conditions apply.
On Grid Always On	No	When enabled, the smart load output stays on all the time while the inverter is on-grid.
Off Grid Immediately Off	No	When enabled, the smart load output turns off at once when the inverter goes off-grid.

On Grid Always On and **Off Grid Immediately Off** have higher priority. They override the other smart load settings.

> **NOTICE** Outside the set time window, the smart load output is not limited by the SOC or voltage conditions. It follows the same discharge behavior as the EPS output.

AC Coupling Function

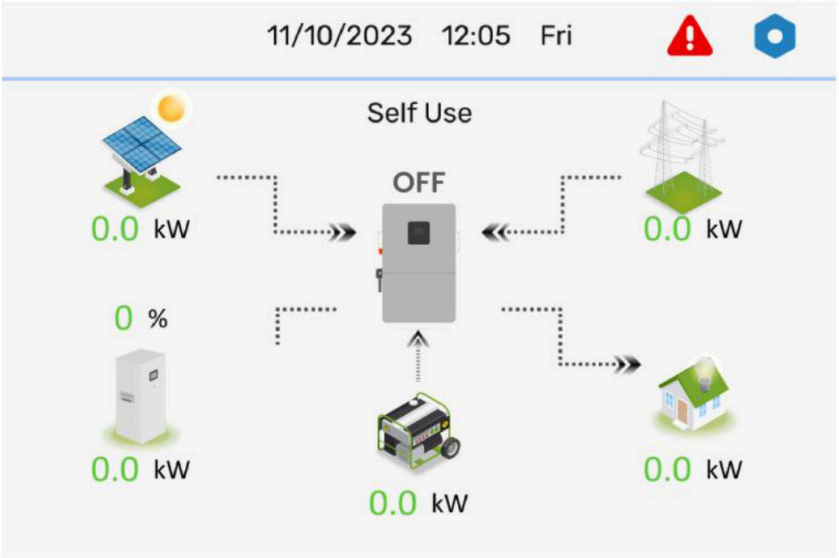
The AC Coupling function lets a PV inverter be connected to the generator port. When the conditions are met, power from that PV inverter can flow into this inverter. Section 7.5 explains the function. Use SOC values for lithium batteries and voltage values for other batteries.



AC Coupling 3/3 screen (default values)

Setting	Default	What it does
Start SOC/V	25%	The battery SOC or voltage point that turns AC coupling on.
Stop SOC/V	85%	The battery SOC or voltage point that turns AC coupling off.
Backlash SOC/V	5%	The backlash (hysteresis) value for the AC coupling Start and Stop SOC or voltage points.

Accessing Generator Data from the Main Screen

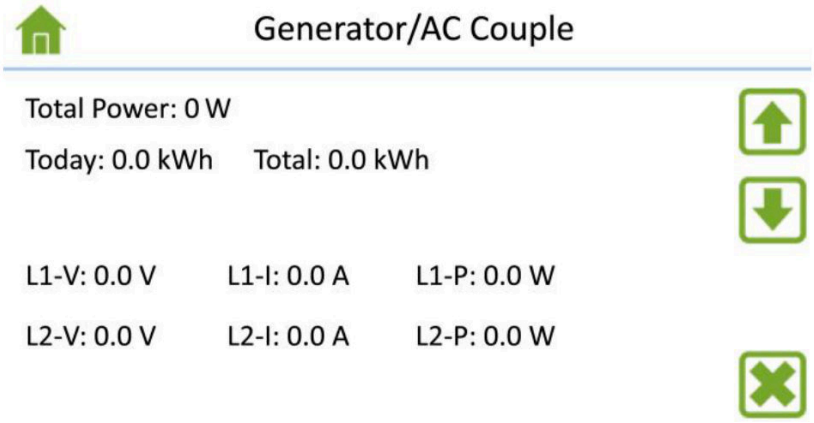


Main screen with the generator icon (bottom center)

Once the generator function is turned on, a generator icon appears on the main screen. Tap this icon to open the Generator Details Page.

Generator Details Page

The **Generator/AC Couple** page shows live readings from the generator. What this page shows can vary with the setup of each system.



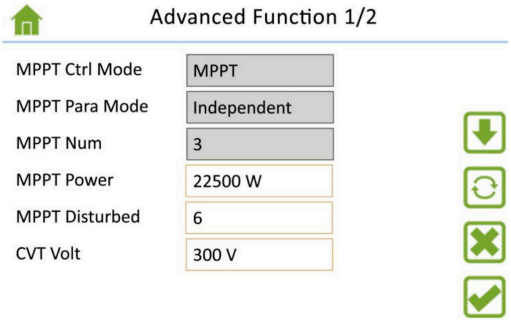
Generator/AC Couple details page

- **Total Power:** the total generator power output right now.
- **Today:** the energy generated today.
- **Total:** the total energy generated to date.
- **L1 / L2:** the voltage, current and power of each line.

9.10 Advanced Settings Menu

The **Advanced Function** menu has two screens. Screen 1/2 holds the PV tracking (MPPT) settings. Screen 2/2 holds the parallel settings, the Wi-Fi baud rate and the EMS communication settings.

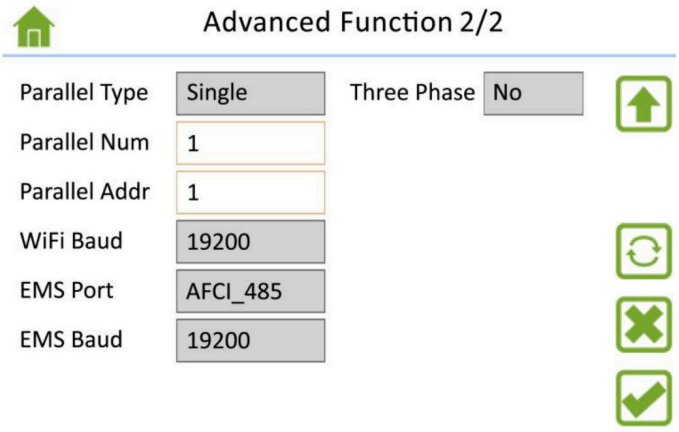
Advanced Function, screen 1/2: PV tracking (MPPT)



Advanced Function, screen 1/2

Setting	What it does
MPPT Ctrl Mode	Picks how the PV inputs are tracked. **MPPT** = maximum power point tracking. **CVT** = constant voltage tracking.
MPPT Para Mode	**Independent** = each MPPT is controlled on its own. **Parallel** = all PV channels are controlled together.
MPPT Num	The number of PV strings. The screen shows 3.
MPPT Power	Sets the MPPT power limit. The screen shows 22500 W.
MPPT Disturbed	Sets the MPPT perturbation step (the size of each small change the tracker makes while it looks for the maximum power point). The screen shows 6.
CVT Volt	Sets the voltage used in CVT (constant voltage tracking) mode. The screen shows 300 V.

Advanced Function, screen 2/2: parallel, Wi-Fi and EMS

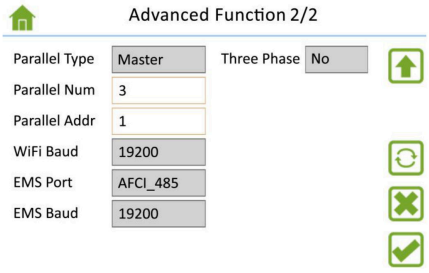


Advanced Function, screen 2/2 (single inverter)

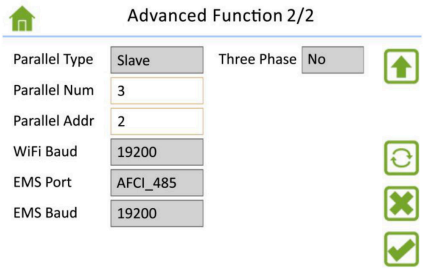
Setting (screen name)	Default	What it does
1. Parallel settings		
Parallel Type	Single	Sets the parallel operating mode of the system. The screens in this section show **Single**, **Master** and **Slave**.
Parallel Num	1	Sets the total number of inverters in the system (master + slaves).
Parallel Addr	1	Sets the address of this inverter in the system. The master inverter is always address 1. The slave inverters follow in order (2, 3, ...).
2. Wi-Fi and EMS communication settings		
WiFi Baud	19200	Sets the baud rate for communication between the inverter and the data logger.
EMS Port	AFCI_485	Picks the spare RS485 port on the inverter that is used for EMS (Energy Management System) communication. You can change it to suit the actual use.
EMS Baud	19200	Sets the communication baud rate for the EMS port.
3. Three-phase setting		
Three Phase	No	Turns on three-phase group parallel operation of the inverters. It is used only for a three-phase parallel system (section 6.9). The split-phase example below uses **No**.

Example: parallel system with three inverters

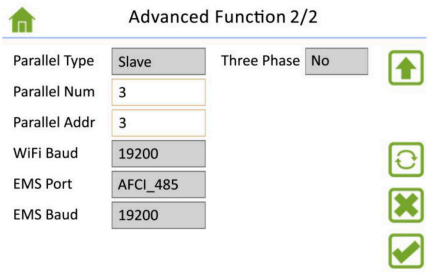
Wire the parallel system and set the DIP switches first (section 6.9). The screens below then show how to set up a parallel system with one master inverter and two slave inverters. Every setting must be the same on all units except **Parallel Type** (Master on one unit, Slave on the rest) and **Parallel Addr**. Parallel Addr must be different on each unit.



(1) Master inverter



(2) Slave inverter 1



(3) Slave inverter 2

Setting	(1) Master inverter	(2) Slave inverter 1	(3) Slave inverter 2
Parallel Type	Master	Slave	Slave
Parallel Num	3 (total number of inverters)	3	3
Parallel Addr	1	2	3
WiFi Baud	19200	19200	19200
EMS Port	AFCI_485	AFCI_485	AFCI_485
EMS Baud	19200	19200	19200
Three Phase	No	No	No
Role	This inverter is the master unit. It controls and synchronizes the whole system.	This inverter works as Slave 1, with address 2.	This inverter works as Slave 2, with address 3.

NOTICE

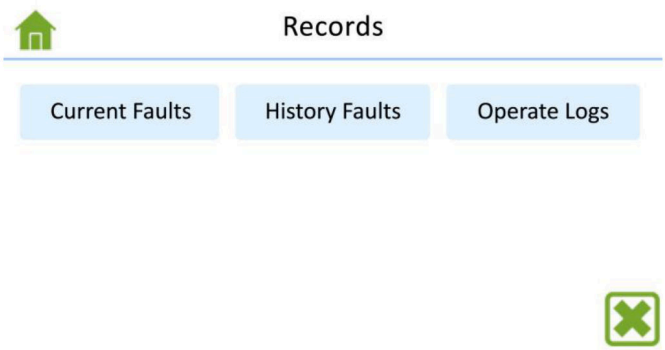
1. All inverters must have the same WiFi Baud rate and the same EMS Baud rate.
2. Make sure the wiring and grounding between all units are correct.
3. Make sure all inverters have the same firmware version.
4. Communication cables (for example RS485 or CAN) must be connected securely.

9.11 Record Menu

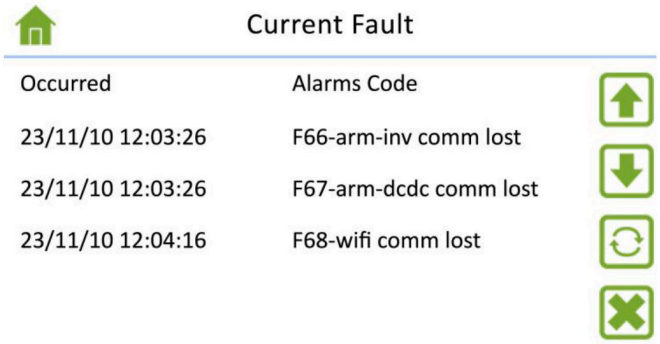
The **Records** page holds the fault logs and the operation log. It has three buttons:

- **Current Faults:** shows the active faults and alarms of the inverter. Each entry gives the time stamp, the fault code and a description of the current problem.
- **History Faults:** shows the fault and alarm history of the inverter. It records the start time, the recovery time and the details of each fault event.
- **Operate Logs:** lists user operations, such as changing the inverter mode, turning the system on or off, or changing system settings. Use these logs to track system changes and user actions.

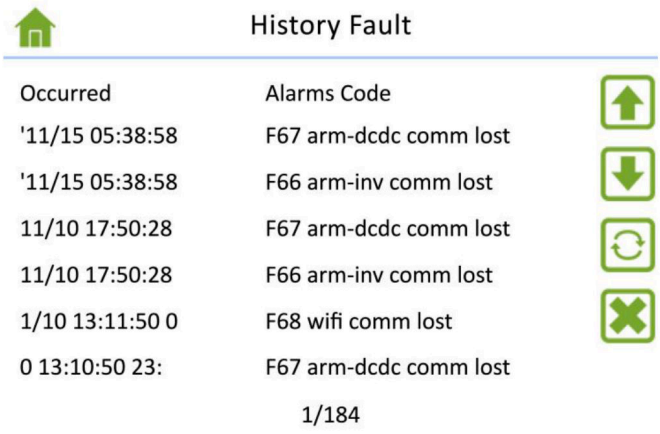
The fault and alarm codes are explained in chapter 10 (Troubleshooting).



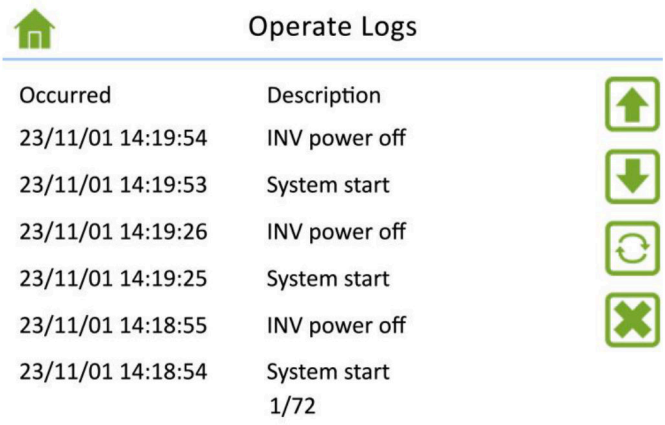
Records page



Current Fault page (example entries)



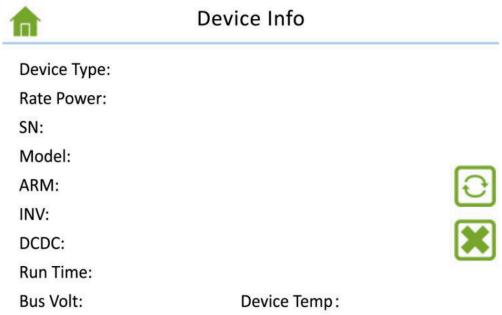
History Fault page (example entries; the time column on this factory sample screen is garbled)



Operate Logs page (example entries)

9.12 Device Info Setup Menu

The **Device Info** page shows basic information about the inverter. It is useful for system diagnostics, for keeping track of firmware versions, and for technical support.



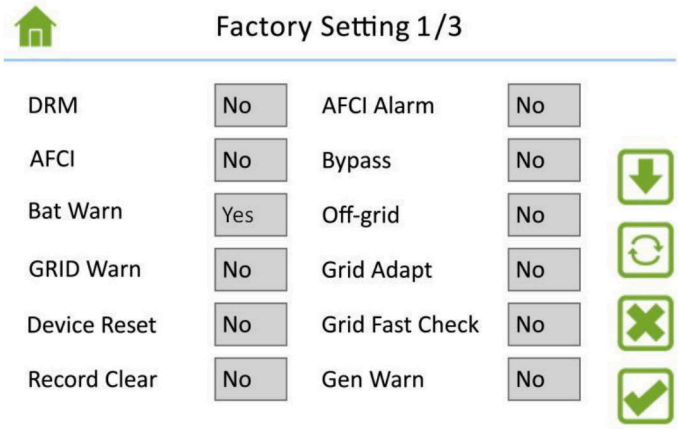
Device Info page

Screen label	What it shows
Device Type	The category or class of the inverter.
Rate Power	The rated power of the inverter.
SN	The serial number: the unique number given to each unit.
Model	The model number of the inverter.
ARM / INV / DCDC	The software (firmware) versions now installed: ARM, INV and DCDC.
Run Time	The total running time of the inverter since it was first started.
Bus Volt	The bus voltage of the inverter.
Device Temp	The temperature of the inverter.

9.13 Factory Setting Menu

The **Factory Setting** menu has three screens.

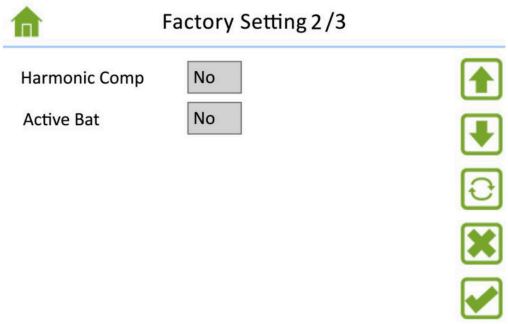
Factory Setting, screen 1/3



Factory Setting, screen 1/3

Setting (screen name)	Default	What it does
DRM	No	When enabled, the inverter can receive external control signals. An external DRM (demand response mode) controller must be connected.
AFCI	No	Arc-fault detection. This function works only when the arc-fault module is enabled.
Bat Warn	Yes	When enabled, the screen shows the battery alarms the inverter detects, such as the battery-disconnected alarm and the DC battery over-voltage alarm.
GRID Warn	No	When enabled, the screen shows the grid alarms the inverter detects, such as the grid over-voltage alarm and the grid over-frequency alarm.
Device Reset	No	Restores the factory defaults. All user settings and basic control settings go back to the factory values. The fault records, alarm records and energy statistics are cleared at the same time.
Record Clear	No	Clears all records: the fault and alarm records and the energy statistics.
AFCI Alarm	No	Clears AFCI (arc-fault) alarms by hand.
Bypass	No	When enabled, the inverter switches its output to bypass after the power conversion system (PCS) shuts down.
Off-grid	No	When enabled, the system switches itself to off-grid mode to supply the loads. Use it where the customer requires that no power flows back into the grid (anti-backflow).
Grid Adapt	No	For areas with a poor (weak or unstable) grid. Contact Fantastech (574-218-1973, derek@fantastech.org) before you turn it on. Do not change it without approval.
Grid Fast Check	No	When enabled, this setting is linked to the EPS Enable setting.
Gen Warn	No	Turns on the generator voltage alarm. When a generator mode is selected and the generator port detects no voltage, the inverter reports **Gen Voltage Abnormal**.

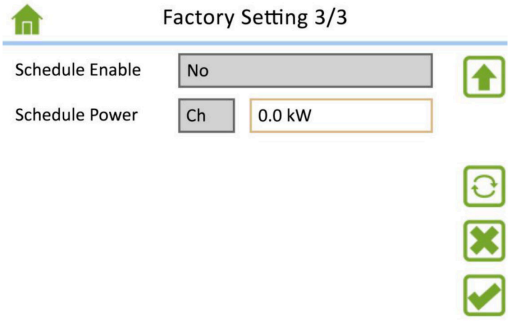
> **FANTASTECH RULE** On off-grid jobs keep **GRID Warn** at **No**. With no grid connected, it would only show false grid alarms.



Factory Setting, screen 2/3

Factory Setting, screen 2/3

Setting	Default	What it does
Harmonic Comp	No	When enabled, the inverter compensates for harmonic currents on the grid side. This reduces waveform distortion and improves power quality.
Active Bat	No	When the battery goes into sleep or lock state because of under-voltage, the system draws power from the PV or the grid to wake the battery up and restore its ability to charge and discharge.



Factory Setting, screen 3/3

Factory Setting, screen 3/3

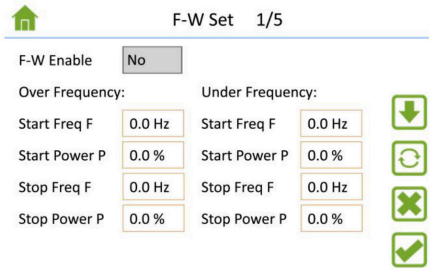
Setting	Default	What it does
Schedule Enable	No	Controls the inverter power for testing (debugging). When enabled, the inverter runs on the charge/discharge power schedule you set. This function has a higher priority than the normal operating modes. Turn it off when testing is done.
Schedule Power	Ch 0.0 kW	Sets charge or discharge (left box) and the scheduled power value in kW (right box).

9.14 Grid Curve (Puerto Rico)

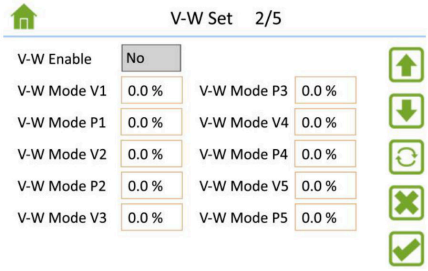
These five screens hold the grid-support curves used for grid certification and utility review in Puerto Rico. Contact Fantastech (574-218-1973, derek@fantastech.org) or your installer before you change any setting here.

- **F-W Set (1/5):** frequency-watt curve. Over-frequency and under-frequency start and stop points (frequency in Hz, power in %).
- **V-W Set (2/5):** volt-watt curve, points V1–V5 and P1–P5.
- **V-Q Set (3/5):** volt-var curve, points V1–V5 and Q1–Q5.
- **P-Q Set (4/5):** watt-var curve, points P1–P5 and Q1–Q5.
- **P-F Set (5/5):** points P1–P5 and F1–F5.

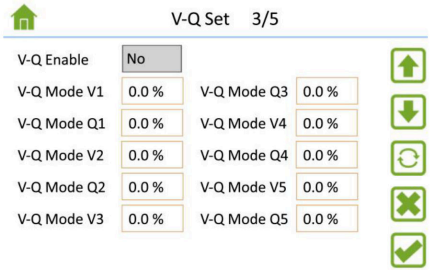
Each screen has its own **Enable** setting. On the screens shown, every curve is set to **No** and every point is 0.0.



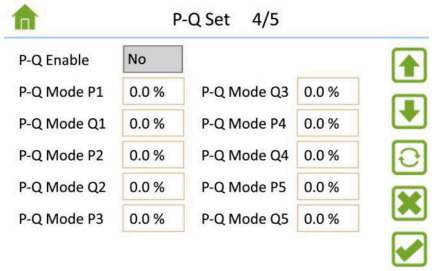
F-W Set, screen 1/5



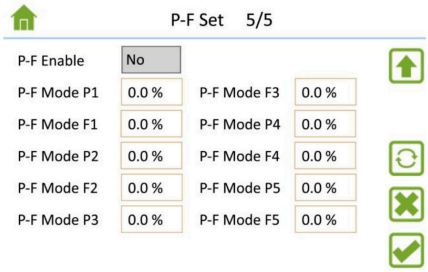
V-W Set, screen 2/5



V-Q Set, screen 3/5



P-Q Set, screen 4/5



P-F Set, screen 5/5

10. Troubleshooting

This chapter helps you find and fix most problems that can occur with the KR-L8K–L15K-SPUS series inverters. It helps you narrow down the source of a problem. Read the troubleshooting steps below.

1. Check the inverter screen for warning or fault messages and fault codes. The **Current Fault** list in the Record menu (section 9.11) shows the active faults and alarms.
2. If a message is shown, write it down before you do anything else.
3. Try the fix given for that code in the tables below.
4. If the problem continues, contact Fantastech (574-218-1973, derek@fantastech.org).

> **NOTE** Short words used in the code names: inv = inverter; bat / BAT = battery; bms = battery management system; volt = voltage; curr = current; OV = over-voltage; OVP = over-voltage protection; UVP = under-voltage protection; OCP = over-current protection; chg = charge; dischg = discharge; Ch&Disch = charge and discharge; Intermed = intermediate; temp = temperature; comm = communication; para = parameter; err = error; SW = software; HW = hardware; EPO = emergency power off; NTC = temperature sensor; RSD = rapid shutdown; GARDA = power grid amplitude quick check alarm.

Fault codes

Code	Description	What to do
F01	inv over volt	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F02	inv under volt	
F03	short circuit	
F04	inv over curr	
F05	over load	The load is too high, or the inverter has derated (lowered its output). Reduce the load. If the problem continues, contact Fantastech.
F07	inv bus over volt	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F08	inv bus under volt	
F09	Insulation	Turn off the AC grid breaker and the inverter's main DC breaker, and unplug all PV string MC4 connectors. Shut down the whole unit and wait 15 minutes for the capacitors to discharge fully. Then restart the inverter and check whether the fault is gone. If the problem continues, contact Fantastech.
F10	leakage self-test fail	Turn off the PV DC breaker, the lithium battery breaker and the generator AC breaker. Power down the whole unit and wait 5 minutes for it to discharge fully and clear the fault. Then restart the inverter. Check whether the problem is gone. If the problem continues, contact Fantastech.
F11	leakage over limit	
F12	relay self-test fail	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F13	inv locked	
F14	busbar buffer fail	
F15	inv over temp	

Code	Description	What to do
F17	parallel total count err	Check that the parallel communication cable is connected correctly. If the problem continues, contact Fantastech.
F18	parallel comm fail	
F19	parallel SW mismatch	
F20	parallel HW mismatch	
F21	parallel model mismatch	
F22	parallel para sync	
F23	parallel master conflict	
F24	parallel slave id conflict	
F25	parallel signal err	
F26	master no slave	
F33	inv flash fault	The inverter's internal communication or data storage has an error. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F34	inv-dc comm fault	
F35	inv phase-locked	
F36	inner para match fail	
F37	self-test fail	This self-test checks the grid-protection limits. It belongs to the Italian grid standard, which a US unit does not use. The unit cannot connect to the grid if the check fails. The causes fall into five groups: grid problems, DC insulation failure, hardware relay faults, setting errors and wiring problems. Turn off the AC grid breaker, the PV DC breaker and the lithium battery breaker. Shut down the whole unit and wait 10 minutes for the bus capacitors to discharge fully. Then turn the unit back on to reset it. This clears about 80% of temporary fault alerts. Check whether the problem is gone. If the problem continues, contact Fantastech.
F38	inv master fault	The inverter's internal communication or data storage has an error. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F39	inv parallel comm fail	
F40	inv parallel signal err	
F41	inv curr DC component high	
F42	inv parallel overload	
F43	inv load curr unbalance	
F44	inv parallel system fault	

Code	Description	What to do
F49	dc bus over volt	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F50	dc bus under volt	
F51	Intermed bus OVP	
F52	Intermed bus UVP	
F53	dischg OCP	
F54	chg OCP	
F55	LLC HW OCP	
F56	dc bat over volt	
F57	dc bat under volt	
F60	bat reverse connect	Check that the positive (+) and negative (–) battery power cables are not reversed.
F63	dc over temp	Make sure the inverter is out of direct sunlight. Make sure it is in a cool, well-ventilated area. Make sure it is mounted vertically and the ambient temperature is below the inverter's upper temperature limit.
F64	Fan fault	First turn off the inverter. Then ask Fantastech whether your software (firmware) version is the latest one. If it is not, upgrade the firmware.
F65	meter comm lost	Check that the meter communication cable is in good condition.
F66	arm-inv comm lost	The inverter's internal communication or data storage has an error. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F67	arm-dcdc comm lost	
F68	wifi comm lost	Check the status lights on the Wi-Fi logger stick (chapter 8 explains the lights). Make sure the stick is pushed all the way in and locked, and that it is joined to the site Wi-Fi. If the problem continues, contact Fantastech.
F69	bms-can comm lost	Make sure the battery you use is compatible with the inverter. Check that the communication cables and ports between the battery and the inverter are connected correctly.
F70	dsp para match err	The inverter's internal communication or data storage has an error. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F71	EPO	If the inverter does not go back to normal, contact Fantastech.
F72	ambient temp high	Make sure the inverter is out of direct sunlight. Make sure it is in a cool, well-ventilated area. Make sure it is mounted vertically and the ambient temperature is below the inverter's upper temperature limit.
F73	bms-485 comm lost	Make sure the battery you use is compatible with the inverter. Check that the communication cables and ports between the battery and the inverter are connected correctly.

Code	Description	What to do
F74	arm flash error	The inverter's internal communication or data storage has an error. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F75	NTC disconnect	
F76	dsp para set fail	
F81	dc-inv comm lost	
F82	dc flash fault	
F83	dc fault locked	
F86	PV1 over volt	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F87	PV2 over volt	
F88	PV1 over curr	
F89	PV2 over curr	
F90	PV1 reverse connect	Check that the PV cables are connected correctly.
F91	PV2 reverse connect	
F93	PV3 over volt	The inverter has a fault. Turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
F94	PV3 over curr	
F95	PV3 reverse connect	Check that the PV cables are connected correctly.
F96	DC insulation	If the inverter does not go back to normal, contact Fantastech.

Warning codes

Code	Description	What to do
W01	Over load	The load is too high, or the inverter has derated (lowered its output). Reduce the load. If the problem continues, contact Fantastech.
W02	INV over temp	Make sure the inverter is out of direct sunlight. Make sure it is in a cool, well-ventilated area. Make sure it is mounted vertically and the ambient temperature is below the inverter's upper temperature limit.
W03	Grid over frequency	The power grid is abnormal. The inverter goes back to normal operation by itself after the grid recovers. If the problem continues, contact Fantastech.
W04	Grid under frequency	
W05	Grid over volt	
W06	Grid under volt	
W07	Gen volt abnormal	Measure the generator voltage with a multimeter. A voltage or frequency outside the accepted range triggers this alarm. The generator must match the inverter's rated grid voltage and frequency: 120/240 Vac split phase or 208 Vac, 60 Hz (section 3.3.4 Specification). Power off the whole unit and wait 10 minutes for the bus capacitors to discharge fully. Then restart the inverter. Check whether the problem is gone. If the problem continues, contact Fantastech.
W08	Gen freq abnormal	Check that the frequency setting of the generator matches the frequency setting of the inverter. If they match, power off the whole unit and wait 10 minutes for the bus capacitors to discharge fully. Then restart the inverter. Check whether the problem is gone. If the problem continues, contact Fantastech.

Code	Description	What to do
W09	DC BAT over volt	If the lithium battery has an internal fault: turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter and the lithium battery back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
W10	DC BAT under volt	
W11	BAT none-connected (battery not connected)	
W12	DC over temp	Make sure the inverter is out of direct sunlight. Make sure it is in a cool, well-ventilated area. Make sure it is mounted vertically and the ambient temperature is below the inverter's upper temperature limit.
W13	grid rapid inspection	The power grid is abnormal. The inverter goes back to normal operation by itself after the grid recovers. If the problem continues, contact Fantastech.
W14	BAT Ch&Disch limit	If the lithium battery has an internal fault: turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter and the lithium battery back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
W16	RSD disconnected	Check that the RSD (rapid shutdown) switch is on. Power off the whole unit and wait 5 minutes for it to discharge fully and clear the fault. Then restart the inverter. Check whether the problem is gone. If the problem continues, contact Fantastech.
W19	Grid lost phase	The power grid is abnormal. The inverter goes back to normal operation by itself after the grid recovers. If the problem continues, contact Fantastech. Note: GARDA = power grid amplitude quick check alarm.
W20	Grid long time OV	
W21	Grid phase error	
W22	GARDA	
W23	Grid DC component rapid detection	
W24	Grid phase rapid inspection	
W25	PV1 under volt	If the PV has an internal fault: turn off the PV, grid and battery. Wait 5 minutes, then turn the inverter and the lithium battery back on. Check whether the problem is gone. If the problem continues, contact Fantastech.
W26	PV2 under volt	
W27	PV3 under volt	

NOTICE

If the inverter's information panel does not show a fault light, go through this list. Make sure the present state of the installation lets the unit work properly.

- Is the inverter in a clean, dry place with enough ventilation?
- Are the DC input breakers turned on (closed)?
- Are the cables the right size and short enough?
- Are the input and output connections and wiring in good condition?
- Are the configuration settings correct for your installation?
- Are the display panel and the communication cable connected correctly and undamaged?

For more help, contact Fantastech (574-218-1973, derek@fantastech.org). Be ready to describe your system installation and to give the model and serial number of the unit.

11. Maintenance

11.1 Power On the Inverter for the First Time

DANGER Follow these steps, in this order, to turn on the inverter. This is the same order as section 7.1.4.

1. Turn on the battery breaker(s).
2. Press the power button down to ON. Wait for the LED indicator to turn on (this may take a few minutes).
3. Turn on the PV DC disconnect switch (turn the PV switch to the ON position).
4. Turn on the AC circuit breaker between the inverter GRID port and the power grid. Then turn on the AC circuit breaker between the inverter LOAD port and the emergency (backup) loads, and the GEN breaker if a generator is connected.
5. Send the startup command by hand on the inverter screen: **Power On/Off** on Work Mode page 2/4 (section 9.6). For safety, you can set **Auto Power On** after this first power-on so the inverter starts by itself.
6. The inverter should now start running.

11.2 Power Off the Inverter

DANGER

- Power off the inverter before any operation or maintenance work. Otherwise, electric shock may occur.
- Delayed discharge: after power off, wait at least 5 minutes until the internal components have discharged (the label on the unit says 5 minutes).

1. Turn off all AC breakers (GRID, GEN and LOAD). This is the start-up order in reverse, the same as section 7.1.5.
2. Turn off the built-in PV switch of the inverter.
3. Press the power button to the OFF position.
4. Turn off the battery breaker between the inverter and the battery.

11.3 Removing the Inverter

WARNING Make sure the inverter is powered off. Wear proper PPE before any work.

1. Disconnect all cables: DC, AC, communication, the communication module and PE (ground).
2. Remove the inverter from the mounting plate.
3. Remove the mounting plate.
4. Store the inverter properly. If it will be used later, store it as section 4.3 Storage requires.

11.4 Disposing of the Inverter

If the inverter can no longer work, dispose of it according to the local disposal rules. Do not throw the inverter away with household waste.

> **WARNING** Make sure the inverter is powered off. Wear proper PPE before any work.

12. Appendix

12.1 Routine Maintenance

Item	What to do	How often
System cleaning	Check the heat sink, air intake and air outlet for foreign matter or dust.	Once every 6–12 months
PV switch	Turn the DC switch on and off ten times in a row to make sure it works properly.	Once a year
Electrical connections	Check that the cables are connected securely. Check whether any cable is broken or has exposed copper core.	Once every 6–12 months
Sealing	Check that all terminals and ports are sealed properly. Reseal a cable hole if it is not sealed or is too big.	Once a year

12.2 Limited Warranty

KOOR All-in-One inverters are warranted by Fantastech Energy Solution LLC for 10 years from the date of installation. The full terms are on the Limited Warranty — Terms page at the front of this manual, and the registration card is at the back.

The warranty does not cover any product or product part that has been modified or damaged by any of the following:

- Installation or removal (for example: batteries of the wrong voltage, batteries connected backward, water or rain damage to electronics, preventable damage to solar wires).
- Alteration or disassembly.
- Normal wear and tear.
- Accident or abuse.
- Unauthorized firmware or software updates, or changes to the software code.
- Corrosion.
- Repair or service by an unauthorized repair facility.
- Operation or installation that goes against the KOOR product instructions.
- Fire, floods or acts of nature.
- Shipping or transportation.
- Incidental or consequential damage caused by other components of the power system.
- Any product whose serial number has been altered, defaced or removed.

WARNING Fantastech Energy Solution LLC expressly disclaims any responsibility for performance problems caused by improper installation. Installers and users are solely responsible for following the proper installation procedures in the documentation provided. Fantastech Energy Solution LLC disclaims any liability for changes to the installation that might cause electrical malfunctions or any other problems related to the KOOR product.

Notes

Warranty Registration Card

KOOR All-in-One Hybrid Inverter · KR-L8K-SPUS · KR-L10K-SPUS · KR-L12K-SPUS · KR-L15K-SPUS

FANTASTECH LIMITED WARRANTY

KOOR All-in-One Hybrid Inverter and KOOR batteries — from the date of installation

10 YEARS

Fill this card out when the system is turned on, keep a copy with your invoice, and send it back to Fantastech. Registering takes two minutes, and it is what lets us handle a claim fast if you ever need one.

Photocopy this page, or fill it in and send a photo. Mail the card to Fantastech, or e-mail a clear photo of it to derek@fantastech.org.

OWNER

Customer name

Phone

E-mail

Install address — street, city, state, ZIP

EQUIPMENT

Inverter model

Inverter serial number (S/N)

Battery model

Battery serial number(s)

Fantastech order number

Purchase date

Dealer / installer

Commissioning date

SEND THE FINISHED CARD TO FANTASTECH

E-mail — fastest

derek@fantastech.org

Snap a picture of this card with your phone and send it. Include a picture of your invoice.

Mail

Fantastech Energy Solution LLC
3645 S 1200 W
Goshen, IN 46528

Phone

574-218-1973

Call us with the serial number and we will fill it in for you.

KOOR

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